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Novel Statistical Learning Approaches for Open Banking-Type Data



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Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

(Andrej Svetlošák)

To everyone who supported and helped me on this journey.

Lay Summary

Open banking legislation provides a framework under which, given the customer permission, financial institutions share information about customers, such as account information and transactions, with each other and, in some cases, with third parties. This paradigm shift in the way financial institutions can access information has far-reaching implications. For example, financial institutions in the UK have access to information on all UK accounts of their current and potential customers. This gives them a detailed overview of an individual's personal finances, and allows them to offer and provide services with an unprecedented level of personalisation and accuracy. Open banking, with all its promises to innovate and improve the financial sector, brings with it a host of new challenges that need to be addressed by novel statistical learning techniques. In this work, we develop ways to use newly available open banking-type data for personal finance analysis and market segmentation. The first statistical method developed here allows the comparison of the finances of a particular individual with the finances of a comparison group based on common characteristics. This can be done both in a given time period as well as over time. The identification of market segments is the focus of the second main topic of this thesis. The aim of is to assign individuals to homogeneous customer groups. Existing state-of-the-art methods can provide cluster solutions. However, they suffer from two major drawbacks. Using all available information existing methods will most likely create the same number of groups for each variable under consideration. For example, if age and income are used for customer segmentation, the number of customer groups with similar income will be the same as the number of groups of people with similar age. This seems rather arbitrary, and not reflective of reality. Furthermore, existing state-of-the-art models are computationally intensive for large datasets. To fully harness the richness of open banking-type data novel statistical learning tools are needed. The here proposed Reign-and-Conquer clustering segmentation algorithm allows the number of customer groups to vary across variables, while at the same time being computationally feasible. This thesis contributes to the development of new methods for solving statistical learning problems in an open banking context. The time-varying comparative mean value as well as the Reign-and-Conquer clustering algorithm presented here are tested and illustrated on both simulated and real data. We apply the methods to open banking-type datasets and highlight their uses and benefits for a new era of personal finance data analysis.

Abstract

This thesis develops novel statistical learning methods motivated by open banking-type data as specified in [European Commission \(2018\)](#) legislation. Open banking legislation provides a framework under which, given the customer permission, financial institutions share certain information about customers, such as data on accounts and transactions, among themselves, and in some cases, with third parties. The scale and nature of this newly accessible data poses challenges for private, non-profit and government institutions. This thesis introduces new statistical methods that address some of the identified challenges in the analysis and use of open banking-type data.

First, we propose an approach for comparing a specific reference subject (e.g. a customer of a financial institution or a user of a financial application) to a comparison group, defined via covariate truncation. The time-varying comparative mean value is developed here as a statistical method that learns about the dynamics governing how the response of a reference subject differs from that of a comparison group. We take advantage of local polynomial regression to provide a smooth estimator for the time-varying comparative mean value. By contrasting a specific customer against similar customers, our method offers interesting diagnostics that can be used by financial institutions to recommend personalised services, such as financial advice and products.

Another contribution of this thesis is the development of a clustering/market segmentation method that takes advantage of the positive attributes of model-based clustering, while addressing some of its pitfalls. Standard model-based clustering is likely to result in the same number of clusters per margin. This seems to be a rather fictitious assumption for a wide range of situations to model. Model-based clustering is also computationally intensive for large datasets. The here developed Reign-and-Conquer clustering addresses both of these drawbacks by modelling margins separately, allowing for different number of clusters on each margin, whereupon the joint is reconstructed using an game theory inspired algorithm. Since the method specifies only marginal models, it is partially parallelisable and therefore computationally appealing for large datasets, such as open banking-type data. Practical application and overall good performance are demonstrated on both artificial and real data scenarios.

This thesis concludes with a discussion of the here introduced methods, their potential extensions and opportunities for future research.

Acknowledgment

This thesis was supervised by Miguel de Carvalho, Raffaella Calabrese, Manuel Peleteiro and Bronius Baniulis, whose diligent, patient and constructive feedback not only shaped and enriched the here presented work, but also furthered my personal and professional development. Warmest thanks goes to Gabriel Martos Venturini, who co-authored Chapter [4](#) of this thesis. Sincere thanks to the participants of the Financial Health Check Data Fest 2017-2018, the International Workshop on Statistical Modelling 2019, the Credit Scoring and Credit Control Conference XVI, the 17-th conference of the International Federation of Classification Societies 2022, the 7-th edition of the workshop on Models and Learning in Clustering and Classification 2022 and The International Conference of the ERCIM WG on Computational and Methodological Statistics 2018, 2019 and 2022. Their interest, feedback and helpful suggestions have enhanced the research conducted and presented in this thesis. Likewise, many thanks to the staff of the School of Mathematics and the Business School of the University of Edinburgh. Our discussions and input have been instrumental in shaping my doctorate.

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Part I

Preparations

Chapter 1

Introduction and Motivation

This Chapter introduces open banking, which is the motivation and starting point for the concepts developed in this thesis.

1.1 Open Banking

In the following, we offer a snapshot of the fields of open banking (OB) and fintech, focusing on the aspects related to the motivation for our research. The rapidly developing literature on these areas is difficult to review in a few paragraphs, given the interest in the subject from a variety of fields ranging from computer science (Gomber et al., 2018; Wang et al., 2020), financial services (Mention, 2019), regulatory technologies (Buckley et al., 2020) to healthcare (Stranieri et al., 2021). Fintech and OB are seen as game-changing and disruptive innovations (Lee & Shin, 2018; Nicoletti et al., 2017; Omarini, 2018), providing opportunities and challenges for financial institutions, consumers, academics, regulators and governments alike.

The OB digital revolution is an initiative spearheaded by several countries, such as Australia, the US, the UK and the EU, with the latter embracing OB in the form of PSD2 legislation (Brodsky & Oakes, 2017; European Commission, 2018; He et al., 2020). The introduction of OB goes hand in hand with the currently expanding field of fintech. There is no universally accepted definition of fintech, but as (Mention, 2019, p. 9) points out, "Whatever it is, fintech is here to stay".

Fintech and OB are transforming the financial services sector (Nicoletti et al., 2017), with one of the main objectives being to enhance customer experience (Gomber et al., 2018). This has led to a surge of interest in designing customer-centric fintech services (Breidbach et al., 2019), with personalisation of these services being key to the success of OB.

Trendsetters such as Facebook (Cheng & Zagat, 2019) are already seizing upon the opportunities stemming from subject-to-group comparison in an OB context, i.e. differentiating between a reference subject and similar customers. These new technologies are aimed directly at consumers and therefore need to be intuitive and easy to interpret. In addition to industry demand,

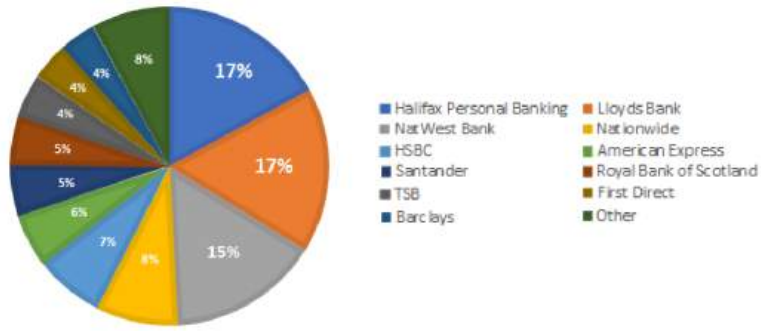


Figure 1.1: MD Data: Overview of financial institutions in the analysed MD data.

developments in fintech and OB have been accelerated by innovations in machine learning, artificial intelligence and related fields (Mention, 2019).

However, little has been done so far to address these new challenges from a statistical perspective. This thesis aims to fill this methodological and applicational gap by providing novel statistical learning methods that deliver useful, intuitive and easily interpretable OB-based results that can be used by institutions and clients alike.

1.2 Money Dashboard OB-type Data

Money Dashboard (MD) is a UK financial services provider that supplied the OB-type data for the purposes of this thesis. Among other services, MD provides customers with an overview of their categorised spending, budgeting and other personal accounting information. MD provides these services via an application. Further details on the business lines of MD can be found at

www.moneydashboard.com

This thesis analyses data of 10,689 MD application users in the years 2017 and 2018. This dataset was selected from a larger database after data wrangling and consultations with experts. The data consist of detailed transactional information such as transaction amount, financial service provider, account type, date, vendor, postcode of transaction occurrence, as well as high- and lower-level categorisation of the transaction created by MD. There are 68 UK financial institutions in the data. Their shares of the total number of analysed accounts are depicted in Figure 1.1. The three main types of accounts in the data are current accounts, credit card accounts, and savings accounts. Figure 1.2 shows the prevalence of account types in the data. Demographic information, such as age and country of residence, is also available in the dataset. An overview of these can be seen in Figure 1.3.

The data used for analysis in thesis provides unique information on the daily financial behaviour of registered users. With its informational content and structure the data is considered OB-type.

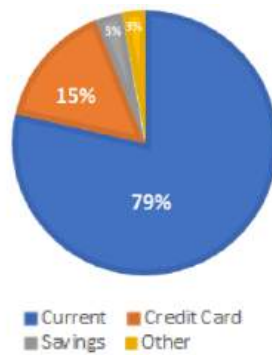


Figure 1.2: MD Data: Overview of account types in the analysed MD data.

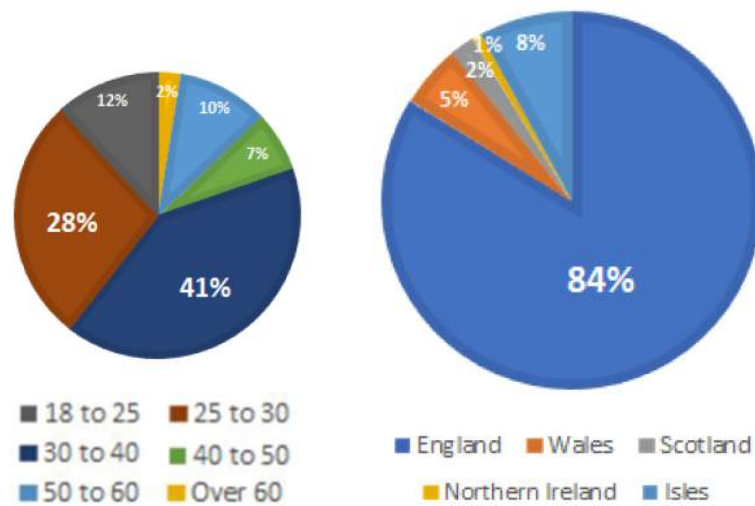


Figure 1.3: MD Data: Demographic information (left) and country of residence (right) of MD users in the 2017 sample.

1.3 Analysis of Personal Finances

Terminology

The terminology used to assess the state of an individual's finances, also referred to as financial health, varies considerably in the literature. There is a lack of the lack of generally accepted terminology and definitions, as [Fergusson et al. \(1981\)](#); [Prawitz et al. \(2006\)](#); [Xiao \(2016\)](#) all point out. There are various terms used for this purpose, such as: economic well-being ([Williams, 1993](#)), material well-being ([Joo, 2008](#)), financial wellness ([Back & DeVaney, 2004](#)), subjective well-being ([Van Praag et al., 2003](#)), financial well-being ([Breen, 1991](#)), financial satisfaction ([Joo & Grable, 2004](#)), financial stress ([Kim & Garman, 2003](#)), financial strain ([Melzer, 2011](#)) and financial distress ([Prawitz et al., 2006](#)).

The used terminology usually carries a positive or negative connotation, as [Prawitz et al. \(2006\)](#) points out. Recent examples of this are papers by [Melzer \(2011\)](#) and [Bhutta \(2014\)](#). Both authors analyse the impact of payday loans on household finances. Melzer, who claims that there is a negative relationship between the availability and use of payday loans and household finances uses the term "financial strain". On the other hand, Bhutta, who finds no empirical evidence for such a link, uses the term "financial health".

Most authors focus only on certain aspects of personal financial health. For example, [Fergusson et al. \(1981\)](#) focus on changes in consumption and ownership and their impact on material well-being; [Andersen et al. \(2016\)](#) analyse Danish household debt; and [Bhutta \(2014\)](#) uses credit ratings to track changes in household financial health. The latter is particularly noteworthy as the author himself points out that "credit scores are just one dimension of financial health" ([Bhutta, 2014](#), p. 231).

Other authors, such as [Joo \(2008\)](#) and [Gerrans et al. \(2014\)](#), try to develop a unified concept of financial health based on combining knowledge from various fields, such as: economics, business, finance, psychology, sociology and psychology. The problem with these approaches is the lack of empirical studies to support the proposed frameworks. Adding to the ambiguity of the used terminology are cases where authors use the same term but define it differently. An example is [Xiao \(2016, p. 36\)](#), who offers two different definitions of the same term on the same page: "Financial well-being refers to a financial status in that a consumer or family has adequate resources to live a comfortable life" and "Doing financially well also refers to a financial status that is better than the average compared to a reference group".

Last but not least, there are concepts closely related to financial health: financial literacy and financial capacity. Even though we will not further discuss these concepts the rest of research presented in this thesis, they are relevant for creating a complete picture of individual and/or household finances. Financial literacy is defined as "peoples' ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt and pensions" ([Lusardi & Mitchel, 2014, p. 4](#)), while according to [Xiao \(2016, Chapter 1\)](#) financial

capability can be considered as a combination of financial literacy and financial behaviour to achieve financial well-being. These decisions are usually based on an individual's confidence.

Objective and Subjective Measures of Personal Finances

In this section, we broadly define the notions of objective and subjective measures of financial health and provide current examples. The OB-type data described in Section 1.2 contains information about individual transactions that provide information about individual financial behaviour. Therefore, we focus on the definitions used in the field of behavioural finance.

Summarising (Xiao, 2008, 2016), we can generally define objective measures as indicators based on quantitative variables and subjective measures as indicators based on qualitative data. Examples of objective measures are total income, consumption, expenditure, debt, asset value and combinations of these variables, such as net worth and various financial ratios. Subjective indicators are usually obtained through surveys. Typical assessments relate to satisfaction with one's financial status, satisfaction with income, satisfaction with the availability of financial resources and satisfaction within individual categories of consumption and savings. Subjective measures give a direct assessment of individuals provided by the subjects themselves. The problem with such assessments are individual behavioural biases, which can lead to respondents over- or under-reporting certain values. Objective measures of financial status do not suffer from behavioural biases. However, the disadvantage of these measures is the lack of information about risk attitudes and preferences of individuals and households.

Depending on available data and research goals, objective and subjective measures can be used separately, or they can be combined to assess financial health. It can be argued that the more data becomes available, the more accurate the assessments can be made. However, appropriate sampling methods are key for drawing usable conclusions. Mainly for this reason, authors such as Van Praag et al. (2003), Back & DeVaney (2004), Joo (2008), Xiao (2008, 2016), Lusardi & Mitchel (2014) and Gerrans et al. (2014), as well as institutions such as the Money Advice Service (2013) use approaches that combine objective and subjective measures. Since this thesis focuses on quantitative OB-type data only objective measures are relevant for this body of work.

Financial Ratios

We now briefly discuss financial ratios as an objective measure of personal finances. According to Winger & Frasca (2000), a financial ratio is an index that can measure current financial strength and its development over time. There are many different ratios. Table 1.1 provides a summary of the most commonly used ratios based on Lawrence et al. (1992), Greninger et al. (1996), Maki (2002), May & Tudela (2004), Bucks et al. (2006), Harness et al. (2008) and OECD (2020).

Ratio	Formula	Ratio	Formula
Liquidity	$\frac{\text{Liquid Assets}}{\text{Monthly Expenditure}}$	Housing Expenses (1)	$\frac{\text{Renter's Expenses}}{\text{Gross Income}}$
Debt to Service	$\frac{\text{Total Loan Payments}}{\text{Disposable Income}}$	Housing Expenses (2)	$\frac{\text{Homeowner's Expenses}}{\text{Gross Income}}$
Consumption to income	$\frac{\text{Monthly Consumption}}{\text{Monthly Income}}$	Asset Allocation (1)	$\frac{\text{Liquid Assets}}{\text{Net Worth}}$
Expense to income	$\frac{\text{Total Monthly Expenditure}}{\text{Monthly Income}}$	Asset Allocation (2)	$\frac{\text{Net Investment Assets}}{\text{Net Worth}}$
Debt to Income	$\frac{\text{Total Debt Payments}}{\text{Income After Tax}}$	Asset Allocation (3)	$\frac{\text{Foreign Investment}}{\text{Total Investment}}$
Debt to housing wealth	$\frac{\text{Housing Debt}}{\text{Value of all Property}}$	Asset Allocation (4)	$\frac{\text{Financial Assets}}{\text{Total Assets}}$
Insolvency and credit	$\frac{\text{Nonmortgage Debt Payments}}{\text{Income After Tax}}$	Tax Burden (1)	$\frac{\text{Payroll Taxes}}{\text{Gross Income}}$
Net worth to income	$\frac{\text{Net worth}}{\text{Income}}$	Tax Burden (2)	$\frac{\text{Payroll Taxes} + \text{Property Taxes}}{\text{Gross Income}}$
Loan to Value	$\frac{\text{Value of Mortgage}}{\text{Purchase Price of House}}$	Inflation Protection (1)	$\frac{\% \text{ Change in Net worth}}{\text{Inflation Rate}}$
House Value	$\frac{\text{Current Value of House}}{\text{Original Purchase Price of House}}$	Inflation Protection (2)	$\frac{\% \text{ Change in Investment Assets}}{\text{Inflation Rate}}$

Table 1.1: Summary of most common financial ratios.

Although financial ratios are commonly used by academics, regulatory institutions and financial institutions they were not constructed with having OB-type data in mind and therefore play only a minor role in this thesis.

Financial Comparison and Clustering

One of the main motivations of the here presented research is the development of statistical learning tools for the evaluation of personal finances, in particular methods tailored for comparing individuals with similar population groups and finding homogeneous groups of individuals.

On an individual level, the ability to compare how the personal finances of a specific individual compare to those of people with similar characteristics (peers) allows individuals and financial institutions to identify how the personal finances and financial behaviour of a specific customer differ from those of their peers. This helps tailoring personalised financial advice, products and solutions. An example of such developments in the market is Facebook’s recently filed patent ([Cheng & Zagat, 2019](#)), that is embracing subject-to-group comparison to provide services in an OB-like context. Although there is industry interest in subject-to-group methods, there have been no academic methodological developments to date.

The idea of comparing similar individuals can be extended to a larger scale. Identifying groups in the population allows institutions to better address the needs of each market segment. A simple example are financial institutions such as banks identifying groups of interest and

specifically tailoring their products and services to these groups. At the level of regulators and non-governmental organisations, market segmentation allows policies and services to be designed to maximise the intended impact on the market and/or specific population groups. An example of the latter would be the identification of financially vulnerable individuals, and the provision of advice before their financial problems accumulate to a level that is almost unmanageable. The Money Advice Service has a model for this purpose, shown in Figure 1.4. Various measures and methods can be used for market segmentation. Both objective and subjective measures of financial health are suitable for this purpose. Methodologically, most clustering approaches such as model-based clustering (Fraley & Raftery, 2002) or similarity-based clustering (Hastie et al., 2009; Kaufman & Rousseeuw, 1987; MacQueen, 1967) can be used to create a market segmentation. As we discuss in Section 2.1, model-based methods are much more flexible than similarity-based methods. For this reason, we choose model-based approaches as the starting point for the market segmentation method developed in Chapter 4. Although model-based clustering is better suited to develop a customer segmentation method for OB-type data, this group of methods suffers from a drawback that limits its potential to deliver clustering solutions that reflect reality. In Chapter 4 we show that the number of clusters will most likely be identical across all measures. This poses a problem when it comes to using model-based approaches to obtain meaningful market segmentation. For example, if information on age, income and savings is used as model input, the number of clusters with people of similar age, the number of clusters with people of similar income and the number of clusters with people of similar savings will most likely be identical. When working with very detailed data containing many measures, such as OB type data, this problem is exacerbated. Chapter 4 provides details of this problem and introduces a solution in the form of the proposed Reign-and-Conquer clustering method.

1.4 Open Problems and Main Contributions

The main contributions of this thesis are centred around the following questions, which we have identified through the study and development of methods aimed at fully harnessing the potential of OB-type data, such as the MD data described in Section 1.2.

Contribution 1

Studies evaluating personal finances, discussed in Section 1.3, focus on comparing selected population groups with others in order to find, evaluate and explain observed differences between them. These group-to-group comparisons are usually conducted using well-studied methods such as hypothesis testing and regression with group indicators.

The introduction of OB-type data opens a new era of personalisation of financial services based on individual-specific analysis. Group-to-group comparison methods are not suitable

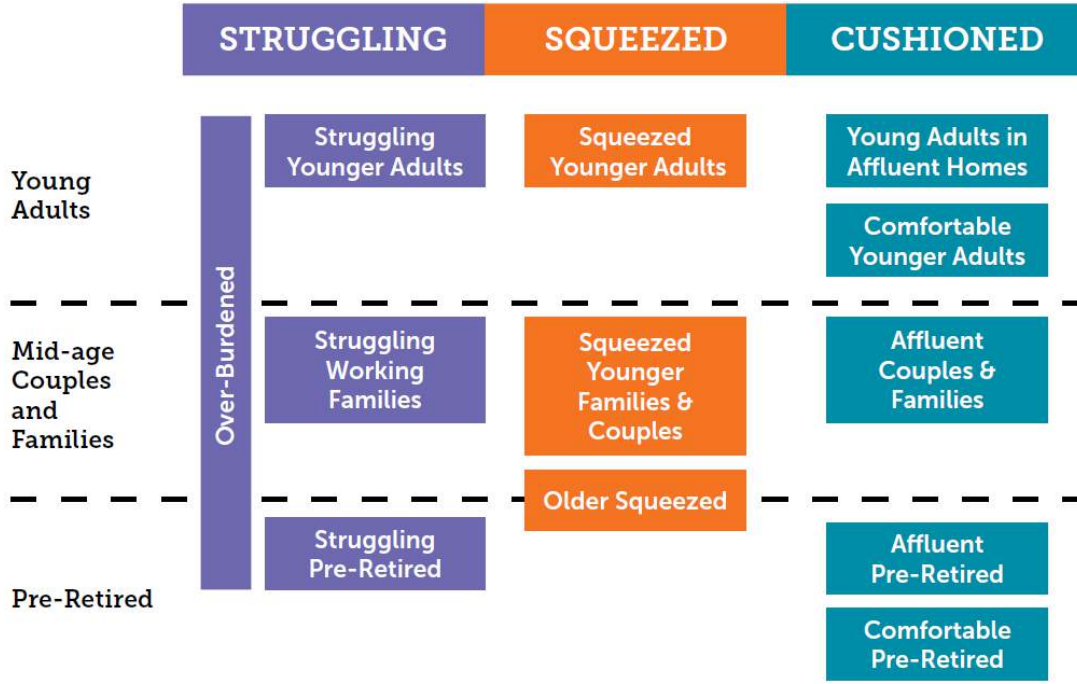


Figure 1.4: The Money Advice Service segmentation model.

for this type of analysis and little has been done to address this issue with sound statistical methodology. Chapter 3 fills this methodological and applicational gap by developing a statistical learning approach for comparing a specific individual (reference subject) to a group of similar individuals (comparison group) defined by covariate truncation centred at the covariate values of the reference subject. The *comparative mean value* is the expected value of a response conditioned on the comparison group. The similarity between the reference subject and the comparison group is controlled by a homogeneity parameter set by the user of the methodology. Contrasting the comparative mean value with the reference subject's response provides diagnostics on how the reference subject's response differs from the average response of the comparison group, thus providing a subject-to-group comparison. More robust extensions of the time-varying comparative mean value are discussed in Section 3.5.

To track the dynamics governing the comparative mean value Section 3.2.2 introduces the *time-varying comparative mean value*. Taking advantage of local polynomial smoothing (Fan & Gijbels, 1996), in particular the Nadaraya–Watson kernel smoother (Watson, 1964), we construct a *smooth time-varying version of the comparative mean value*. Nonparametric smoothing provides us with a predictor and makes it easier to detect trends and understand the results of the analysis.

The developed concepts are tested using multiple simulation studies. We show that the estimators recover the true time-varying comparative mean value well.

Besides pioneering statistical subject-to-group comparison, the time-varying comparative mean value can be seen as a truncated covariate regression, which we discuss in Section 3.2.3.

At the time of publication of Chapter 3, Svetlošák et al. (2021), the time-varying comparative mean value was the first method for a truncated covariate regression.

Contribution 2

Clustering is at the heart of customer segmentation methods. We provide an overview of selected methods from the literature in Section 2.1. Clustering can be divided into two categories: similarity-based methods (i.e. k -means and k -medoids) and model-based methods (i.e. mixture models), with only a few bridging the worlds, such as Duan & Dunson (2021) and Rigon et al. (2023).

In Chapter 4 we propose a novel model-based approach for cluster analysis that lies at the interface of model-based clustering and similarity-based clustering. The developed method addresses two main problems.

As we show in Section 4.1, joint mixture models are likely to induce the same number of components on each margin, which is the number of joint components. In other words, the number of clusters on each margin is the same as the number of clusters on the joint distribution. This seems restrictive and may not reflect reality captured by the data. This is especially true for OB-type data, as the following example shows. Suppose we have data on age, income and expenditure. Modelling them jointly will most likely lead to obtaining the same number of age groups in the population as there are groups with similar income and groups with similar expenditure. Since OB-type data is much more detailed and complex than the example given here, it is essential to develop a model that allows differing number of clusters on the margins, which is what the model in Chapter 4 does.

The second issue with mixture models are computational costs. Mixture models suffer from the well-known curse of dimensionality, which limits their use in high-dimensional settings Bouveyron & Brunet-Saumard (2014).

The proposed method, which we call *Reign-and-Conquer Clustering*, solves the two aforementioned problems in the following way. Each of the margins are modelled with a separate mixture model. This allows the number of components to differ across the margins. At the same time, this approach makes the method partially parallelisable and thus helps to reduce computational costs.

Estimating the margins separately is the first step of the method. The other two steps of the algorithm (explained in Section 4.2) use similarity-based methods to first construct protoclusters from which the final clusters on the joint distribution are obtained. Taken together, the three steps bridge similarity-based and model-based clustering approaches.

The model partitions the sample space in a way that filters out regions with residual mass, where the minimum requirement is set by the user or determined via a data-driven approach. This results in the model focusing on identifying dominant clusters in the data.

Furthermore, in Section 4.4 we conceptualise steps two and three of the algorithm using game

theory. With this reformulation, the proposed method contributes to the literature on game theory-inspired clustering.

At the time of submission, Reign-and-Conquer clustering (de Carvalho et al., 2022) was the first method to identify and directly address the problem of mixture models generating the same number of components on each margin. The concept also addresses the curse of dimensionality and combines the worlds of similarity-based and model-based clustering with an extension into game theory.

1.5 Structure and Organization of the Thesis

The rest of this thesis is structured as follows. Section 2 summarises statistical learning methods needed for the development of the proposed methods in Part II of the thesis.

Chapter 3, published as Svetlošák et al. (2021), introduces the concept of the comparative mean value. First, a time-invariant version of the comparative mean value and its estimator are devised, followed by a time-varying version of the comparative mean value and a smooth time-varying comparative mean value constructed by modifying the Nadaraya-Watson kernel smoother. Links to existing literature and methods are explored before the constructs are tested and empirically applied to a OB-type dataset provided by MD. The Chapter concludes with a discussion of the results, comments and possible extensions of the proposed methods.

Chapter 4, submitted as de Carvalho et al. (2022), introduces the three-step Reign-and-Conquer clustering algorithm, first introduced in a two-dimensional setting and then extended to d -dimensions. This is followed by discussing how to fit and partially parallelise the method, making it computationally appealing. In Section 4.4, the algorithm is reformulated using game theory to further enrich the framework Reign-and-Conquering clustering provides. Application examples and performance comparisons with other methods on simulated and real datasets are given in Sections 4.5.1 and 4.5.2. The Chapter concludes with final observations and concluding remarks.

This thesis closes with a discussion of the here developed methods, algorithms, obtained results and possible future research that can be based on the here presented concepts. This is followed by supporting information and supplementary materials.

Chapter 2

Preparations on Statistical Learning

This Chapter offers a high level overview on ideas, concepts and method from statistical learning, which are entailed the main methodological contributions of this thesis.

2.1 Unsupervised Learning

Cluster analysis aims to group objects into clusters so that objects within a cluster are more closely related to each other than to objects from other clusters (Hastie et al., 2009, Section 14.3). Unsupervised learning aims to detect patterns in unlabelled data without using a classified training set. Following Hastie et al. (2009) this Chapter lays the groundwork for Chapter 4.

Similarity-based Methods

Similarity-based methods aim to maximise similarity within clusters and minimise similarity between clusters. The similarity-based clustering methods discussed in this chapter are all hard clustering approaches, which means that each observation is assigned to exactly one cluster.

Before we introduce some of the most commonly used similarity-based clustering approaches, we need to define similarity measures that are an essential part of this class of methods. Let $x_{i,j}$ for $i = 1, \dots, N$ be elements of variables $j = 1, \dots, p$. The dissimilarity between two elements $x_{i,j}, x_{i',j}$ of the same variable j is measured by a distance function $d_j(x_{i,j}, x_{i',j})$. There is a variety of distance functions, such as the Minkowski distance, the Mahalanobis distance, the Canberra distance and the city block distance. The most common choice is the squared Euclidean distance $d_j(x_{i,j}, x_{i',j}) = (x_{i,j} - x_{i',j})^2$. The distance between two objects i and i'

across all variables is

$$D(x_i, x_{i'}) = \sum_{j=1}^p d_j(x_{i,j}, x_{i',j}).$$

One of the most popular, if not the most popular, similarity-based method is k -means (MacQueen, 1967). It is an iterative algorithm that minimises the average distance of elements to cluster centres. Algorithm 1, adapted from Hastie et al. (2009, p. 510), summarises the k -means algorithm. The rest of the chapter uses the notation $\mathbf{x}_i$ for the i -th observation, which contains information about all p variables.

Algorithm 1 k -means Clustering

1. For a set of observations S , cluster assignment C , set of cluster means $\{\mathbf{m}_1, \dots, \mathbf{m}_k\}$, and $N_k = \sum_{i=1}^N I\{C(i) = k\}$ with I being the indicator function solve

$$\min_{C, \{\mathbf{m}_k\}_{k=1}^K} \sum_{k=1}^K N_k \sum_{C(i)=k} \|\mathbf{x}_i - \mathbf{m}_k\|^2$$

to obtain means of currently assigned clusters

$$\bar{\mathbf{x}}_S = \arg \min_{\mathbf{m}} \sum_{i \in S} \|\mathbf{x}_i - \mathbf{m}\|^2.$$

2. Given the current $\{\mathbf{m}_k\}_{k=1}^K$ obtained in step 1 solve

$$\min_{C, \{\mathbf{m}_k\}_{k=1}^K} \sum_{k=1}^K N_k \sum_{C(i)=k} \|\mathbf{x}_i - \mathbf{m}_k\|^2$$

by assigning each cluster observation to the closest mean

$$C(i) = \arg \min_{1 \leq k \leq K} \|\mathbf{x}_i - \mathbf{m}_k\|^2.$$

3. Iterate step 1 and 2 until cluster assignments do not change.
-

It is worth noting that Algorithm 1 uses the squared Euclidean distance instead of the regular Euclidean distance. Without squaring, the optimisation problem becomes the considerably more difficult Weber problem, for which there is no closed form solution (Bradley et al., 1996, p. 371). However, squaring the Euclidean distance also comes at a price. The greater the distance of the observations from the cluster centres, the more importance is given to them. This makes k -means prone to the influence of outliers. A more robust alternative to k -means is the k -medoids algorithm, which we also discuss in this chapter.

The popularity of k -means leads to the algorithm being constantly revisited, modified and improved. Recent examples of improvements to k -means include Arthur & Vassilvitskii (2007), Fränti & Sieranoja (2019) and Sinaga & Yang (2020).

It can be shown that certain finite mixture model approaches can be modified to be equivalent to k -means clustering. In particular, modifying the classification expectation maximisation

algorithm (Celeux & Govaert, 1995) with a uniform spherical Gaussian distribution as the probability model leads to an equivalent problem as k -means (Fraley & Raftery, 2002, p. 614).

Like most other clustering methods, k -means requires the number of clusters as input. We discuss some of the criteria for determining the number of clusters present in the data in Section 2.2.1.

Replacing means as cluster centres with medians leads to another widely used clustering algorithm, k -medoids. Since the cluster centres are medians and not means, k -medoids is more robust in comparison to k -means. Moreover, k -medoids allows the practical use of a variety of similarity measures. This makes k -medoids more flexible and allows better detection of clusters with high variance and/or different shapes. However, compared to k -means, k -medoids is much more computationally intensive, limiting its use on large samples. Just like k -means, k -medoids is largely popular and has been constantly revisited and modified. Recent modifications include Park & Jun (2009), Juanying & Yanan (2016), and Yu et al. (2018).

Another widely used unsupervised learning method is hierarchical clustering (Ward, 1963). Unlike k -means and k -medoids, it does not require the user to specify the number of clusters as input for the algorithm. However, the user has to select a dissimilarity measure based on pairwise dissimilarities of observations from different groups. Hierarchical clustering also requires the user to choose a linkage function that determines how the dissimilarity is computed. There is no golden rule for choosing a dissimilarity measure and linkage function. The choice should be made based on the nature of data being analysed as linkage can influence the formation of clusters. For an overview see Ding & He (2002). Even though the number of clusters is not needed as input for the algorithm, one has to decide how many clusters there should be in the end, e.g. using one of the heuristics described in Section 2.2.1.

The agglomerative (bottom-up) version of the algorithm starts with each observation being a group on its own. Groups with smallest dissimilarities get merged until a hierarchy is created in which all observations forming one group at the top. The splitting (top-down) approach starts with all observations in one group, which are then gradually split into smaller subgroups until each observation is a group of its own. In both cases, the algorithm creates a hierarchy with $N - 1$ (for N observations). This hierarchy is usually represented by a binary tree called a dendrogram. Figure 2.1 shows an example of a dendrogram created by agglomerative hierarchical clustering applied to the well-known Iris dataset (Fisher, 1936). The distance between two clusters G, H is measured by $d(G, H) = \max_{i \in G, i' \in H} d_{ii'}$, and a complete linkage was chosen. A variety of papers have been published on the subject of hierarchical clustering. Examples include Castro et al. (2004) and Bien & Tibshirani (2011).

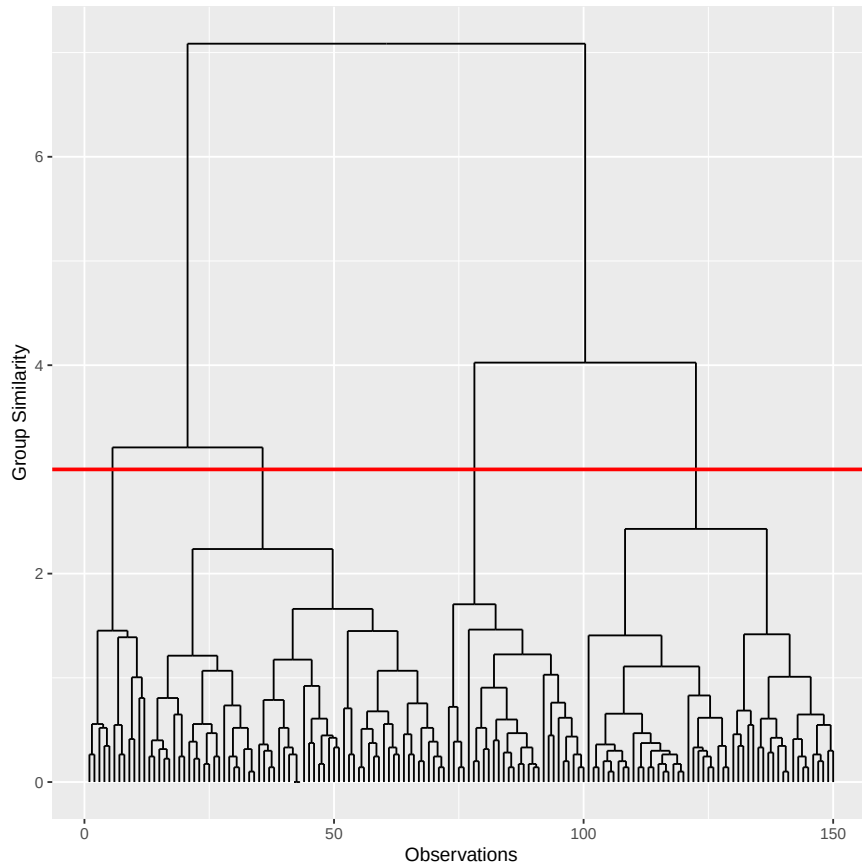


Figure 2.1: Dendrogram: Agglomerative hierarchical clustering of Iris data with complete linkage. The horizontal line depicts the true number of clusters in the data.

Model-based Methods

Another large class of methods is model-based clustering. Unlike similarity-based methods, they are based on a formal statistical model, hence the name model-based (Stahl & Sallis, 2012, pp. 324-325). The idea for these models came from Tiedeman (1955). Since then, the literature on model-based approaches has grown enormously, as has their popularity and use. Examples of seminal works on mixture models for clustering are Day (1969), Wolfe (1970), and Scott & Symons (1971). Mixture models have applications in a variety of fields, including medicine, genetics, behavioural sciences, economics, management, environmental sciences, chemistry, engineering, computer science and robotics. See e.g. Muthén (2002), Khansari-Zadeh & Billard (2011), Durango-Cohen et al. (2013), Miettunen et al. (2016), McLachlan & Peel (2000), Chen & Ludtke (2021), and Zou et al. (2022).

The objective is to group observations by fitting a weighted sum of probability distribution functions that estimates the joint distribution of the observations. Let X be a random variable

with distribution F . The finite mixture model is defined as

$$f_X(x \mid K, \boldsymbol{\Theta}, \boldsymbol{\Pi}) = \sum_{k=1}^K \pi_k q(x \mid \boldsymbol{\theta}_k), \quad (2.1)$$

where q is a density function with parameters $\boldsymbol{\Theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K)$ and weights $\boldsymbol{\Pi} = (\pi_1, \dots, \pi_K)$, with $\pi_k \in (0, 1]$ such that $\sum_{k=1}^K \pi_k = 1$. When q is a Gaussian density, we obtain a Gaussian mixture model (GMM). Mixture models have proven to be very powerful tools for density estimation and clustering. One of the reasons for their popularity is that GMMs are dense on the space of all density functions under the L_1 metric, which means that they can estimate any density arbitrarily well (Li & Barron, 1999). Each component of the sum represents a cluster. Mixture models are a soft clustering method, i.e. observations can belong to multiple clusters. Modelling densities allows the making of probabilistic statements and uncertainty quantification of an object belonging to a cluster. Adding to the advantages, solutions from mixture models can be used as predictors for variables within the same model (Muthén, 2002).

This statistical approach for modelling densities makes mixture models more flexible. For illustration, we use k -means, k -medoids and a GMM to cluster simulated data described in Example 2.1.1.

Example 2.1.1 (Illustration of k -means, k -medoids and a Gaussian mixture model). We generate data using three bivariate Gaussian distributions with means $\boldsymbol{\mu}_1 = (0, 0)$, $\boldsymbol{\mu}_2 = (5, 12)$, $\boldsymbol{\mu}_3 = (-3, 30)$ and covariance matrices

$$\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}_3 = \begin{bmatrix} 4 & -6 \\ -6 & 25 \end{bmatrix}.$$

Figure 2.2 (a), (b) and (c) show the clusters and their geometry created by k -means, k -medoids and a GMM respectively. Parts (d), (e) and (f) show the misclassified observations in orange. In the sample with $N = 600$, k -means misclassified 38 observations, k -medoids with Euclidean distance misclassified 34 observations, while GMM misclassified only 6 observations. Although we used an squared Euclidean distance as a measure for k -medoids, thus the circular geometry of the clusters, the algorithm performs better than k -means because the clusters have high variance and k -medoids is more robust than k -means. GMMs are not limited by a similarity measure and by modelling the joint density by default, they take variance into account and can better capture the shape of the clusters. This advantage allows them to outperform similarity-based methods in many scenarios, but especially in high variance scenarios such as that of example 2.1.1. Due to their flexibility we choose model-based clustering approaches as our starting point for developing a customer segmentation method suitable for OB-type data.

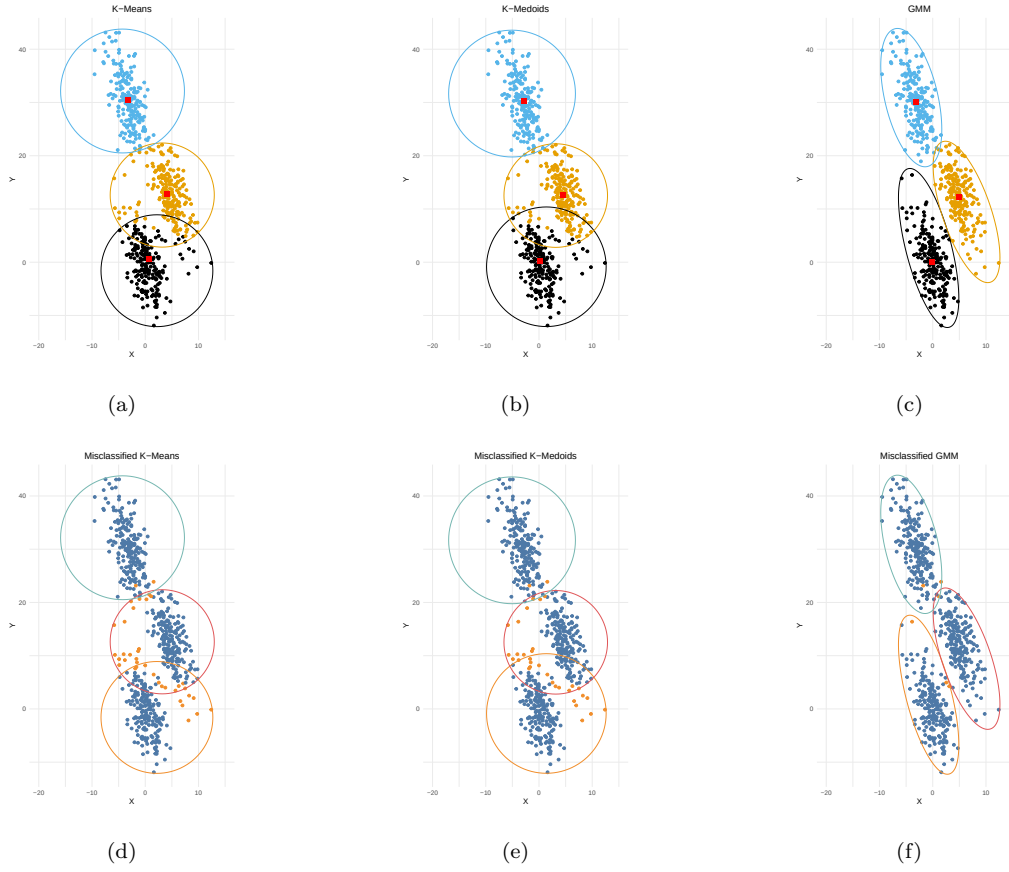


Figure 2.2: Example 2.1.1: (a) fit and geometry of clusters created by k -means, (b) fit and geometry of clusters created by k -medoids, (c) fit and geometry of clusters created by a GMM, with cluster centres in red, (d) misclassified points by k -means in orange, (e) misclassified points by k -medoids in orange, (f) misclassified points by GMM in orange.

A standard approach to learn about Θ, Π , and K is via log-likelihood maximisation. Let $(x_i, y_i)_{i=1}^N$ be a random sample from distributions $x \sim F_X$ and $y \sim F_Y$. The maximisation problem is

$$\max_{\Theta, \Pi} \ln\{f_X(x \mid K, \Theta, \Pi)\} = \max_{\Theta, \Pi} \sum_{i=1}^N \ln \left\{ \sum_{k=1}^K \pi_k q(x \mid \theta_k) \right\}.$$

However, due to the complexity of the log-likelihood, it is common to fit models using the expectation-maximisation algorithm (EM) (Dempster et al., 1977). EM is a general iterative method for finding the parameters that maximise the log-likelihood. For a detailed description of the EM algorithm, see McLachlan & Krishnan (2007). Although the EM algorithm is an attractive alternative to direct maximisation of the log-likelihood, it has several drawbacks. First, EM is an algorithm that attempts to find the global maximum of the log-likelihood in the parameter space. If multiple local maxima exist, there is a possibility that the algorithm will converge to one of them, leading to suboptimal results. This problem can be mitigated by using multiple starting points for the algorithm and selecting the local maximum to which most of the starting points converge, see e.g. Celeux & Govaert (1992). This increases the certainty

of the estimates, but does not solve the inherent problem. The EM algorithm is also prone to overfitting. "Achieving the global maxima while fitting a model to observed data will generally result in some amount of overfitting" (Andrews, 2018, p. 161). Another potential pitfall is the existence of singularities in the log-likelihood. In the presence of singularities, the resulting distributions can be degenerate.

Bayesian methods open up another world for the approximation and estimation of mixture models. Bayesian methods use Bayes' theorem to perform inference. Due to their construction, they do not suffer from singularity and degeneracy problems. The maximum a posteriori (MAP) estimator was proposed by Fraley & Raftery (2007). It is a Bayesian regularisation approach that uses a dispersed but proper conjugate prior to find the posterior mode via an EM algorithm. Another major advantage of Bayesian approaches is the ability to incorporate background knowledge from previous experiments or expert opinions. This prior information shapes the posterior distribution via Bayes' theorem. There is a wealth of literature on how to choose an appropriate prior. For details on Bayesian inference, see Gelman et al. (2021).

For various reasons, Bayesian model-based approaches suffer from identifiability issues. As mentioned earlier, mixture models are prone to overfit the data. Another cause of identifiability issues arises from the "invariance of the log-likelihood to relabelling of components" (Stephens, 2000, p. 62), also known as label switching. While the two problems have been addressed separately in the literature, e.g. by Andrews (2018) and Stephens (2000), it is much harder to find a common solution for overfitting and label switching.

Due to the drawbacks of the EM algorithm, there is growing interest in Gibbs samplers and Markov chain Monte Carlo (MCMC) methods (Brémaud, 2001, Section 11). One type of powerful MCMC methods are dimension jumping algorithms, also known as reversible jump MCMC methods (RJ MCMC) (Green, 1995). These methods are a generalisation of the Metropolis-Hastings MCMC algorithm (MH) (Metropolis et al., 1953). The RJ MCMC algorithm is particularly useful in cases where the dimension of the parameter space may change between iterations (Gelman et al., 2021, Section 12.3). The idea is to introduce additional random variables, which allows the matching of all parameter space dimensions so that the MH algorithm can be applied. RJ MCMC is often used for Bayesian model selection.

Another widely used method for Bayesian model selection is the Bayesian Information Criterion (BIC) (Schwarz, 1978). The criterion is $BIC = k \ln(N) - 2 \ln(\hat{L})$, where $\hat{L}$ is the maximum likelihood estimate of the likelihood L . Since the BIC contains a penalty, models with a lower BIC are preferential. Other methods for Bayesian model selection are Akaike's information criterion (AIC) $AIC = 2k - 2 \ln(\hat{L})$, the consistent Akaike information criterion (CAIC), Bayesian Ying-Yang (BYY) harmony learning, harmony empirical learning criterion (BYY-HEC), harmony data smoothing criterion (BYY-HDS), and the minimum description

length (MDL). For an overview of selection criteria see [Richardson & Green \(1997\)](#), [Fraley & Raftery \(2002\)](#), [Baudry et al. \(2010\)](#) and [Gassiat & Van Handel \(2013\)](#).

Non-Local Priors

Recently introduced non-local priors (NLP) ([Fúquene et al., 2019](#)) propose a solution to overfitting. NLPs address the identifiability problem by encouraging parsimony by enforcing separation between the models under consideration. Let $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ be a sample of independent observations, where $\mathbf{y}_i \in \mathbb{R}^p$ for $i = 1, \dots, n$. To simplify the notation, we adjust Equation (2.1) to

$$f(\mathbf{y}_i | \boldsymbol{\vartheta}_K, M_K) = \sum_{j=1}^K \pi_j f(\mathbf{y}_i | \boldsymbol{\theta}_j). \quad (2.2)$$

In the above equation, $f(\mathbf{y} | \boldsymbol{\theta}_j)$ are component densities of a K -component model M_K , with parameters $\boldsymbol{\theta}_j \in \Theta$, and weights π_j . The complete model parameters are $\boldsymbol{\vartheta}_K = (\boldsymbol{\theta}, \boldsymbol{\Pi}) \in \Theta_K$, with $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K)$ and $\boldsymbol{\Pi} = (\pi_1, \dots, \pi_K)$. It is assumed that the mixture in Equation (2.2) is generically identifiable, see e.g. [Teicher \(1963\)](#), [Yakowitz & Spragins \(1968\)](#), [Grün & Leisch \(2008\)](#) and [Allman et al. \(2009\)](#).

A continuous prior density $f(\boldsymbol{\vartheta}_K | M_K)$ is a NLP if and only if

$$\lim_{\boldsymbol{\vartheta}_K \rightarrow \mathbf{t}} f(\boldsymbol{\vartheta}_K | M_K) = 0,$$

for any $\mathbf{t}$ in the parameter space Θ_K , so that $f(\mathbf{y} | \mathbf{t}, M_K) = f(\mathbf{y} | \boldsymbol{\vartheta}_{K'}, M_{K'})$ for some $\boldsymbol{\vartheta}_{K'} \in \Theta_{K'}$ and $K' < K$. The NLP gives $\boldsymbol{\vartheta}_K$ a vanishing density if the mixture contains redundant components, so that Equation (2.2) is equivalent to a mixture with K' components. Priors that do not satisfy the properties of NLPs are called local priors (LP).

Let $d_{\boldsymbol{\vartheta}}(\boldsymbol{\vartheta}_K)$ be a continuous penalty function that converges to 0 either as $\pi_j \rightarrow 0$ for any $j = 1, \dots, K$ or as $\boldsymbol{\vartheta}_i \rightarrow \boldsymbol{\vartheta}_j$ for any $i \neq j$. The general NLP class is defined by

$$f(\boldsymbol{\vartheta}_K | M_K) = d_{\boldsymbol{\vartheta}}(\boldsymbol{\vartheta}_K) f^L(\boldsymbol{\vartheta}_K | M_K). \quad (2.3)$$

In Equation (2.3), $f^L(\boldsymbol{\vartheta}_K | M_K)$ is an arbitrary LP under the condition that $f(\boldsymbol{\vartheta}_K | M_K)$ is proper. To obtain a more practically useful form of Equation 2.3 we examine $f^L(\boldsymbol{\vartheta}_K | M_K) = f^L(\boldsymbol{\theta}_K | M_K) f^L(\boldsymbol{\Pi} | M_K)$ and $d_{\boldsymbol{\vartheta}}(\boldsymbol{\vartheta}_K) = d_{\boldsymbol{\theta}}(\boldsymbol{\theta}) d_{\boldsymbol{\Pi}}(\boldsymbol{\Pi})$, where

$$d_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = \frac{1}{C_K} \prod_{1 \leq i < j \leq K} d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j), \quad (2.4)$$

and $d_{\boldsymbol{\Pi}}(\boldsymbol{\Pi}) \propto \prod_{i=1}^K \pi_i^r$ for $r > 0$. In the above equation $C_K = \int f^L(\boldsymbol{\theta} | M_K) \prod_{1 \leq i < j \leq K} d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) d\boldsymbol{\theta}$ is the normalisation constant. Using a symmetric Dirichlet $f(\boldsymbol{\Pi} | M_K) = \text{Dir}(\boldsymbol{\Pi}; q) \propto d_{\boldsymbol{\Pi}}(\boldsymbol{\Pi})$

$\text{Dir}(\mathbf{\Pi}; q - r)$, where $q > 1$ and $r \in [q - 1, q)$, yields

$$f(\boldsymbol{\vartheta}_K \mid M_K) = d_{\boldsymbol{\theta}}(\boldsymbol{\theta}) f^L(\boldsymbol{\theta} \mid M_K) \text{Dir}(\mathbf{\Pi}; q).$$

In the case $\boldsymbol{\theta}_i = (\boldsymbol{\mu}_i, \Sigma_i)$ one can define moment penalties $d(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j) = (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)' \mathbf{A}^{-1} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) / g$, where $\mathbf{A}$ is a symmetric positive definite matrix and $g > 1$ is a prior dispersion parameter. Moment penalties allow C_K from Equation (2.4) to have a closed form, which simplifies estimation. In case the mixture model in Equation (2.2) is a GMM it is appealing to set the prior $f(\boldsymbol{\theta} \mid M_K)$ to

$$d_{\boldsymbol{\theta}}(\boldsymbol{\theta}) f^L(\boldsymbol{\theta} \mid M_K) = \frac{1}{C_K} \prod_{1 \leq i \leq j \leq K} \frac{(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)' \mathbf{A}_{\Sigma}^{-1} (\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)}{g} \prod_{j=1}^K N(\boldsymbol{\mu}_j \mid 0, g \mathbf{A}_{\Sigma}) \text{IW}(\Sigma_j \mid \nu, S),$$

where IW is the inverse-Wishart distribution and (g, ν, S) are hyperparameters. A good choice for $\mathbf{A}_{\Sigma}^{-1}$ is $\mathbf{A}_{\Sigma}^{-1} = (1/K) \sum_{j=1}^K \Sigma_j^{-1}$, which is a symmetric positive definite matrix related to the L_2 distance between Normal distributions.

This setting simplifies the interpretation and estimation of C_K and is used in the estimation of margins in the Reign-and-Conquer algorithm developed in Chapter 4. For further details on NLP mixtures and their estimation see [Fúquene et al. \(2019\)](#).

2.2 Nonparametric Smoothing

In this section we discuss nonparametric smoothing methodology mainly relevant for Chapter 3 published as [Svetlošák et al. \(2021\)](#). Throughout this section we use the following notation. Let $(x_i, y_i)_{i=1}^N$ be a random sample from distributions $x \sim F_X$ and $y \sim F_Y$. The observations x_0 are query points forming a grid between which the smoothers interpolate and estimate a function $f(X)$.

K -nearest Neighbours

One of the simplest and most common smoothing methods is k -nearest neighbours. It is a method that fits a linear regression over $k + 1$ observations defined by a query point x_0 and the k points closest to it

$$\mathbb{E}\{y \mid x \in N_k(x_0)\},$$

where $N_k(x)$ is the set of k nearest points to x_0 . The k -nearest neighbours estimator is

$$\hat{f}(x) = \frac{1}{k} \{y_i \mid x_i \in N_k(x)\}.$$

The estimator fits linear regressions between query points x_0 . Besides smoothing k -nearest neighbours is used for classification in binary group problems. The method can be used to find

a decision boundary between two groups of observations that are in the same dataset. One of the positive features of k -nearest neighbours is its possible use for regression and parameter selection. The method requires only one parameter, namely the number of neighbours in consideration k . Linear regression requires p parameters, which are the coefficients of the variables, and in the case of when an intercept is included the parameter number increases to $p + 1$. Unlike linear regression, k -nearest neighbours is a nonparametric method and therefore independent of assumptions about the distribution of the data. This gives the method an additional advantage in that it is flexible and can capture the nature of the data without being constrained by assumptions about the observed data.

Given the locality of the method, k -nearest neighbours displays low bias but high variance. As a smoothing method, k -nearest neighbours has a major drawback. The estimator fits straight lines between query points so that the resulting function is discrete. This means that the k -nearest neighbours fit is discontinuous and therefore not smooth. This problem is solved by weighted kernels, which are also described in this section.

Local Regression

In the following, we present methods that are first introduced in a univariate setting and then extended to the $\mathbb{R}^p$ case.

Kernel smoothing methods are flexible regression approaches that estimate a regression function $f(X)$. Unlike some other nonparametric methods, such as k -nearest neighbours, kernel regression methods provide smooth estimates of $f(X)$. This is done by fitting simple models using observations at the location of each query point x_0 . Kernels are functions $K_\lambda(x_0, x) = K(|x - x_0|/h_\lambda(x_0))$, where h_λ , which is a width function, defines the neighbourhood of the query point x_0 . Larger values of λ expand the neighbourhood, resulting in lower variance but higher bias in the estimates. Small values of λ lead to low bias but high variances. The kernel width λ is usually selected via cross-validation of some form. Least-squares based validation is considered optimal for the choice of λ (DasGupta, 2008, Section 3.10.2).

To provide smoothness to the estimators, weights are assigned to each observation in $h_\lambda(x_0)$. Observations near the query point x_0 are assigned larger weights, that smoothly approach 0 as the distance from x_0 increases. In this sense, kernel smoothing estimators are weighted moving (rolling) averages.

However, some kernel methods, such as the Nadaraya-Watson kernel-weighted average smoother (Nadaraya, 1964; Watson, 1964), suffer from boundary problems. As the estimator moves towards the boundary, it tries to use unavailable data from outside the variable domain. This leads to a bias in the estimation near the boundaries. For the same reason, there bias can also occur inside the interior of the domain of X . This happens when the observations are too unevenly distributed. Local linear regression and higher degree local polynomial regression mitigate this problem.

Local linear regression locally fits straight lines. At each query point x_0 , the following weighted least squares problem is solved

$$\min_{\alpha(x_0), \beta(x_0)} \sum_{i=1}^N K_\lambda(x_0, x_i) \{y_i - \alpha(x_0) - \beta(x_0)x_i\}^2.$$

The resulting estimate is $\hat{f}(x_0) = \hat{\alpha}(x_0) + \hat{\beta}(x_0)x_0$. This solution helps with the boundary bias that lower-degree kernel methods suffer from. This correction comes at the cost of a small increase in variance. However, local linear regression suffers from its own bias problem. It tends to be biased in regions where the true function is curved in the inner regions of the domain. This problem is often referred to as *trimming the hills* or *filling the valleys* (Hastie et al., 2009, Section 6.1).

Local polynomial regression is a generalisation of kernel smoothing methods and polynomial regression. It addresses drawbacks of local linear regression by locally fitting low-degree polynomials at each query point x_0 . The reduction of the trimming the hills bias of local linear regression comes at the cost of higher variance, which increases with polynomial degree $d \in \mathbb{N}$. This means that although local polynomial regression is less biased in the interior of the domain, it generally performs worse than local linear regression at the boundaries.

The following describes the problem to be solved in fitting local polynomials of degree d

$$\min_{\alpha(x_0), \beta_j(x_0), j=1, \dots, d} \sum_{i=1}^N K_\lambda(x_0, x_i) \left\{ y_i - \alpha(x_0) - \sum_{j=1}^d \beta_j(x_0) x_i^j \right\}.$$

The solution to this weighted least squares problem is $\hat{f}(x_0) = \hat{\alpha}(x_0) + \sum_{j=1}^d \hat{\beta}_j(x_0) x_i^j$. Note that in the limiting case ($d = 0$), local polynomial regression is equivalent to the Nadaraya-Watson kernel-weighted average smoother

$$\hat{f}(x_0) = \frac{\sum_{i=1}^N K_\lambda(x_0, x_i) y_i}{\sum_{i=1}^N K_\lambda(x_0, x_i)}.$$

A common choice of kernel for local regression smoothers is the Epanechnikov quadratic kernel $K_\lambda(x_0, x) = K(|x - x_0|/h_\lambda(x_0))$, with

$$K(t) = \begin{cases} \frac{3}{4}(1 - t^2), & \text{if } |t| \leq 1, \\ 0, & \text{else.} \end{cases}$$

Other commonly used kernels are the Gaussian kernel and the tri-cube kernel.

Using the global surface temperature data (Hartmann et al., 2018), Figure 2.3 illustrates the presented smoothing methods with different bandwidths.

Having introduced local regression in a single covariate setting, we now discuss its extension to $\mathbb{R}^p$. The basic idea and structure of the methods remain the same. The main difference is that the kernels and weights are now p -dimensional. We focus on local polynomial regression first, as it generalises kernel smoothing methods. Let $b(X)$ be a vector of polynomial terms with degrees smaller than d . For each query point $x_0 \in \mathbb{R}^p$, we need to solve the least squares problem

$$\min_{\beta(x_0)} \sum_{i=1}^N K_\lambda(x_0, x_i) \{y_i - b(x_i)^T \beta(x_0)\}^2.$$

The resulting estimate is $\hat{f}(x_0) = b(x_0)^T \hat{\beta}(x_0)$. In the multivariate case local linear regression now fits locally p -dimensional polynomials of degree $d = 1$, i.e. hyperplanes. The Nadaraya-Watson weighted-kernel estimator is a limiting case of local polynomial regression of degree $d = 0$ and therefore still locally fits constants.

In all the here mentioned cases, the weights are provided by p -dimensional kernels. Bandwidth selection can be done via cross-validation (Li & Racine, 2004).

Bias problems that exist in the univariate case become even more dominant as the edges of the domains become more complex and the proportion of points on the edges increases with increasing d . Therefore, the trade-off between bias and variance becomes substantial.

For more details on the local regression methods discussed here, see Wand (1995) and Fan & Gijbels (1996).

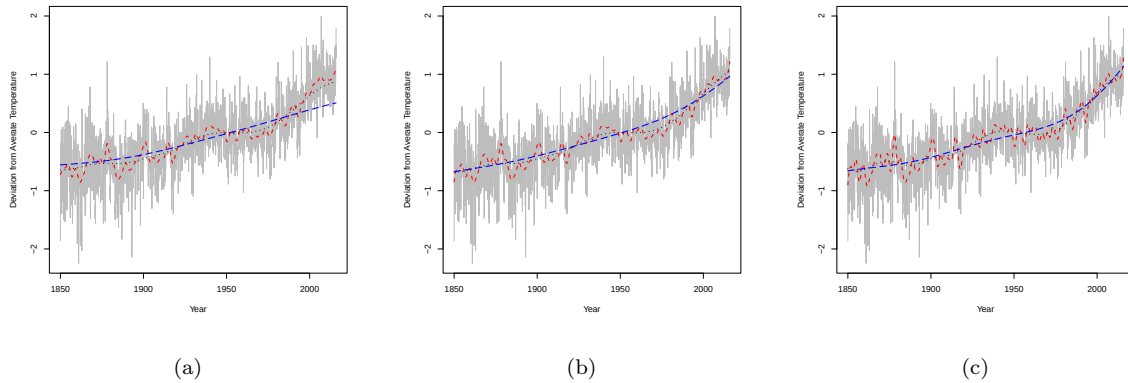


Figure 2.3: Global surface temperature data: (a) Nadaraya-Watson Smoother with bandwidths 1, 10, and 30, (b) local linear regression with bandwidths 1, 10, and 30, (c) local polynomial regression with bandwidths 1, 10, and 30.

2.2.1 Notes and Comments

This section contains miscellanea of final remarks relevant for Part II that were not addressed elsewhere in Section 2.

Both similarity-based methods and mixture models usually require the number of clusters

or components as input. The literature on this topic is extensive. Examples of selection criteria for the number of clusters or components are the within groups sum of squares (WSS), gap statistic, and the average silhouette width (ASW). An overview of selection criteria can be found in [Hu & Xu \(2003\)](#) and [Charrad et al. \(2014\)](#).

While most unsupervised clustering methods require the number of clusters/components as input, there is a growing field of research on methods that estimate the number of clusters within the model. Examples are repulsive point processes ([Beraha et al., 2022](#); [Xie & Xu, 2020](#)), group-sort-fuse procedures ([Manole & Khalili, 2021](#)) and NLP mixtures. We take advantage of the latter for modelling margins in the Reign-and-Conquer algorithm developed in Chapter 4.

Having covered a selection of commonly used unsupervised clustering methods in Section 2.1, it is worth noting that some existing approaches draw on game theory. For a detailed overview of game theory see [Gibbons \(1992\)](#) and [Maschler et al. \(2020\)](#). In the seminal paper [Bulò & Pelillo \(2009\)](#), the authors show that the concept of a cluster is equivalent to the notion of an equilibrium in an evolutionary game. The work generated a lot of interest and was followed by papers such as [Hou et al. \(2022\)](#) and [Hou et al. \(2023\)](#). Clustering based on game theory has found many practical applications, an overview of which can be found in [Bulò & Pelillo \(2017\)](#). While the contributions presented in Chapter 4 also have some connections to game theory, our approach and strategy are rather different from those of the latter papers.

Evaluation methods for unsupervised clustering approaches include average silhouette width [Rousseeuw \(1987\)](#), the Dunn index ([Dunn, 1974](#)), the Davies-Bouldin index ([Davies & Bouldin, 1979](#)), and connectivity ([Handl & Knowles, 2005](#)).

Part II

Main Contributions

Chapter 3

Subject-to-Group Statistical Comparison for Open Banking-Type Data

This Chapter develops the concept of the time-varying comparative mean value. Open banking (OB) creates an opportunity for financial institutions to offer more personalised services by better differentiating between a specific customer (reference subject) and similar customers (comparison group). We propose the time-varying comparative mean value as a statistical method that learns about the dynamics governing how the response of a reference subject differs from that of a comparison group, defined via covariate truncation. The proposed model can be regarded as a time-varying truncated covariate regression model of which a smooth version is devised by resorting to local polynomial regression. The simulation study suggests that our estimators accurately recover the true time-varying comparative mean value in a variety of scenarios. We showcase our methods using OB-type data from a financial service provider in the UK, with the dataset containing detailed information on customers' accounts across 70 UK financial institutions. By contrasting a specific customer against similar customers, our method offers interesting diagnostics that can be used by financial institutions to recommend personalised services.

3.1 Introduction

In this Chapter we propose an approach for comparing a specific reference subject (e.g. a customer of a financial institution or an user of a financial application) to a comparison

group. Since the proposed methodology is motivated from the recent open banking (OB) paradigm, we start by offering some background on the applied context motivating the methodological contributions.

The digital revolution of OB is impacting a wealth of decision-makers, problem-solvers, and innovators. In particular, under OB financial service providers can access datasets containing detailed financial information on potential customers across several financial institutions. Thanks to OB-type data financial institutions will have the opportunity to offer more personalised services by better differentiating between a specific customer (*reference subject*) and similar customers (*comparison group*). Trendsetters like Facebook (Cheng & Zagat, 2019) are already seizing upon the opportunities stemming from subject-to-group comparison in an OB context, that is, differentiating between a reference subject and similar customers. Yet, thus far no methodological developments have been made that allow for a sound subject-to-group comparison, and one of the goals of this Chapter will be to explore this open problem.

Different fields spreading from computer science (Wang et al., 2020), regulatory technologies (Buckley et al., 2020) and healthcare (Stranieri et al., 2021) are showing interest in OB. The digital revolution of OB is an initiative led by multiple governments, such as Australia, USA, UK, and EU in the form of the PSD2 legislation (Brodsky & Oakes, 2017; European Commission, 2018; He et al., 2020). The introduction of OB goes hand in hand with the currently expanding area of fintech, which aims to enhance customer experience (Gomber et al., 2018); this has led to a vast increase of personalised customer-centric fintech services (Breibach et al., 2019). Fintech and OB are viewed as game-changing and disruptive innovations (Lee & Shin, 2018; Nicoletti et al., 2017; Omarini, 2018), bringing opportunities and challenges for financial institutions, consumers, academics, regulators, and governments alike.

Besides industry demand, the developments in OB and fintech have been accelerated by innovations in machine learning, artificial intelligence, and related fields (Mention, 2019). Yet, little has been done to address these new challenges from a statistical perspective. We aim to fill this methodological and applicational gap by providing a statistical solution to subject-to-group comparison, that provides valuable OB-based financial advice that is intuitive and easy to interpret.

OB-type data has the potential to change how personalised financial services are provided, and creates a necessity to develop subject-to-group comparison methods tailored for these purposes. Unlike subject-to-group comparisons, methods for comparing population groups, such as hypothesis tests, analysis of variance (ANOVA) and regressions with group indicators, are well-studied. The analysis of treatment outcomes between groups is also well studied in methods such as propensity scoring Rosenbaum & Rubin (1983). Recent microfinancial studies utilise new and highly personalised data sources to provide valuable insight on credit scoring (De Cnudde et al., 2019; Óskarsdóttir et al., 2019; Zhu et al., 2020). Yet these methods are conceptualised to compare different population groups and not suitable for identifying and analysing

the differences between a specific customer and a group with similar characteristics, and thus can not harness the full potential of OB-type data and alike. A key goal of this chapter is to propose an approach that can be used for learning about the latter differences.

In this Chapter, we contribute by providing an approach for comparing a specific reference subject to a comparison group defined via covariate truncation, based on what we will refer to as the *time-varying comparative mean value*. While the proposed methodology is motivated from OB data, we underscore that our framework is general, and that thus the proposed methods can be applied to any context where the goal is to conduct a subject-to-group comparison. To the best of our knowledge, this section pioneers the mathematical modelling of subject-to-group comparisons.

Here and below we will use the expression covariate truncation to refer to a truncation centred on the reference subject's covariate value. Conditioned on the comparison group identified by covariate truncation (say, income) the expected value of the response (say, expenditure), is analysed. The homogeneity in terms of the covariate within the comparison group is controlled by a similarity variable (δ below) set by the user; we analyse the properties of the proposed method in terms of this variable. This distinguishes this method from other smoothing methods, such as kriging [Cressie \(1990\)](#), which smooths based on the whole range of the covariate space rather than a specific truncated section. No parametric structure on the studied variables is assumed. Time dependency is also not assumed as the time-varying comparative mean value constructed to compare a an individual to a group of peers and not the effect of covariate changes. A smooth time-varying comparative mean value is also devised by resorting to local polynomial regression ([Fan & Gijbels, 1996](#)). By contrasting the time-varying comparative mean value with the reference subject values, we can identify differences between the reference subject and a group with similar characteristics (peers). By varying the similarity variable (δ) information on the distribution of the response can be obtained. Local polynomial regression has the added benefit of providing a predictor for the comparative mean value. However, unlike certain group-to-group comparison methods such as propensity score matching, the method is not meant to analyse effects of various changes and treatments for the reference subject, but rather to identify differences between the subject and their peers.

A statistical concept that relates to the here proposed time-varying comparative mean value is the so-called F -barycenter ([Hill & Monticino, 1998](#)). For a random variable $X \sim F$, with finite expectation and with strictly increasing F , the F -barycenter of $(a, c]$ is defined as

$$b_F = E\{X \mid X \in (a, c]\}. \quad (3.1)$$

As we will elaborate below, the proposed time-varying comparative mean value is designed for a regression framework, and it can be regarded as an $F_{Y_t|X_t}$ -barycenter, for the pairs of stochastic processes $\{X_t\}$ and $\{Y_t\}$, and with $(a, c]$ centred at the reference subject's covariate.

Other statistical methods that are conceptually related to the here proposed time-varying comparative mean value are the marginal expected shortfall (MES) and the systemic expected shortfall (SES) proposed by [Acharya et al. \(2017\)](#) and further developed in [Cai et al. \(2015\)](#) and [Acharya et al. \(2017\)](#); we will provide further details on these methods in Section 3.2.4. Although the related statistical concepts have some points of contact with our methods, their approach would not be suitable for offering a subject-to-group comparison. The time-varying comparative mean value also relates to regression toward the mean and k -nearest neighbours. Yet the latter two methods converge in probability to a conditional mean given a fixed value of the covariate ([Hastie et al., 2009](#), Section 2.4), whereas here the goal will be on learning about a conditional mean—given the covariate is in an interval induced by covariate truncation. The time-varying comparative mean value is also connected to truncated regression methods, such as those developed in [Efron & Petrosian \(1999\)](#), [Shen \(2010\)](#) and [Ying et al. \(2019\)](#). We show that the time-varying comparative mean value can be seen as a regression with truncated covariates. This is an important distinction to the latter regression methods, as those methods are tailored for truncated responses. To our knowledge, the comparative mean value provides a first attempt to model truncated covariates on a regression framework.

The remainder of this Chapter unfolds as follows. In Section 3.2 we present the comparative mean value (static and time-varying versions), introduce estimation methods, show the connection of the comparative mean value to regression, and provide comments on related statistical concepts. In Section 3.3 we test the estimators of the time-varying comparative mean value using a Monte Carlo simulation. In Section 3.4 we then showcase the method’s use on Money Dashboard personal financial data of UK residents in 2017 and 2018. We track the development of expenditure of selected reference subjects, contrasted with the expenses of peers with similar income. In Section 3.5 we summarise and discuss our main findings.

3.2 Comparative Mean Value

3.2.1 Comparative Mean Value: Construction and Properties

Let $X \sim F_X$ and $Y \sim F_Y$, with F_X and F_Y denoting the distribution functions of X and Y , respectively; let $X_0 = x_0$ and $Y_0 = y_0$ be fixed values for a reference subject. Although below we define the comparative mean value as a general concept, for our applied setup the expression ‘reference subject’ should be understood as representing a customer of a financial institution or an user of a financial application. We are interested in the expected value of the response Y , for subjects that are similar to the reference subject in respect to the covariate X . Specifically, the goal is contrasting the reference subject to similar subjects, whose covariate X lies within a neighbourhood of x_0 , $W_\delta(x_0) \equiv [x_0 - \delta, x_0 + \delta]$. All comparisons below are for $\delta > 0$ such that

for all $x \in W_\delta(x_0)$ it holds that $f_X(x) = dF_X/dx > 0$; the extension to the asymmetric setting, $W_{\delta_1, \delta_2}(x_0) = [x_0 - \delta_1, x_0 + \delta_2]$, is straightforward but notationally burdensome. The modelled object of interest is what we refer to as the *comparative mean value* (at level δ)

$$\mu_\delta = E\{Y \mid X \in W_\delta(x_0)\}. \quad (3.2)$$

As $\delta \rightarrow 0$, there are connections with regression towards the mean, and we explore these links in Section 3.2.3.

The comparative mean value (3.2) obeys similar properties as the F -barycenter (Hill & Monticino, 1998, Lemma 2.2); indeed it follows that:

$$\begin{aligned} \{F_X(x_0 + \delta) - F_X(x_0 - \delta)\}\mu_\delta &= \{F_X(\mu_\delta) - F_X(x_0 - \delta)\}E\{Y \mid X \in [x_0 - \delta, \mu_\delta]\} \\ &+ \{F_X(x_0 + \delta) - F_X(\mu_\delta)\}E\{Y \mid X \in [\mu_\delta, x_0 + \delta]\}, \end{aligned} \quad (3.3)$$

and $E\{Y \mid X \in [x_0 - \delta, \mu_\delta]\} = \mu_\delta$ if and only if $E\{Y \mid X \in [\mu_\delta, x_0 + \delta]\} = \mu_\delta$.

In practice, we estimate μ_δ as follows. Let $n + 1$ be the total number of subjects, for which data $\{X_i, Y_i\}_{i=0}^n$ are available. We define the *window of comparison* as the set $A_\delta = \{i : X_i \in W_\delta(x_0)\}$. This means that we are conditioning on those subjects whose covariates X_i are within a range of δ from the reference value x_0 .

Let $k_{n,\delta}$ be the cardinality of the window of comparison, that is $k_{n,\delta} = |A_\delta|$; although k is a function of both n and δ , to ease notation we omit these dependencies and write $k \equiv k_{n,\delta}$. We define the *comparative sample mean* (at level δ) as

$$\hat{\mu}_\delta = \frac{1}{k} \sum_{i \in A_\delta} Y_i. \quad (3.4)$$

We can see that the estimator in (3.4) first selects subjects whose covariates X_i are within the interval $W_\delta(x_0)$, centred around the reference subject's covariate x_0 . Then the average of the response Y_i for the selected peer group is calculated. Assume:

$$1. \lim_{\delta \rightarrow 0} k = |A_0| \quad 2. \lim_{\delta \rightarrow \infty} k = n + 1 \quad 3. \lim_{n \rightarrow \infty} k = \infty.$$

Under the latter assumptions it can be shown that (3.4) has the following properties

$$1. \lim_{\delta \rightarrow 0} \hat{\mu}_\delta = y_0, \text{ a.s.} \quad 2. \lim_{\delta \rightarrow \infty} \hat{\mu}_\delta = \bar{Y}, \text{ a.s.} \quad 3. \hat{\mu}_\delta \xrightarrow{p} \mu_\delta, \text{ as } n \rightarrow \infty, \text{ for } \delta > 0. \quad (3.5)$$

Due to the weak robustness of the mean it can be argued that outliers can have an effect on the properties of the comparative mean value. In Section 3.5 we suggest the comparative mean quantile method to alleviate this issue. Alternatively, using trimmed means can be an option, e.g. Lugosi & Mendelson (2021).

Next, we devise a time-varying framework extending the methods introduced above.

3.2.2 Time-varying Comparative Mean Value

To track the dynamics governing the comparative mean value, we now develop a time-varying version. Let $\{X_t\}$ and $\{Y_t\}$ be stochastic processes and let $\{X_{0,t} = x_{0,t}\}$ and $\{Y_{0,t} = y_{0,t}\}$ be the fixed values of the reference subject. Our object of interest is now the *time-varying comparative mean value* (at level δ)

$$\mu_{\delta,t} = E\{Y_t \mid X_t \in W_\delta(x_{0,t})\}. \quad (3.6)$$

The time-varying version of properties (3.3) hold for (3.6). Let $n_t + 1$ be the total number of subjects in period t and let

$$\{X_{i,1}, Y_{i,1}\}_{i=0}^{n_1}, \dots, \{X_{i,T}, Y_{i,T}\}_{i=0}^{n_T}$$

be the observed data (say, income and expenditure). We again fix the reference subject's index at $i = 0$. The *time-varying window of comparison* is $A_{\delta,t} = \{i : X_{i,t} \in W_\delta(x_{0,t})\}$. Denoting $k_t = |A_{\delta,t}|$, we assume

$$1. \lim_{\delta \rightarrow 0} k_t = |A_{0,t}| \quad 2. \lim_{\delta \rightarrow \infty} k_t = n_t + 1 \quad 3. \lim_{n_t \rightarrow \infty} k_t = \infty.$$

The estimator of (3.6) is the *time-varying comparative sample mean* (at level δ)

$$\hat{\mu}_{\delta,t} = \frac{1}{k_t} \sum_{i \in A_{\delta,t}} Y_{i,t}. \quad (3.7)$$

For any time period, subjects whose covariates $X_{i,t}$ are within the interval $W_\delta(x_{0,t})$ are selected and the conditional mean of the response Y_t is calculated. Let $\bar{Y}_t = 1/n_t \sum_{i=1}^{n_t} Y_{i,t}$. The estimator $\hat{\mu}_{\delta,t}$ has the following properties

$$1. \lim_{\delta \rightarrow 0} \hat{\mu}_{\delta,t} = y_{0,t}, \text{ a.s.} \quad 2. \lim_{\delta \rightarrow \infty} \hat{\mu}_{\delta,t} = \bar{Y}_t, \text{ a.s.} \quad 3. \hat{\mu}_{\delta,t} \xrightarrow{p} \mu_{\delta,t}, \text{ as } n_t \rightarrow \infty, \text{ for } \delta > 0. \quad (3.8)$$

We now derive a smooth version of the time-varying comparative sample mean so to smooth the dynamics. For this purpose we resort to local polynomial regression (Fan & Gijbels, 1996) as the obtained smooth dynamics naturally extend those that would be obtained via standard regression methods, in the sense that they can be understood as a running weighted regression. When the degree of the polynomial is zero, we get the well-known Nadaraya–Watson estimator (Watson, 1964). Using a nonparametric approach provides two main benefits. First, we obtain a predictor for the time-varying comparative mean value. Second, while subject-to-group comparison contains valuable information, if the empirical results oscillate too much in time it is hard to recognise a clear trend. Smoothing the results makes them more intuitive and easy to understand.

The nonparametric regression model is defined as $\mu_{\delta,t} = m_t(\delta) + \varepsilon_t$, where ε_t is an independent and identically distributed error term with $E\{\varepsilon_t\} = 0$ and $\text{var}\{\varepsilon_t\} = \sigma^2$. The function $m_t(\delta)$ can be estimated by what we refer to as the *smooth time-varying comparative sample*

mean (at level δ)

$$\hat{m}_t(\delta) = \frac{\sum_{i=1}^T K_h(t-i) \hat{\mu}_{\delta,i}}{\sum_{i=1}^T K_h(t-i)}. \quad (3.9)$$

In (3.9), $K_h(\cdot) = K(\cdot/h)/h$, where K is a kernel function and $h > 0$ is a smoothing parameter (bandwidth). As usual, the bandwidth $h = h_T$ is a sequence such that $h \rightarrow 0$ and $hT \rightarrow \infty$, as $T \rightarrow \infty$. It is well known that the choice of the kernel has little effect on the estimates (Wand, 1995, Section 2.7). Yet the choice of bandwidth is an important one, as a poor choice can lead to over-smoothing or under-smoothing. A standard approach to determine h is via cross-validation, with least-squares based validation being regarded as optimal (DasGupta, 2008, Section 3.10.2). We use the method proposed by Li & Racine (2004) and implemented by the authors in the R package `np`. Under fixed n_t and regularity conditions, the estimator in (3.9) inherits the following asymptotic properties from the Nadaraya–Watson estimator (e.g. Racine, 2001, Section 2), as $T \rightarrow \infty$

$$\text{bias}\{\hat{m}_t(\delta)\} \approx m_t^{(2)}(\delta) \frac{h^2}{2} \int_{-\infty}^{\infty} K(v) v^2 dv, \quad \text{var}\{\hat{m}_t(\delta)\} \approx \frac{\sigma^2}{Th} \|K\|_2^2.$$

Here $\|\cdot\|_2$ is the L^2 norm; noticeable in the latter expressions is the trade-off between variance and bias in respect to the bandwidth.

3.2.3 Time-varying Comparative Mean Value as a Truncated Covariate Regression

We now discuss the connection between the time-varying comparative mean value (3.6) and regression, which originates from observing that

$$\mu_{\delta,t} \rightarrow E\{Y_t \mid X_t = x_{0,t}\}, \quad \text{as } \delta \rightarrow 0.$$

For generality, we first focus on the nonparametric specification $Y_t = m_t(X_t) + \varepsilon_t$. For any $\delta \geq 0$, we then have

$$\mu_{\delta,t} = E\{m_t(X_t) \mid X_t \in W_\delta(x_{0,t})\} + E\{\varepsilon_t \mid X_t \in W_\delta(x_{0,t})\}.$$

Denoting by F_t the cumulative distribution function of the covariate X_t , and using Ruiz & Navarro (1996, p. 564), we can provide a general formula for computing the time-varying comparative mean value, only assuming that X_t and ε_t are independent and $E(\varepsilon_t \mid X_t) = 0$. Under these assumptions it can be shown that $E\{\varepsilon_t \mid X_t \in W_\delta(x_{0,t})\} = 0$. Thus, for $\delta > 0$ the time-varying comparative mean value (3.6) is

$$\mu_{\delta,t} = \frac{1}{F_t(x_{0,t} + \delta) - F_t(x_{0,t} - \delta)} \int_{x_{0,t}-\delta}^{x_{0,t}+\delta} m_t(x) dF_t(x). \quad (3.10)$$

Equation (3.10) suggests another route for estimating the time-varying comparative mean value. Yet it would require estimation of the margins and of the conditional mean, followed by integration, and thus it is not pursued further here.

If the association between the response Y_t and the covariate X_t is linear, that is if $Y_t = \alpha_t + \beta_t X_t + \varepsilon_t$, the time-varying comparative mean value (3.6) becomes

$$\begin{aligned}\mu_{\delta,t} &= \alpha_t + \beta_t \mathbb{E}\{X_t \mid X_t \in W_\delta(x_{0,t})\} + \mathbb{E}\{\varepsilon_t \mid X_t \in W_\delta(x_{0,t})\} \\ &= \alpha_t + \beta_t \frac{1}{F_t(x_{0,t} + \delta) - F_t(x_{0,t} - \delta)} \int_{x_{0,t}-\delta}^{x_{0,t}+\delta} x dF_t(x).\end{aligned}\tag{3.11}$$

For specific parametric distributions we can provide closed form expressions for $\mu_{t,\delta}$. In Table 3.1, adapted from Johnson et al. (2005, p. 134) and Ruiz & Navarro (1996, p. 570), we do this for a selection of distributions, under the linear specification $Y_t = \alpha_t + \beta_t X_t + \varepsilon_t$. Noteworthy is the case $X_t \sim \text{Unif}[a, b]$, for which the comparative mean value (3.11) is independent of δ . We depict a linear setting with $X_{i,t} \sim N(\theta_t, \sigma_t)$ in Section 3.3.1, Scenario B.

3.2.4 Comments on Conceptually-Related Methods

Beyond the links with the F -barycenter (3.1) highlighted in the Introduction, we now make some remarks on conceptually-related methods. As discussed in the Introduction, the other concepts relating to the comparative mean value are the MES and the SES. Time-varying version of all these concepts can be easily defined; yet to streamline the discussion we will focus on the static setup, and thus we work with pairs of random variables (X, Y) rather than stochastic processes (X_t, Y_t) . The MES is defined as

$$\text{MES}(p) = \mathbb{E}\{Y \mid X > Q(1 - p)\}, \quad 0 < p < 1,$$

where $Q(p) = \inf\{x : F(x) \geq p\}$; the SES has links with the MES and details on it can be found in Acharya et al. (2017). While keeping in mind the different context in which these concepts are devised, namely the evaluation of risk for financial institutions, the main difference between the MES, the SES and the time-varying comparative mean value (3.6) is the set the expected value is conditioned on. The MES and SES are conditioned on unbounded half-open intervals. However, for our purpose of comparing a reference subject to a group of peers it is more meaningful to condition on a bounded interval around the reference subject covariate.

Ex-ante one could be led to believe that k -nearest neighbours would be tantamount to our time-varying comparative mean value. Yet as we discuss below, k -nearest neighbours is designed for estimating $\mathbb{E}\{Y \mid X = x_0\}$ —similarly to regression toward the mean methods—whereas our approach is tailored for estimating $\mathbb{E}\{Y \mid X \in W_\delta(x_0)\}$. To see this, let $N_k(x_0)$ be the set of k closest values of X_i to x_0 . For k -nearest neighbours it can be shown that (Hastie et al., 2009,

Table 3.1: Parametric examples of the time-varying comparative mean value for linear dependence between Y_t and X_t .

Distribution	Time-varying comparative mean value	Restrictions
$N(\theta_t, \sigma_t^2)$	$\alpha_t + \beta_t \left\{ \theta_t + \sigma_t \frac{\phi\left(\frac{x_{0,t}-\delta-\theta_t}{\sigma_t}\right) - \phi\left(\frac{x_{0,t}+\delta-\theta_t}{\sigma_t}\right)}{\Phi\left(\frac{x_{0,t}+\delta-\theta_t}{\sigma_t}\right) - \Phi\left(\frac{x_{0,t}-\delta-\theta_t}{\sigma_t}\right)} \right\}$	$\delta \neq 0$
Unif(a, b)	$\alpha_t + \beta_t x_{0,t}$	$a < b$ and $\delta \neq 0$
Exp(λ_t)	$\alpha_t + \beta_t \frac{\exp(2\lambda_t \delta)(x_{0,t}+\delta+1/\lambda_t) - x_{0,t} - \delta - 1/\lambda_t}{\exp(2\lambda_t \delta) - 1}$	$\lambda_t > 0$ and $\delta \neq 0$
Power(c_t)	$\alpha_t + \frac{\beta_t}{c_t+1} \frac{\{c_t(x_{0,t}+\delta)+a\}(x_{0,t}+\delta-a)^{c_t} - \{c_t(x_{0,t}-\delta)+a\}(x_{0,t}-\delta-a)^{c_t}}{(x_{0,t}+\delta-a)^{c_t} - (x_{0,t}-\delta-a)^{c_t}}$	$a \leq x_{0,t} - \delta < x_{0,t} + \delta \leq b$ and $c_t > 0$
Cauchy(location = θ_t , scale = σ_t)	$\alpha_t + \beta_t \theta_t + \frac{\beta_t \sigma_t \log(\sigma_t^2 + (x_{0,t} + \delta - \theta_t)^2) - \log(\sigma_t^2 + (x_{0,t} - \delta - \theta_t)^2)}{2 \arctan\left(\frac{x_{0,t} + \delta - \theta_t}{\sigma_t}\right) - \arctan\left(\frac{x_{0,t} - \delta - \theta_t}{\sigma_t}\right)}$	$\sigma_t > 0$ and $\delta \neq 0$
Pareto(scale = σ_t , shape = θ_t)	$\alpha_t + \beta_t \left\{ \frac{\theta_t}{1-\theta_t} \frac{(x_{0,t}-\delta)^{\theta_t} (x_{0,t}+\delta) - (x_{0,t}+\delta)^{\theta_t} (x_{0,t}-\delta)}{(x_{0,t}+\delta)^{\theta_t} - (x_{0,t}-\delta)^{\theta_t}} \right. \\ \left. \frac{x_{0,t}^2 - \delta^2}{\delta^2} \log\left(\frac{x_{0,t}+\delta}{x_{0,t}-\delta}\right) \right\}$	$\theta_t \neq 1$ and $\delta \neq 0$ $\delta \neq 0$ and $x_{0,t} \pm \delta \neq 0$
Laplace(location = θ_t , scale = σ_t)	$\alpha_t + \beta_t \left\{ \frac{\exp(\frac{2\delta}{\sigma_t})(\sigma_t + x_{0,t} - \delta) - \sigma_t - x_{0,t} - \delta}{\exp(\frac{2\delta}{\sigma_t}) - 1} \right. \\ \left. \frac{2\theta_t - (x_{0,t} - \delta - \sigma_t) \exp\left(\frac{x_{0,t} - \delta - \theta_t}{\sigma_t}\right) - (x_{0,t} + \delta + \sigma_t) \exp\left(\frac{\theta_t - x_{0,t} - \delta}{\sigma_t}\right)}{2 - \exp\left(-\frac{x_{0,t} + \delta - \theta_t}{\sigma_t}\right) - \exp\left(-\frac{x_{0,t} - \delta - \theta_t}{\sigma_t}\right)} \right. \\ \left. \frac{\exp(\frac{2\delta}{\sigma_t})(x_{0,t} + \delta - \sigma_t) - x_{0,t} + \delta + \sigma_t}{\exp(\frac{2\delta}{\sigma_t}) - 1} \right\}$	$\sigma_t > 0, \theta_t \leq x_{0,t} - \delta$ and $\delta \neq 0$ $\sigma_t > 0$ and $x_{0,t} - \delta < \theta_t \leq x_{0,t} + \delta$ $\sigma_t > 0, x_{0,t} + \delta < \theta_t$ and $\delta \neq 0$

Section 2.4)

$$\frac{1}{k} \sum_{X_i \in N_k(x_0)} Y_i \xrightarrow{p} E\{Y \mid X = x_0\}, \quad \text{as } n, k \rightarrow \infty,$$

assuming $k/n \rightarrow 0$. Whereas, as noted in (3.5), for the comparative mean value we have

$$\frac{1}{k} \sum_{i \in A_\delta} Y_i \xrightarrow{p} E\{Y \mid X \in W_\delta(x_0)\}, \quad \text{as } n \rightarrow \infty.$$

In words, the time-varying comparative mean value converges in probability to an expected value conditioned on the interval $W_\delta(x_0)$ induced by covariate truncation, while regressions toward the mean methods and k -nearest neighbours converge in probability to an expected value conditioned on a fixed value of the covariate. Additionally, as $n \rightarrow \infty$ we have $k \rightarrow \infty$, but observe that if $\delta \rightarrow \infty$, k is not an intermediate sequence as $k/n = 1$.

Another area of research related to the time-varying comparative mean value is truncated regression. In certain applications it is necessary to truncate the observations and focus only on a selected interval. Based on the results from, [Turnbull \(1976\)](#), [Efron & Petrosian \(1999\)](#) propose the nonparametric maximum likelihood estimate (NPMLE) for doubly truncated data. The authors study a truncated response Y , restricted to a closed interval $[a, c]$ and covariate X . Using the same setting, [Shen \(2010\)](#) provide an alternative derivation of the NPMLE hazard function, and recently [Ying et al. \(2019\)](#) provided estimators for the standard linear model. As shown in Section 3.2.3, the time-varying comparative mean value (3.6) can be thought as a regression, in which the covariate is truncated to the window of comparison. An important distinction is that the time-varying comparative mean value is for truncated covariates, whereas the latter regression methods are tailored for truncated responses. Indeed, to our knowledge the literature covering truncated regression has focused on truncated responses, while the time-varying comparative mean value provides a first attempt to model truncated covariates on a regression framework.

3.3 Simulation Study

3.3.1 Monte Carlo Simulation

In this section we assess the finite sample performance of the time-varying comparative sample mean, as defined in (3.7) and the smooth time-varying comparative sample mean, as defined in (3.9). Before reporting the results of the Monte Carlo simulation, we first illustrate the methods on a single run experiment. For this purpose we generate data

$\{X_{i,1}, Y_{i,1}\}_{i=0}^{n_1}, \dots, \{X_{i,T}, Y_{i,T}\}_{i=0}^{n_T}$ in two scenarios and for two values of $n_t = n$. In Scenario A, we generate independent $X_{i,t}$ and $Y_{i,t}$. In Scenario B, we generate $X_{i,t}$ and the $Y_{i,t}$ are obtained by the following linear dependency structure $Y_{i,t} = \alpha + \beta X_{i,t}$. The parametrisation of both scenarios is available from Table 3.2.

In both scenarios we set the reference subject's values to be $x_{0,t} = E(X_t)$ and $y_{0,t} = E\{Y_t \mid X_t \in W_\delta(x_{0,t})\}$. For Scenario B, the comparative mean value in this case is given by the parametric formulation of (3.11) for the normal distribution found in Table 3.1. Because of our choice of reference value $x_{0,t} = E(X_{i,t})$, and the symmetry of the normal distribution density function the time-varying comparative mean value (3.6) in Scenario B is $\mu_{\delta,t} = E\{Y_t \mid X_t \in W_\delta(x_{0,t})\} = 5 + 2|\sin(t/10) + 1|$.

Table 3.2: Data generating processes used over the simulation study; we use the shape-scale parametrisation of the Gamma distribution.

Data generating process		
Data	Scenario A	Scenario B
Covariate ($X_{i,t}$)	Gamma($\exp(t/100), \exp(t/100)$)	$N(\sin(t/10) + 1 , 5)$
Response ($Y_{i,t}$)	Gamma($\{\sin(t/10) + t/10\}^2/t, 2)$	$5 + 2 X_{i,t}$

Both scenarios are tested for $n_t = 1,000$ and $n_t = 10,000$. In the Monte Carlo experiment we repeat the single run simulation 1,000 times. Two additional scenarios with jumps in the response are presented in the Supplementary Material. As mentioned in Section 3.2.2 cross-validation by Li & Racine (2004) is used to select the smoothing bandwidth. However, to avoid overfitting the asymptotically optimal bandwidth for kernel density methods given by DasGupta (2008, p. 531) is set as a lower bound.

As expected, based on properties listed in (3.8), in Figs 3.1–3.2 the following dynamics can be observed. The number of subjects with $X_{i,t}$ in $A_{\delta,t}$ increases as either δ or n_t increase. For $\delta \rightarrow 0$, only subjects with $X_{i,t} = x_{0,t}$ are in the window of comparison $A_{\delta,t}$. For continuous variables X_t and Y_t this is the case only for the reference subject. Thus, in both scenarios we get $\hat{\mu}_{0,t} = y_{0,t}$. As we slightly increase δ , we get more subjects in the window of comparison $A_{\delta,t}$ and the time-varying comparative sample mean fluctuates around the $\mu_{\delta,t}$. As either δ or n_t increase we get that $\hat{\mu}_{\delta,t}$ approaches $\mu_{\delta,t}$. The single run experiment and the Monte Carlo simulation indicate that both the time-varying comparative sample mean (3.7) and the smooth time-varying comparative sample mean (3.9) model the time-varying comparative mean value (3.6) well. Furthermore, the estimates get closer to the real value as δ and/or n_t increase. This is due to more subjects being in the window of comparison, as can be seen on the x -axis in Figs 3.1–3.2.

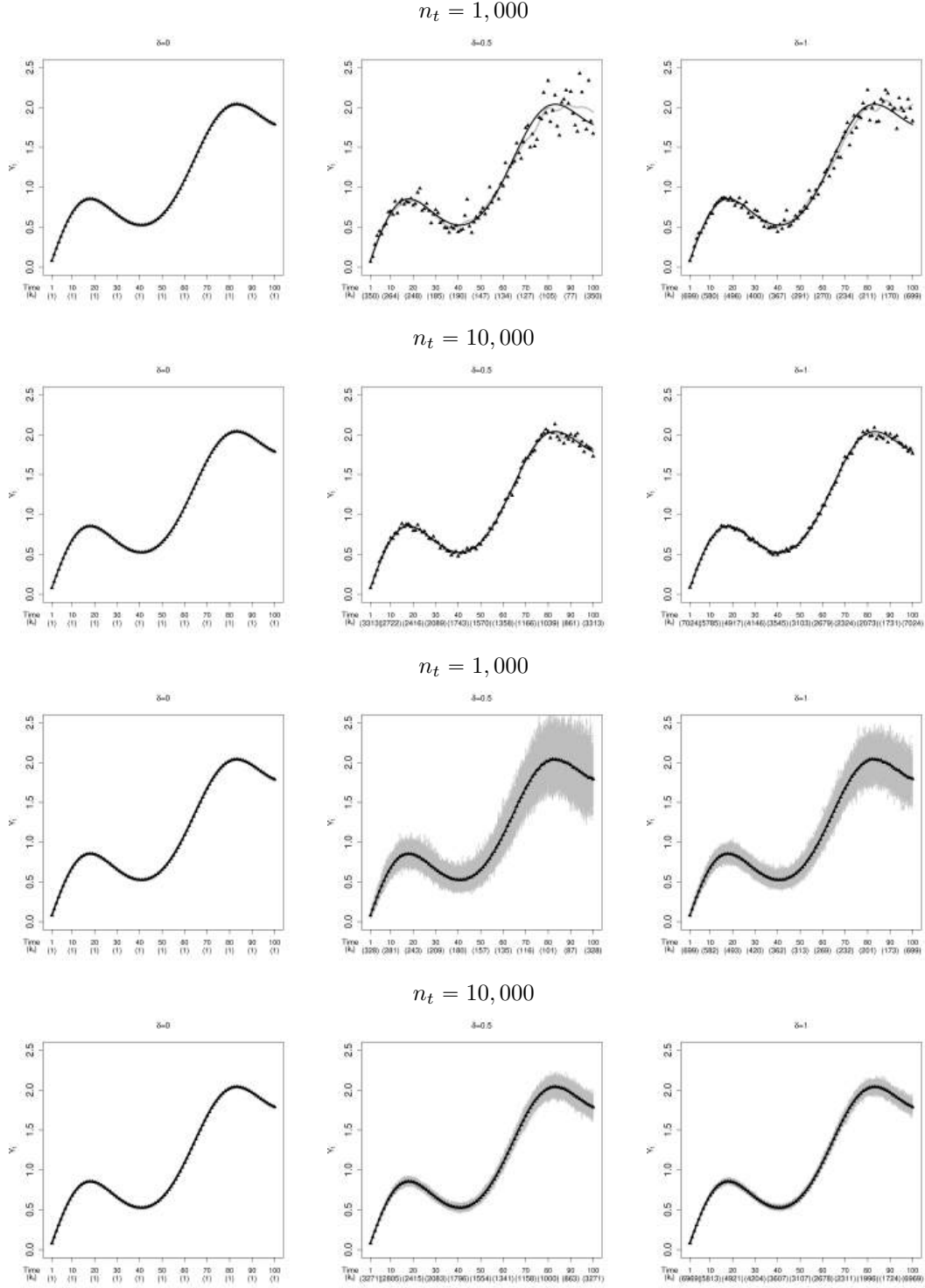


Figure 3.1: Time-varying comparative mean value, Scenario A: Top two rows show the single run experiment, and the bottom two rows show the Monte Carlo simulation. The time-varying comparative sample mean (3.7) is represented by the triangles ($\blacktriangle$). The grey curves represent the customer smooth time-varying comparative sample means (3.9), and the black curve is the true comparative mean value (3.6). The x -axis denotes time, and the number of subjects in the window of comparison, k_t .

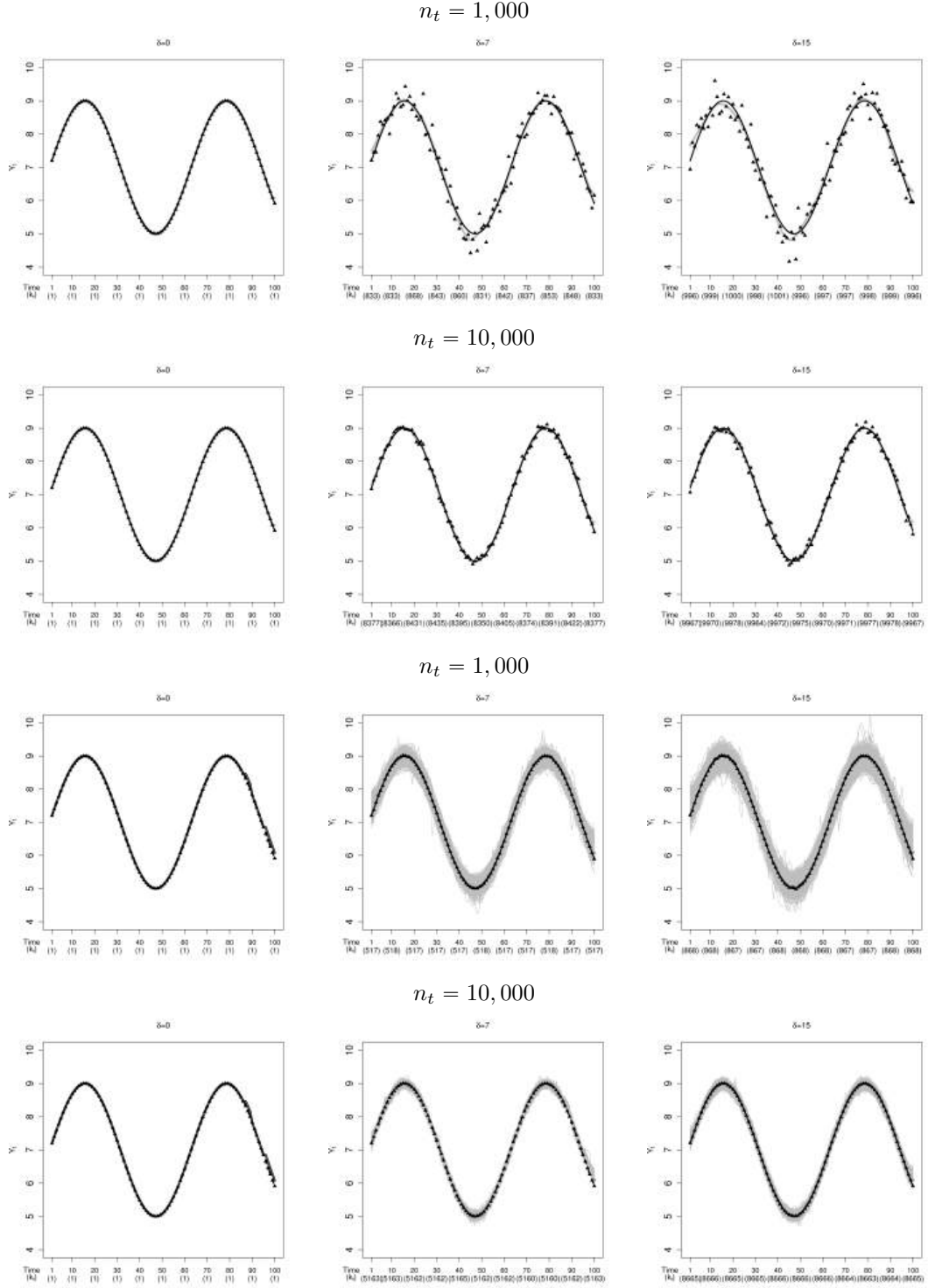


Figure 3.2: Time-varying comparative mean value, Scenario B: Top two rows show the single run experiment, and the bottom two rows show the Monte Carlo simulation. The time-varying comparative sample mean (3.7) is represented by the triangles ($\blacktriangle$). The grey curves represent the customer smooth time-varying comparative sample means (3.9), and the black curve is the true comparative mean value (3.6). The x -axis denotes time, and the number of subjects in the window of comparison, k_t .

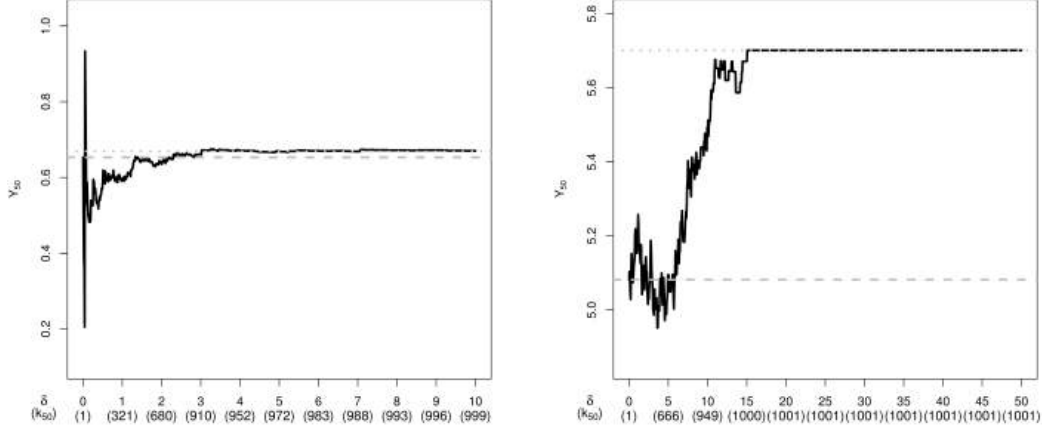


Figure 3.3: Time-varying comparative mean value over all windows of comparison for Scenario A (left) and Scenario B (right): The black curve represents the time-varying comparative sample mean (3.7) as a function of δ , and the dotted line is the total sample mean and the dashed line is the time-varying comparative mean value (3.6) for $t = 50$. The x -axis shows the δ values and the number of subjects in the window of comparison, k_t .

3.3.2 Comparative Mean Value over all Windows of Comparison

We now examine how the time-varying comparative mean value looks like when we consider all windows of comparison. For this purpose we fix the time at $t = 50$ and plot the time-varying comparative sample mean (3.7) as a function of δ . The obtained results, as seen in Fig. 3.3, provide information on the position of the reference subject's response among all comparison groups selected using the window of comparison, as well as the total sample mean.

From (3.8), we know that for $\delta \rightarrow 0$ we have $\hat{\mu}_{0,t} = y_{0,t}$. This means that the position of the time-varying comparative sample mean (3.7) as a function of δ , at $\delta = 0$, provides us information on the position of the reference subject compared to the total sample mean. The greater the absolute slope at small values of $\delta > 0$, the more the reference subject's response deviates from the average in the comparison group. For large positive slopes the subject's response $y_{0,t}$ is considerably lower, than the responses $Y_{i,t}$ of close peers. For small negative slopes, the subject's response is considerably higher, than the response average of close peers. As δ increases more subjects are included in the window of comparison, and the comparative sample mean approaches the total sample mean.

To provide a specific example of interpreting the time-varying comparative mean value over all windows of comparison, we choose to focus on Scenario A. In Fig. 3.3, Scenario A, we can see the reference subject's value is just below the sample mean. Per construction the reference subject's $y_{0,50}$ gives us the time-varying comparative mean value (3.6) at $t = 50$. The difference between the time-varying comparative mean value and the total sample mean is caused by the variance in each scenario, which is particularly noticeable in Scenario B where $\text{var}(Y_{i,t}) = 100$, and disappears at sufficiently high sample sizes. As δ slightly increases from the starting point $\delta = 0$ the time-varying comparative sample mean jumps up and down. This means, there are

subjects with very similar $X_{i,50}$, but very different $Y_{i,50}$ values, compared to the response of the reference subject. The intensity of these fluctuations at small values of δ is due to relatively low number of peers in the window of comparison and the weak robustness properties of the mean. For δ around (0.1, 2.2) the time-varying comparative sample mean stays below the reference subject's $y_{0,50}$; this indicates that the reference subject's response is higher than the average in any other group of peers created by the window of comparison $A_{\delta,50}$ for δ around (0.1, 2.2). Such a subject could be considered to be an outlier in the latter peer groups. After $\delta > 2.2$ the time-varying comparative sample mean as a function of δ aligns with the total sample mean, which is just slightly higher than the time-varying comparative mean value, indicating subjects with similar $X_{i,50}$ values have on average slightly higher $Y_{i,50}$ values.

3.4 Empirical Analysis

3.4.1 Data Description

We showcase the usefulness of the comparative mean value by considering three reference subjects, and comparing their expenditure to subjects with similar income.

The concepts and methods devised here were motivated by a collaboration with a UK financial services provider, Money Dashboard (MD) and the implementation of the time-varying concept in financial analysis platforms and financial applications. This is naturally connected to the high demand for personalised financial advice in the UK. According to the [Financial Conduct Authority \(2017, p. 10\)](#), in the last 12 months 6% of UK adults have received financial advice on personal investments, with “45% of those who have not received advice in the last 12 months report that they have had regulated financial advice related to investments, saving into a pension or retirement planning in the past”.

MD supplies its customers (in this Section users) with a summary of their registered accounts at any financial institution in the UK via an application. Further details on the business lines of MD can be found at

www.moneydashboard.com

The data here analysed data consist of demographic information on the users and of transaction records of all registered accounts across 70 financial institutions in the UK. These transactions carry information on vendor, location and purpose of the transaction. Based on their structure and information content the MD data can be considered to be of OB-type (details on OB can be found in [Section 1.1](#)). The MD dataset is unique in detail and provides a rare in-depth look into the daily financial behaviour of UK customers.

One drawback of the MD sample is that it consists of a self-selected group of MD application users. This has some implications on the variable distributions. Because we focus on specific

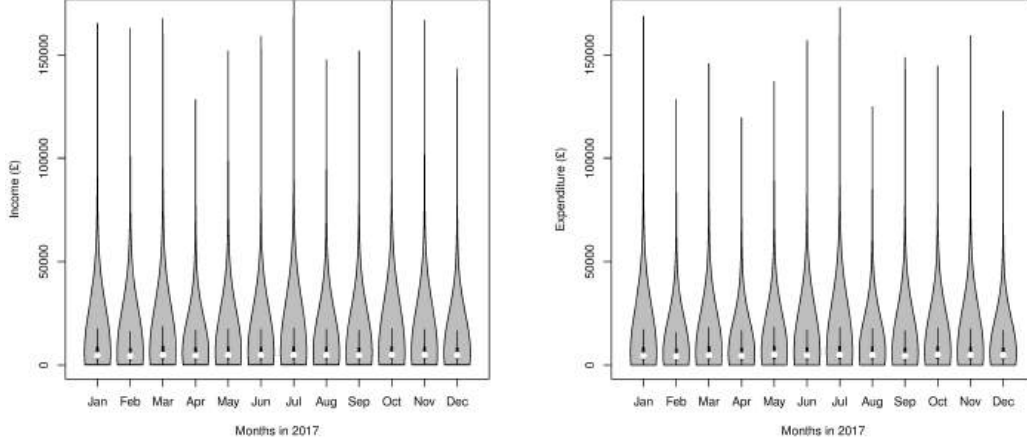


Figure 3.4: Violin plot of income (left) and expenditure (right) over 2017.

users, as long as we do not draw conclusions for the whole UK population from our results, our analysis and findings are not affected.

The used data consist of anonymised transaction records of 10,689 MD application users for 2017 and 2018. Using these information we categorise the transactions and aggregate income $X_{i,t}$ and expenditure $Y_{i,t}$, on a monthly basis, of any user i in month t . We initially focus on 2017. In Section 3.4.3 we include the 2018 data in our analysis.

The violin plots in Fig. 3.4 depict the monthly distributions of income and expenditure in the analysed 2017 data, and on average income and expenses are balanced out. However, the heavy tails of both the income and expenditure distributions in time are noticeable. This is highlighted by the difference in the average income of £10,293.29 and the median income of £4,767.615. The same goes for expenditure, where the mean is £9,923.63 and the median is £4,750.55. As users utilise MD services to manage multiple accounts, the data consist of a self-selected group of customers with sufficient funds and capability to manage them. In comparison with the total UK population, this leads to upward shifted distributions in the MD sample. For example, the [HM Revenue and Customs \(2018\)](#) obtained a mean monthly income of £2,891.66 and median monthly income of £2,033.33 for 2017.

3.4.2 Time-varying Comparative Mean Expenses

After having gained an overview of the MD data we next apply the time-varying comparative sample mean (3.7) with expenditure being the response ($Y_{i,t}$) and income being the covariate ($X_{i,t}$). We select three distinct users (reference subjects) with different financial behaviour from the dataset, whose expenditure ($y_{0,t}$) can be seen in Fig. 3.6, and whose income ($x_{0,t}$) we use as reference covariates the peer selection in (3.7). The peer selection is based on income and the expected value of expenditure, conditioned on the selected peer group is calculated as in

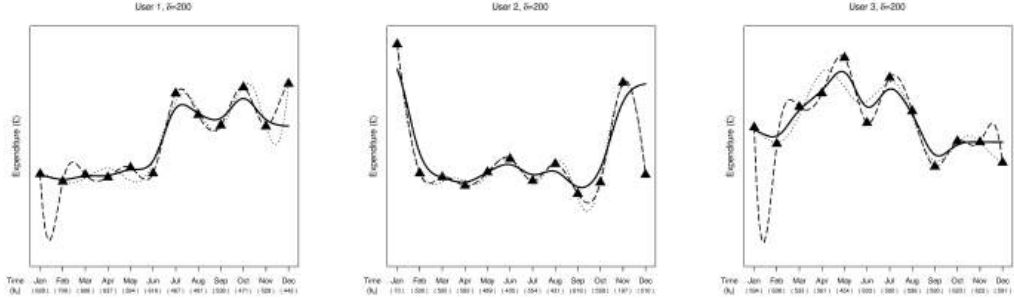


Figure 3.5: Smooth time-varying comparative mean expenditure: The time-varying comparative mean expenditure (3.7) is represented by the triangles ($\blacktriangle$), and the smooth comparative mean expenditure is represented by solid (kernel, (3.9)), dashed (Gaussian process), and dotted (P-spline) lines. The x -axis denotes time, and the number of subjects in the window of comparison, k_t .

(3.7). Using the the asymptotically optimal bandwidth for kernel density methods, given by DasGupta (2008, p. 531), we provide the smooth time-varying comparative sample mean (3.9). Then the results are compared to the response of the reference subject. In other words, we will be tracking how a user’s expenses develop in comparison to peers, who have similar income.

Fig. 3.5 shows the time-varying comparative mean expenses (3.7), and the smooth time-varying comparative mean expenditure (3.9) for three different users at $\delta = \pounds 200$; for privacy reasons, we are not allowed to display the y values nor on Fig. 3.5 nor on any subject-specific fit presented below. For comparison, we also present in Fig. 3.5 the smooth time-varying comparative mean expenditure that is learned by other two popular methods, namely Gaussian processes (Rasmussen & Williams, 2006) and P-splines (Eilers & Marx, 1996); the results from all smoothers are essentially equivalent. The value of δ was chosen to allow the peers to have very similar, but not identical income as the reference subject. In each panel of Fig. 3.5 users with income in the range of $\pm \pounds 200$ from the reference user’s income are selected and the time-varying comparative mean expenditure (3.7) is given. As can be seen in Fig. 3.5, smoothing provides a natural way to track the dynamics of the time-varying comparative mean expenditure over time.

In Fig. 3.6, we contrast the smooth time-varying comparative mean expenditure (3.9) with the expenditure of the reference subjects for three different values of δ . To give an example of how to interpret the obtained results, we will focus on the case $\delta = \pounds 200$. The smooth time-varying comparative mean expenditure (3.9) represents the mean expenditure of users with income $\pm \pounds 200$ in range of the reference user’s income. The points in Fig. 3.6 are the total expenses of the reference users in each month of 2017. We can notice how the three reference users considerably differ in their spending behaviour. With the exception of March, user 1 spends consistently more than the average of users with the same income $\pm \pounds 200$. The opposite case is user 2, who tends to spend less than the average of users with similar income. Exceptions are January and November, when the user’s spending is aligned with peers. For user 3 spending is lined up with that of the peer average, with the exceptional deviations in

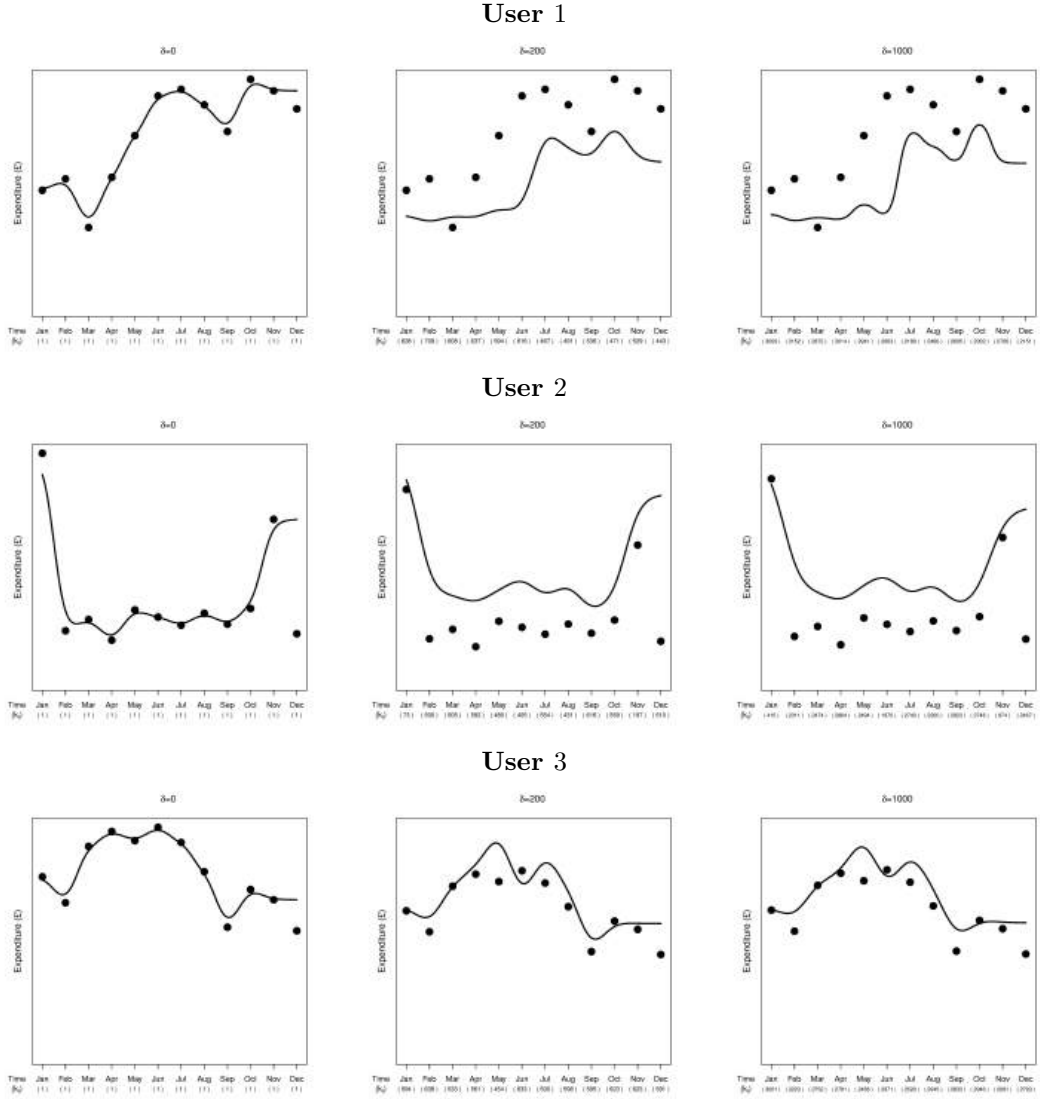


Figure 3.6: Time-varying comparative mean expenditure analysis: The smooth time-varying comparative mean expenditure is represented by the solid line; the dots ($\bullet$) are the user expenses $y_{0,t}$. The x -axis denotes time, and the number of subjects in the window of comparison, k_t . Each column corresponds to a different window of comparison.

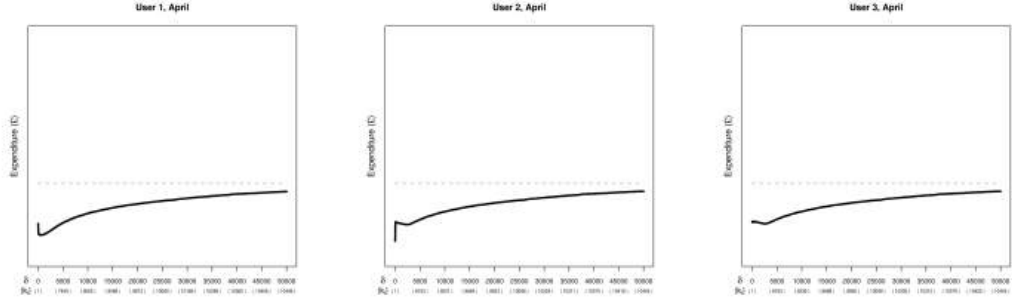


Figure 3.7: Comparative mean expenditure over all windows of comparison: The comparative mean expenditure (3.7) is represented by the black curve. The grey dotted curve represents the sample mean of expenditure in the given month. The x -axis shows the δ values and the number of subjects in the window of comparison, k_t .

February, September and December of 2017.

Given the information in Fig. 3.6, one can provide guidance on personal finance to the respective users. For illustration, because user 1 tends to spend consistently more money than peers with similar income, the user might be considered at risk of immediate or future financial problems. One can provide advice on how to reduce spending or occasionally send notifications to such users to remind them of their tendency to overspend. For user 2, who spends consistently less than peers with similar income, one can focus on advice on savings and investments. And last but not least, advisers can send user 3 warnings or recommendations when the user's spending behaviour starts to deviate from the peer average. The guidance also needs to take account the financial situation within the peer group, as even customers with similar income might be on average overspending or underspending. However, using the comparative mean value one is able to identify group behaviour of the comparison group, behaviour of reference subjects and detect customers with different behavioural patterns in comparison to peers.

Functions measuring the difference between the reference subject value and the time-varying comparative mean value are a possibility for identifying each of the three cases mentioned above. However, the usage of absolute measures requires separating cases when the reference subject's spending is above or below the time-varying comparative mean value so to distinguish between cases of overspending and underspending.

For a more complete picture of how the comparative mean expenditure changes with different values of δ , in Fig. 3.7 we select a specific month and plot the resulting comparative mean expenditure (3.7) with the corresponding values of δ .

The interpretation of Fig. 3.7 is as follows. The starting position of the comparative mean expenditure at $\delta = 0$ in respect to the total sample mean provides information on the reference subjects' position in the total distribution of expenditure. It is noticeable, that despite of the very different expense behaviour of our reference subjects, all three reference subjects spend less than the total sample on average. As the distribution of total expenses is heavy-tailed, most users have expenses below the total sample average.

The slope of the comparative mean expenditure as a function of δ tells us how much the

reference subjects differ from peers with similar income. In absolute, the greater the slope at low values of δ , the more the specific user's response deviates from that of the average in the peer group. For large positive slopes the user is spending considerably less than peers with very similar income. The smaller a negative slope, the more the user spends in comparison to their closest peer groups. As we keep increasing δ more people are included in the window of comparison and the comparative mean expenditure approaches the total sample mean. To illustrate, we turn back to Fig. 3.7. As discussed previously, we already know that user 1 tends to spend consistently more than peers with similar income. For April 2017, we can clearly see this in Fig. 3.7 on the very steep negative slope around $\delta \approx 0$. Furthermore, for any $\delta < \pounds 5,000$ the user's expenditure, seen at $\delta = 0$, stays above mean peer expenditure. This means that user 1 spends more than the average of any peer group selected based on similar income of $\pm \pounds 5,000$ of the user's income. As mentioned previously, this can become or already be a financial problem for the user. The opposite case is user 2, who tends to spend less than peers. This is in line with the slope of the comparative mean expenditure at low δ values in Fig. 3.7. A very small increase of δ around $\delta \approx 0$ increases the comparative mean expenditure considerably. This indicates that users with very similar income spend much more than user 2. As we keep increasing δ the comparative mean expenditure dips a bit, but stays above the reference subject's value. This means the user 2 spends less than any peer group selected using the window of comparison. Last but not least user 3 who spends roughly the same as the average of peers. We can see the comparative mean expenditure does not notably change at any delta of approximately $\delta < \pounds 2,000$. At around $\delta \approx \pounds 3,000$ the comparative mean value drops below the user's value. This indicates more customers with lower expenditure in the window of comparison. For values of approximately $\delta > \pounds 5,000$ this difference disappears and the comparative mean expenditure stays above the user's response.

A commonly used non-statistical approach for personal financial analysis are financial ratios (e.g. see [Bae et al., 2005](#); [Greninger et al., 1996](#); [Melzer, 2017](#); [OECD, 2020](#)) a review of which can be found in [Harness et al. \(2008\)](#). We present the expense to income ratios for users 1–3 in the Supplementary Material. However, we would like to point out that financial ratios were constructed for other purposes than for subject-to-group comparison pioneered in this section.

3.4.3 Further Empirical Analysis

Here we provide further illustrations of the proposed approach, with a particular focus on monitoring:

1. How the estimates of smooth comparative mean expenses get updated once new data arrives?
2. Is there any major impact on the subject-specific fits as the database structure evolves over time?

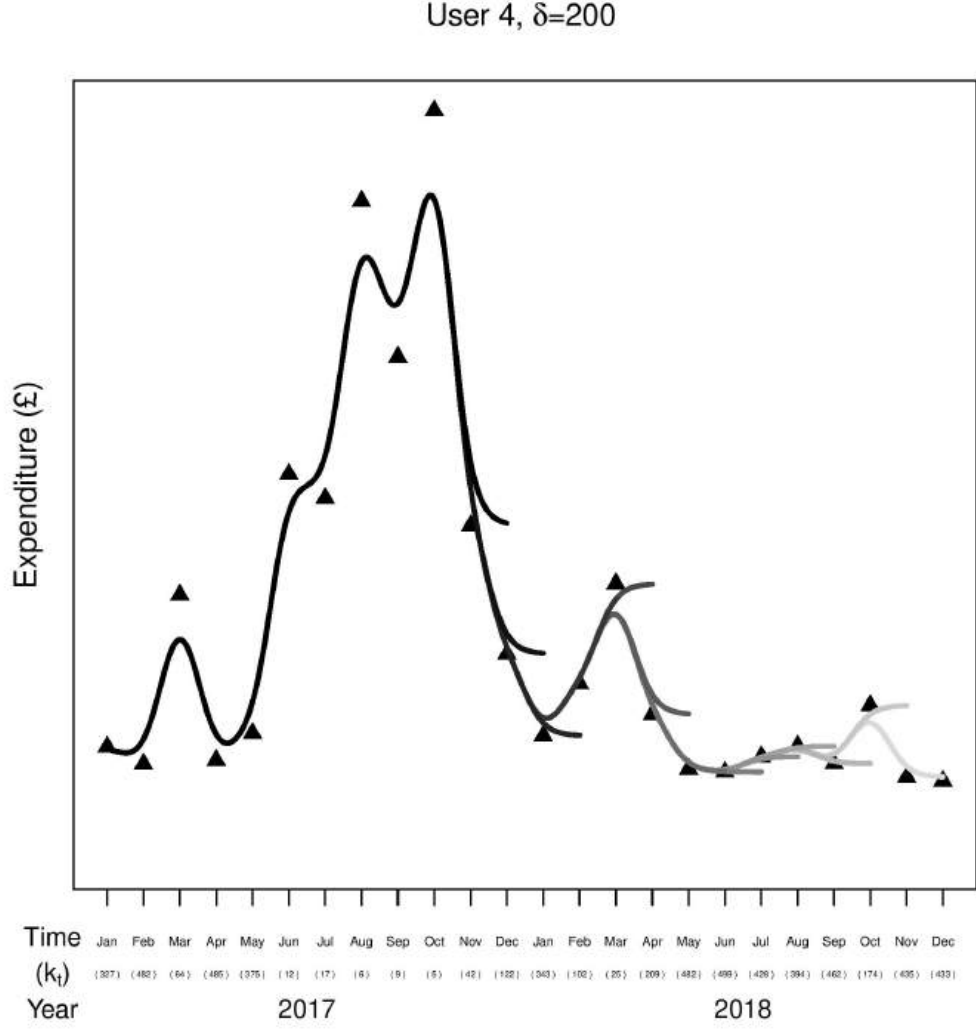


Figure 3.8: Real-time smooth time-varying comparative mean value analysis: Starting with January 2018 new data points are added and the smooth comparative mean expenditure (3.7) is fit. The lighter the curves the more data points are considered. The x-axis shows the number of subjects in the window of comparison k_t , and months between January 2017 and December 2018.

In terms of (1), in fields such as Econometrics the ability of a method to be coherent over real-time—in the sense of not revising estimates, once new data arrives—is key, and it has been a subject of wide interest (see, for instance [Orphanides & Van Norden, 2002](#), and references therein). In terms of (2), the data analysed below are similar to that from Section 3.4.1, but with minor changes in transaction categorization, and consisting of two years of data (2017–2018). As it can be seen from Fig. 3.8 revisions are moderate except at the end of the observation period, which is mostly explained by the well-known boundary–bias challenges faced by the Nadaraya–Watson estimator ([Wand, 1995](#), Section 5.5).

3.5 Discussion and Closing Remarks

Unlike methods for comparing population groups and treatments (e.g. t -test, ANOVA, and propensity score matching), subject-to-group comparison has received little attention in Operational Research and in Statistics, but as this Chapter puts forward the latter comparison is of the utmost importance in an OB setup. In this section, we introduced the time-varying comparative mean value, which is a statistical method for learning about the dynamics of differences between the response of a reference subject and that of a comparison group defined by covariate truncation around the reference subject's covariates. The covariate truncation is controlled by a similarity variable (δ) set by the user. By varying δ one can extract information on the distribution of the response and the position of the reference subject in it. No parametric structure on the response and/or covariates is assumed. The method can be viewed as a time-varying truncated covariate regression model, and local polynomial regression is used to define its smooth version. A simulation study showed the estimators recover the true time-varying comparative mean value well.

As discussed in Section 3.2.4, the time-varying comparative mean has links to statistical concepts such as the F -barycenter, MES and SES, regression toward the mean and k -nearest neighbours. There are also links to regression methods for truncated random variables. Yet, unlike the latter regression methods which are designed for truncated responses, the time-varying comparative mean value deals with truncated covariates. To our knowledge the time-varying comparative mean value provides a first methodological approach to truncated regression on an interval centred around a reference covariate.

We showcase the usefulness and insights obtained by the time-varying comparative mean value on data provided by a UK service provider, MD, who motivated the development of the here proposed methods. Because for each customer in the sample of 10,689 customers, the data covers all transactions in 2017 and 2018 for all accounts registered by the customers across 70 financial institutions in the UK, the data can be considered to be of OB-type.

In the analysis we compare the spending behaviour of selected reference customers with the spending behaviour of customers with similar income as the individual reference customers. Based on the diagnostics obtained by such analyses the financial services provider can offer advice and services specifically tailored for the financial needs of the reference customer. Because the insights obtained by the time-varying comparative mean value are also simple to read and communicate, when implemented into a financial services platform or an financial application, customers can contrast their own finances, e.g. expenses or savings, against customers they consider to be financial peers, e.g. people with on similar income or age.

Although we apply the time-varying comparative mean value to personal finance, due to its generality it can be used in other contexts, such as comparing one patient to patients with similar characteristics to identify patient-to-group differences.

To allow for a more granular definition of peers the time-varying comparative mean value can be defined using a window of comparison with multiple explanatory variables. While here we have focused on a window of comparisons based on the single covariate framework, the extension of the proposed methods for multiple covariates is feasible and will be explored elsewhere.

Tracking how the dynamics of the comparative mean value may change when there are potential jumps or breaks in the data is a natural challenge that one may face in some applications. Versions of the proposed time-varying comparative value that resort to models for curves with jumps (e.g. [Gijbels et al., 2004](#); [Kang, 2020](#)) or to wavelets ([Abramovich et al., 2000](#))—rather than local polynomial methods—may be more natural for those applications. In general, any interpolation method can be used for constructing the time-varying comparative mean value. Their usefulness depends on the context and properties of the given data. An overview of interpolation methods can be found in [Simonoff \(2012\)](#).

To identify customers at risk, it would also seem natural contrasting the reference subject against a high quantile of the response, such as expenses, say on a window of comparison based on subjects with similar covariates, such as income. This would lead to comparative quantile-based approaches, which would have links with quantile regression ([Koenker, 2005](#)). Specifically, the time-varying comparative quantile could be defined as

$$q_\delta = Q_{Y_t}\{p \mid X_t \in W_{\delta,t}(x_{0,t})\}, \quad 0 < p < 1.$$

This is another natural avenue for future research. Such an approach would also have the benefit of providing a more robust estimate due to the robustness properties of quantiles.

Chapter 4

Reign-and-Conquer: Parallelisable Model-Based Approach for Marginal and Multivariate Clustering

This chapter develops a clustering method that takes advantage of the sturdiness of model-based clustering, while attempting to mitigate some of its pitfalls. First, we note that standard model-based clustering likely leads to the same number of clusters per margin, which seems a rather artificial assumption for a variety of datasets. We tackle this issue by specifying a finite mixture model per margin that allows each margin to have a different number of clusters, and then cluster the multivariate data using a strategy game-inspired algorithm, which we call Reign-and-Conquer. Second, since the proposed clustering approach only specifies a model for the margins—but leaves the joint unspecified—it has the advantage of being partially parallelisable; hence, the proposed approach is computationally appealing as well as more tractable for moderate to high dimensions than a ‘full’ (joint) model-based clustering approach. A battery of numerical experiments on artificial data indicate an overall good performance of the proposed methods in a variety of scenarios, and real datasets are used to showcase their application in practice.

4.1 Introduction

Context and Motivation

Clustering is an unsupervised learning approach for the task of partitioning data into meaningful subsets. The huge literature on cluster analysis is difficult to survey in a few sentences, but a concise description of well-known approaches is offered by [Hastie et al. \(2009\)](#), [Everitt et al. \(2011\)](#), and [King \(2015\)](#). Examples of mainstream methods for clustering data include model-based ([Bouveyron et al., 2019](#)), similarity-based ([Kaufman & Rousseeuw, 1987](#); [MacQueen, 1967](#)), and hierarchical clustering ([Hastie et al., 2009](#), Section 14.3). In this chapter we propose a novel model-based approach for cluster analysis that lies at the interface of model-based clustering (i.e., via mixture models) and similarity-based clustering (i.e., via $\mathcal{K}$ -means and $\mathcal{K}$ -medoids). The proposed approach aims to benefit from the flexibility and soundness of model-based clustering, while attempting to mitigate Pitfalls 1 and 2 below. Model-based clustering is a fast-evolving and intradisciplinary research topic as can be seen from the recent Handbook on Mixture Analysis ([Fruhwirth-Schnatter et al., 2019](#)) as well as the survey papers of [Melnykov & Maitra \(2010\)](#), [McNicholas \(2016\)](#), [Gormley et al. \(2023\)](#), and the references therein.

Pitfall 1: The “Single $\mathcal{K}$ Problem”

The idea of thinking of a cluster as a component of a mixture model has a long tradition in cluster analysis, that has its roots in Tiedeman’s work in 1955 ([McNicholas, 2016](#)). Despite the resilience and flexibility of this paradigm, it is often unnoticed that multivariate model-based clustering may induce the same number of clusters on each margin. For many applied contexts of interest, such as customer segmentation using OB-type data described in Section 1.2, it is unnatural to believe that all margins should have exactly the same number of components—and hence the same number of marginal clusters. To appreciate this issue, let’s revisit the Gaussian finite-mixture model,

$$f(\mathbf{x}) = \sum_{k=1}^{\mathcal{K}} \pi_k \phi_d(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \mathbf{x} = (x_1, \dots, x_d), \quad (4.1)$$

where $\phi_d(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the density function of a d -dimensional multivariate Normal distribution with mean $\boldsymbol{\mu}_k = (\mu_{k,1}, \dots, \mu_{k,d})$ and variance-covariance matrix $\boldsymbol{\Sigma}_k$, with diagonal elements $(\sigma_{k,1}^2, \dots, \sigma_{k,d}^2)$. The marginal distributions stemming from (4.1) are

$$f_j(y) = \sum_{k=1}^{\mathcal{K}} \pi_k \phi(y; \mu_{k,j}, \sigma_{k,j}^2), \quad (4.2)$$

for $j = 1, \dots, d$. As can be seen from (4.2) model-based clustering as in (4.1) implies that all margins have $\mathcal{K}$ clusters per margin, except if $\mu_{k,j} = \mu_{k',j}$ and $\sigma_{k,j} = \sigma_{k',j}$ for some k' and k .

Since in practice it is challenging to learn from data if this (i.e., $\mu_{k,j} = \mu_{k',j}$ and $\sigma_{k,j} = \sigma_{k',j}$) holds exactly, we will refer to this challenge as the “single $\mathcal{K}$ problem”.

Pitfall 2: Curse of Dimensionality, with $O(d^2)$ Parameters as $d \rightarrow \infty$

The Gaussian mixture model in (4.1) has $(\mathcal{K} - 1) + \mathcal{K}d + \mathcal{K}d(d + 1)/2$ parameters, and hence the number of parameters increases quadratically with d . This shortcoming is well known to limit the scope of application of model-based clustering on high-dimensional data (Bouveyron & Brunet-Saumard, 2014). Some approaches have been developed with the aim of providing a more parsimonious specification, and hence as byproduct this chapter will also contribute to that literature. A key paper on parsimonious model-based clustering is that of McNicholas & Murphy (2008) who suggest a latent Gaussian model that can be regarded as a mixture of factor models.

Summary of Main Contributions

The main contributions of this chapter are as follows:

- We pioneered the development of a model-based solution for the “single $\mathcal{K}$ problem” outlined in (4.2), by specifying an individual finite mixture models for each of the margins, but making no assumptions on the joint distribution. The sample space is then partitioned via a strategy game-inspired algorithm, which can be used for clustering data, both marginally as well as in a multivariate fashion.
- We develop a computationally appealing and partially parallelisable model-based approach that bypasses the need to learn about $\mathcal{K}d(d + 1)/2$ parameters used in the covariance matrices $\Sigma_1, \dots, \Sigma_{\mathcal{K}}$ required for a ‘full’ (joint) Gaussian model-based clustering approach.
- The proposed data-driven approach for partitioning the sample space, automatically sieves regions that only have a residual amount of mass—via a minimum entry-level requirement that is specified by the user or set in a data-driven manner. In addition, we assess numerically the proposed methodologies and ascertain the reliability of their clustering performance in a battery of numerical experiments.
- As a byproduct, the proposed method contributes to the literature on game-inspired clustering approaches that followed from the seminal paper of Bulò & Pelillo (2009) (e.g., Hou et al., 2022). As will be shown below the proposed approach differs however significantly from that of the previous paper—both in terms of scope (the focus of Bulò & Pelillo is on hypergraph clustering) as well as on the specificities of the game-underlying the proposed clustering approach.

Structure and Organisation of this Chapter

The remainder of this chapter unfolds as follows. In Section 4.2 we introduce the probabilistic framework underlying the partition of the sample space which will be the building block of the proposed clustering approach to be introduced in Section 4.3. Section 4.4 outlines a conceptualization of a variant of the proposed partitioning approach by reinterpreting it as a strategy game. Experiments with artificial and real data, including an analysis of OB-type data described in Section 1.2, are conducted in Sections 4.5 and 4.6, respectively. Final observations closing remarks are given in Section 4.7.

4.2 Reign-and-Conquer Partitioning

4.2.1 The Probabilistic Framework

A key goal in this section is to devise a partition of the sample space of the joint distribution that is meaningful in a sense to be made more clear below. The proposed framework entails three steps, and to streamline the presentation we first focus on the bivariate setting. Comments on the multivariate extension are given in Section 4.2.2, and Section 4.4 outlines a game-theoretical variant of the proposed approach. This section does not yet consider data nor estimation, it rather focuses on a probabilistic setup for partitioning a sample space; comments on learning from data based on the principles below are given in Section 4.3. Here and below, no assumption whatsoever is made on the joint density, and we model each margin as a mixture model. Keeping in mind that any density can be approximated by a mixture of Normals, given enough components, the latter assumption is relatively mild. Furthermore, this assumption is supported by the following observation. Regardless of the joint density $f_{X,Y}(x,y)$ its expectation is $(E(X), E(Y)) = (\int x f_X(x) dx, \int y f_Y(y) dy)$. In other words, the joint expectation is invariant to the dependence structure. The same is true for the component centers (μ_X, μ_Y) . Even though modelling the margins separately does not provide us with the dependence structure of the joint it allows us to learn about the joint expectation and the component centers. This is what step 1 of the Reign-and-Conquer clustering is designed for so that step 2–3 can reconstruct the joint clusters.

Step 1: Margins

(Model-Based Clustering)

Let $X \sim f_X$ and $Y \sim f_Y$, where

$$f_X(x | K, \Theta) = \sum_{k=1}^K \pi_k p(x | \theta_k), \quad f_Y(y | L, \Psi) = \sum_{l=1}^L \omega_l q(y | \psi_l). \quad (4.3)$$

Here, p and q are density functions, with parameters $\Theta = (\theta_1, \dots, \theta_K)$ and $\Psi = (\psi_1, \dots, \psi_L)$; in addition K and L are the number of clusters respectively associated with the margins X and Y .

Step 2: Reign**(Similarity-Based Joint Protocluster Allocation)**

We first divide the sample space of (X, Y) , to be denoted by Ω , via a partition that is based on the set of all marginal cluster means

$$\begin{cases} \mu_X = \{\mu_X^{(1)}, \dots, \mu_X^{(K)}\}, \\ \mu_Y = \{\mu_Y^{(1)}, \dots, \mu_Y^{(L)}\}, \end{cases} \quad \text{where} \quad \begin{cases} \mu_X^{(i)} = E(X \mid \boldsymbol{\theta}_i) = \int x p(x \mid \boldsymbol{\theta}_i) dx, \\ \mu_Y^{(j)} = E(Y \mid \boldsymbol{\psi}_j) = \int y q(y \mid \boldsymbol{\psi}_j) dy. \end{cases} \quad (4.4)$$

Specifically, each point $(\mu_X^{(i)}, \mu_Y^{(j)})$ in the Cartesian product

$$\mu_X \times \mu_Y = \{(\mu_X^{(1)}, \mu_Y^{(1)}), \dots, (\mu_X^{(K)}, \mu_Y^{(L)})\}, \quad (4.5)$$

has a corresponding Voronoi cell $A_{i,j}$ for $i = 1, \dots, K$ and $j = 1, \dots, L$. We refer to the Voronoi cells $A_{1,1}, \dots, A_{K,L}$ as *protoclusters* as they define a first partition of Ω , and call the sites, $\mu_X \times \mu_Y$, as *protocluster centers*.

Step 3: Conquer**(Final Joint Cluster Allocation)**

After dividing Ω we conquer. That is, Step 3 identifies low density protoclusters to be conquered by high density regions, hence refining the naive partition of Ω from Step 2. To avoid including in the resulting partition regions that have a residual amount of mass, a minimum entry-level requirement is chosen to which we refer to as the sieve size $u \in [0, 1]$. Let

$$D_u \equiv \{(i, j) : P(A_{i,j}) \leq u\}, \quad (4.6)$$

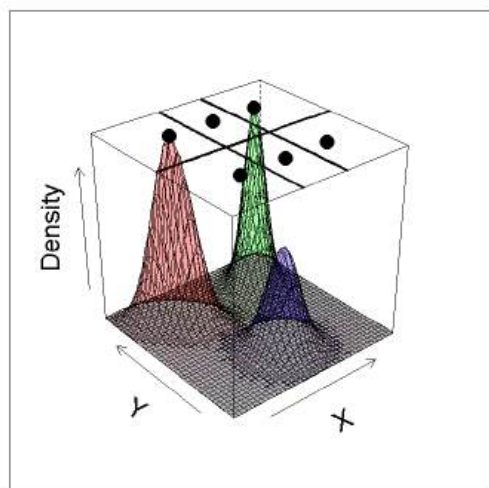
be the indices of the protoclusters that have low mass and that hence will be conquered for a given sieve size. The final sample space partition corresponds to the Voronoi cells $B_{i,j}$ associated with the protocluster centers of the conquerors, i.e., $(\mu_X^{(i)}, \mu_Y^{(j)})$ with $(i, j) \in D_u^c$. To assess how the number of final clusters depends on the sieve size, we define the *conquering function* as

$$C(u) = |D_u^c| = KL - |D_u|, \quad (4.7)$$

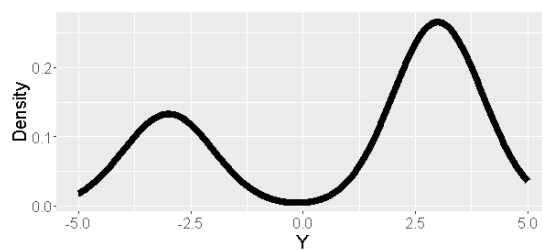
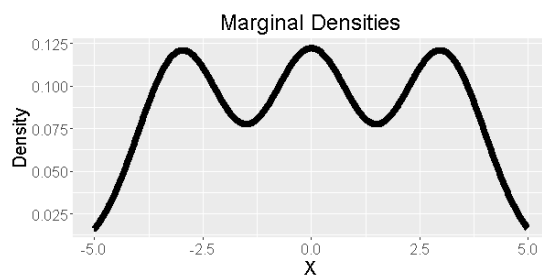
where $|A|$ denotes the cardinality and A^c is the complement of the set A .

Example 4.2.1 illustrates the main concepts and ideas of the sample space partitioning approach discussed above.

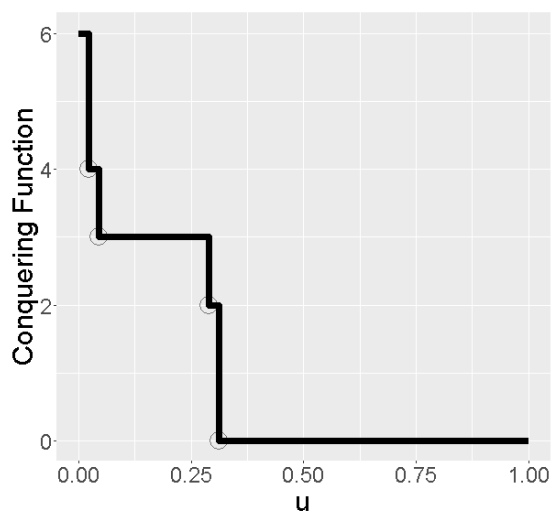
Example 4.2.1 (Reign-and-conquer partitioning on a mixture of three bivariate Normal distributions). In Figure 4.1 (a) we depict a mixture of three bivariate Normal distributions as in Equation (4.1), with means $\boldsymbol{\mu}_1 = (-3, 3)$, $\boldsymbol{\mu}_2 = (3, 3)$, $\boldsymbol{\mu}_3 = (0, -3)$, mixing probabilities



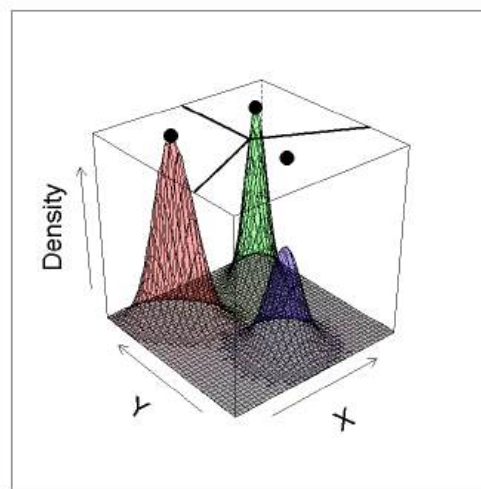
(a)



(b)



(c)



(d)

Figure 4.1: Reign-and-Conquer Partitioning for Example 4.2.1. (a) protocol clusters, protocol cluster centers ($\bullet$), and joint density. (b) Marginal densities. (c) Conquering function. (d) The Voronoi cells of the conquerors for $u = 0.1$.

$\pi_1 = \pi_2 = \pi_3 = 1/3$, and covariance matrices:

$$\Sigma_1 = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad \Sigma_2 = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}, \quad \Sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

This set-up leads to a different number of clusters per margin as can be seen in Figure 4.1 (a) and (b). Specifically, in the Y -margin there are two clusters with centers in $\mu_Y^{(1)} = 3$ and $\mu_Y^{(2)} = -3$ (i.e., $L = 2$), while in the X -margin there are 3 clusters with centers in $\mu_X^{(1)} = 3$, $\mu_X^{(2)} = 0$ and $\mu_X^{(3)} = -3$ (i.e., $K = 3$). In Figure 4.1 (c), we also display the conquering function, and in Figure 4.1 (d) we depict the Voronoi cells of the conquerors corresponding to the sieve size $u = 0.1$. If we were to regard protoclusters from Step 2 as ‘territories,’ what the Reign-and-Conquer partition does at Step 3 is to conquer low density cells and lets the dominant regions of mass conquer them.

4.2.2 d -Dimensional Extension and Theoretical Properties

The approach from Section 4.2.1 extends naturally to a D -dimensional context as follows. For the margins, we now consider $X_1 \sim f_1, \dots, X_d \sim f_d$ with

$$f_j(x \mid K_j, \Theta_j) = \sum_{k=1}^{K_j} \pi_{k,j} p_j(x \mid \theta_k), \quad (4.8)$$

where the notation in (4.8) extends that in (4.3), with $j = 1, \dots, d$. In particular, (4.8) implies that the first margin (X_1) has K_1 clusters, that the second margin has K_2 clusters, and so on. The partition underlying the reign step is now formed by the Voronoi tessellation $\{A_{\mathbf{i}} : \mathbf{i} \in I\}$, with $\mathbf{i} = (i_1, \dots, i_d)$, $I = \{1, \dots, K_1\} \times \dots \times \{1, \dots, K_d\}$, and where the $A_{\mathbf{i}}$ cell corresponds to the protocluster center $(\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)})$, with $\mu_{X_j}^{(k)} = E(X_j \mid \theta_k) = \int x p_j(x \mid \theta_k) dx$. The final clusters yield from the conquering step correspond to the Voronoi cell $B_{\mathbf{i}}$ associated with the protocluster centers of the conquerors, i.e., $(\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)})$ with $\mathbf{i} \in D_u^c$, where

$$D_u \equiv \{\mathbf{i} \in I : P(A_{\mathbf{i}}) \leq u\}, \quad (4.9)$$

for $u \in [0, 1]$. The conquering function is more generally defined in the d -dimensional setting as

$$C(u) = |D_u^c| = K_1 \times \dots \times K_d - |D_u|. \quad (4.10)$$

The conquering function is characterized by a set of properties summarized in the next theorem.

Theorem 4.2.2. *The conquering function, $C(u)$ as defined in (4.10), obeys the following properties:*

- a) *It is nonincreasing.*

b) It is continuous from the left.

c) It is bounded below by $C(1) = 0$ and above by $C(0) = \prod_{j=1}^d K_j$.

d) It integrates to one, i.e., $\int_0^1 C(u) du = 1$.

Proof. See Appendix B.1. □

It can be noticed that the conquering function from Example 4.2.1 verifies all the claims of Theorem 4.2.2 as can be seen from Figure 4.1 (c). In addition to Theorem 4.2.2 it can be shown that the conquering function is a step function that has a finite number of steps provided that $K_1, \dots, K_d$ are finite. See Appendix B.2.

4.3 Learning from Data

We now devise an algorithm based on the probabilistic framework from Section 4.2. The goal is to allocate observations in a dataset $\{\mathbf{x}_l\}_{l=1}^n$, with $\mathbf{x}_l = (x_{l,1}, \dots, x_{l,d})^\top$ in $\mathbb{R}^d$, into a set of meaningful classes—both in terms of the margins as well as the joint. Using the notation from Section 4.2.2, we introduce the R2C (Reign-and-Conquer Clustering) algorithm.

Algorithm 2 R2C (Reign-and-Conquer Clustering)

Step 1. Margins: Fit the j th marginal density in (4.8) using $\{x_{1,j}, \dots, x_{n,j}\}$, for $j = 1, \dots, d$, so to learn about $\{(K_j, \mu_{X_j}^{(1)}, \dots, \mu_{X_j}^{(K_j)})\}_{j=1}^d$.

Step 2. Reign: Learn about the protoclusters $\{A_i : i \in I\}$ of $\{(\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)}) : i \in I\}$.

Step 3. Conquer: Learn about the Voronoi cell of the conquerors, $\{B_i : i \in D_u^c\}$, that is that of $\{(\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)}) : i \in D_u^c\}$, and allocate the l -th observation to a cluster using the encoder

$$\text{Enc}(l) = \arg \min_{(i_1, \dots, i_d)} \|\mathbf{x}_l - (\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)})\|^2. \quad (4.11)$$

If we were to regard protoclusters from Step 2 as ‘territories,’ what R2C does at Step 3 is to let the mass dominant regions conquer the low density ones. The R2C algorithm warrants some further comments:

- **Step 1. Margins:** To learn about $\{(K_j, \mu_{X_j}^{(1)}, \dots, \mu_{X_j}^{(K_j)})\}_{j=1}^d$ several approaches can be taken. We use the NLP (non-local prior) for mixtures approach of Fúquene et al. (2019), but alternatively one could use, for example, RJ MCMC (Reversible Jump Markov Chain Monte Carlo) (Green, 1995). We opt for NPLs as they are designed to enforce parsimony by penalizing mixtures with a redundant number of components, and they bypass the need for complicated algorithms such as RJ MCMC. Determining K_j is a well-studied yet open problem, and an overview of the literature in this can be found in Richardson & Green (1997), Fraley & Raftery (2002), and Baudry et al. (2010).

- **Step 2. Reign:** To compute the protoclusters $\{A_i : i \in I\}$ of $\{(\mu_{X_1}^{(i_1)}, \dots, \mu_{X_d}^{(i_d)}) : i \in I\}$, we resort to the cluster centers from Step 1.
- **Step 3. Conquer:** To learn about the cells of the conquerors, we need to learn about $D_u = \{i \in I : P(A_i) \leq u\}$ —and this implies estimating $P(A_i)$. Several approaches can be taken, and here we opt for the simplest one—the maximum likelihood estimator (MLE). To avoid burdening the notation, we introduce the MLE on the bivariate case, but the details extend easily to the multivariate setting. Let $\{\mathbf{x}_l\}_{l=1}^n = \{(x_{l,1}, x_{l,2})\}_{l=1}^n$ and note that the number of points falling on the protoclusters, $n_{i,j} = |\{\mathbf{x}_l \in A_{i,j}\}_{l=1}^n|$ is Multinomial distributed, that is,

$$\mathbf{n} \sim \text{Multinomial}(\mathbf{p}), \quad (4.12)$$

where $\mathbf{n} = (n_{1,1}, \dots, n_{K,1}, \dots, n_{1,L}, \dots, n_{K,L})$ and $\mathbf{p} = (p_{1,1}, \dots, p_{K,1}, \dots, p_{1,L}, \dots, p_{K,L})$. Hence, the MLE is $\hat{\mathbf{p}} = \mathbf{n}/n$ and Bayesian inference can also be easily conducted.[†] Following the principles from Section 4.2, this estimate implies conquering protoclusters centered at (μ_k, μ_l) , for which

$$\frac{n_{k,l}}{n} < u, \quad \text{for } u \in (0, 1]. \quad (4.13)$$

The estimated regions of the conquerors $B_{k,l}$ are obtained by the Voronoi tessellation on the remaining (μ_k, μ_l) so that $(k, l) \notin D_u \equiv \{(i, j) : P(A_{i,j}) < u\}$.

To a certain extent, the R2C algorithm combines the paradigms of model-based clustering and similarity-based clustering. Indeed, Step 1 consists of a marginal model-based clustering approach. In addition, just as in similarity-based clustering methods, such as k -means, Step 3 entails an encoder (Hastie et al., 2009, Section 14.3), which determines to which cluster observation $\mathbf{x}_l$ belongs to. The Euclidean norm in (4.11) can be replaced by any preferred norm. For example using the Mahalanobis norm would take also the spread of the cluster into account, additionally to the distance from the cluster center.

Some comments on the implementation and computing are in order. In terms of implementation, as mentioned earlier, to avoid including in the resulting partition of the sample space regions that have a residual amount of mass, a minimum entry-level requirement $u \in [0, 1]$ should be set by the user. That value might be set at a fixed low level (say, $u = 0.1$), so that all resulting clusters have at least that mass. Alternatively, data-driven approaches for setting u , based on the fitted conquering function, are also explored in Section 4.5. In terms of computing, Step 1 can be parallelised into D cores, and so to speed up the computations parallel computing was

[†]Bayesian inference can be conducted by assuming a Dirichlet prior over the unit simplex on $\mathbb{R}^{KL}$, i.e., $\mathbf{p} \sim \text{Dirichlet}(\mathbf{a})$, where $\mathbf{a} = (a_{1,1}, \dots, a_{K,1}, \dots, a_{1,L}, \dots, a_{K,L})$. Dirichlet–Multinomial conjugacy then implies that posterior inferences can be obtained from $\mathbf{p} \mid \mathbf{n} \sim \text{Dirichlet}(\mathbf{a} + \mathbf{n})$. Finally, another alternative would be to specify a model for the joint distribution. Yet, given that we only need to learn about the $p_{i,j} = P(A_{i,j})$, and since we prefer to avoid specifying a copula that may not accurately describe the joint distribution, we opt for the above-described likelihood-based approaches.

implemented with the R package `parallel` (R Development Core Team, 2022). Steps 2 and 3 involve the computation of Voronoi tessellations from out of $a \equiv |\mu_{X_1} \times \cdots \times \mu_{X_D}| = K_1 \times \cdots \times K_D$ protocluster centers and from $a - |D_u|$ protocluster centers of the conquerors. While parallel algorithms could have been employed also for higher-performance computation of Steps 2 and 3 (e.g., Peterka et al., 2014) we have opted for the simple partially parallelisable approach in Algorithm 2.

We close this section with a simple yet important comment. While Step 3 of the R2C algorithm leads to multivariate clustering of $\{\mathbf{x}_l\}_{l=1}^n$, marginal clustering can be made directly from Step 1 via the posterior probabilities

$$\hat{Z}_{l,k,j} = \frac{\pi_{k,j} p_j(x_{l,j} | \boldsymbol{\theta}_k)}{\sum_{k=1}^{K_j} \pi_{k,j} p_j(x_{l,j} | \boldsymbol{\theta}_k)}. \quad (4.14)$$

Note that $\hat{Z}_{l,k,j} \in [0, 1]$ estimates the cluster membership labels of the l th observation on the j -th margin, which are defined as $Z_{l,k,j} = 1$ if the l -th observation on the j -th margin $x_{l,j}$ belongs to the k -th component, or $Z_{l,k,j} = 0$ otherwise.

4.4 An Outline of a Game Theory Conceptualization

4.4.1 A Game of Thrones—Starting Point

This section outlines an alternative way to look into the sample space partitioning approach from Section 4.2 as a game. To streamline the presentation we focus on the bivariate case; the extension to the multivariate case is a matter of adjusting notation. More specifically, the game to be considered starts at Step 2 of Reign-and-Conquer partitioning (Section 4.2), players are to be understood as KL ‘Kings’ owning the protocluster ‘territories’ ($\{A_{i,j}\}$) and who decide whether or not they will attack their neighbors. To make matters concrete, think of Figure 4.1 (a) as representing the protocluster ‘territories’ of $KL = 6$ Kings, who have to decide whether or not they attempt to conquer the territories of their neighbors. If a territory is attacked by two Kings, they might have to share the conquered territory.

The neighboring structure of players can be represented via a $KL \times KL$ adjacency matrix $\mathbf{M}$, and it can be visualized using a (undirected) graph $\mathcal{G} = (N, \mathcal{E})$, where $\mathcal{E}$ is a set of edges representing a neighboring relation. The outcome of the game is an element in S (to be defined in Section 4.4.2), and it can be visualized with a directed graph $\mathcal{G} = (N, \mathcal{E})$, where $\mathcal{E}$ is a set of directed edges or arrows representing attacks. To build intuition surrounding these ideas and concepts, let’s revisit Example 4.2.1. Figure 4.2 (a) depicts the graph corresponding to the

neighboring structure of the $KL = 6$ Kings, and the corresponding adjacency matrix is

$$\mathbf{M} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

The directed graph in Figure 4.2 (b) depicts an example of attack decisions compatible with the outcome from Figure 4.1 (d). Indeed, for example, we can think of the outcome in Figure 4.1 (d) as the consequence of players (1, 1) and (2, 2) attacking player (2, 1) and sharing the conquered territory, and so on.

4.4.2 Representation, Equilibrium, and Open Challenges

Below, an ‘attack’ is denoted with a ‘1’, and ‘not to attack’ with a ‘0’. The (normal form) game of interest is given by the triple $G = (N, \{S_i\}_{i \in N}, \{U_i\}_{i \in N})$, where:

- $N = \{(i, j) : i = 1, \dots, K, j = 1, \dots, L\}$ is the set of players (‘Kings’).
- $S_{i,j}$ is the pure set of strategies of King (i, j) ,

$$S_{i,j} = \{\text{who to attack, keeping in mind that only neighbors can be attacked}\},$$

$$S_{i,j} \subseteq \{0, 1\}^{KL},$$

and $S = \prod_{(i,j) \in N} S_{i,j}$ is the set of all vectors of strategies, where ‘ $\times$ ’ is the Cartesian product.

- $U_{i,j}(\mathbf{s})$ is the payoff of King (i, j) , with $\mathbf{s} = (\mathbf{s}_{i,j})_{(i,j) \in N}$, with $U_{i,j} : S \rightarrow \mathbb{R}$.

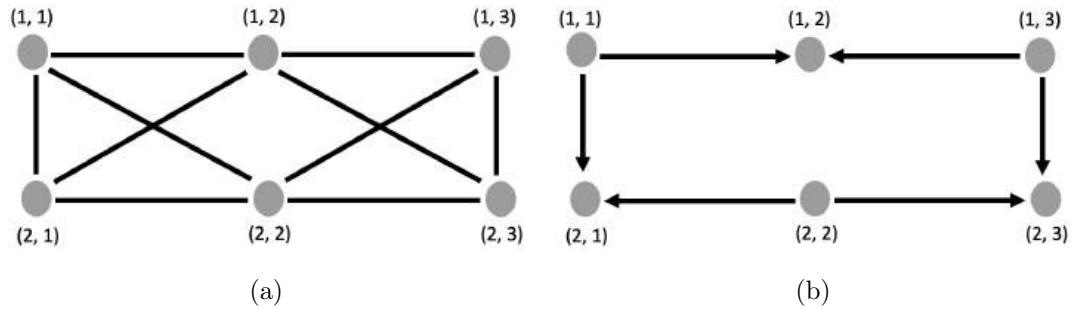


Figure 4.2: Revisiting Example 4.2.1. (a) Neighboring structure corresponding to Figure 4.1 (a). In (b) is depicted a directed graph with an instance of attack decisions compatible with the outcome from Figure 4.1 (d). In both charts the nodes represent players (‘Kings’) $(1, 1), \dots, (2, 3)$.

By construction, the strategy set of each player is finite and hence this is said to be a finite game. While a Nash equilibrium for this game may not exist over pure strategies, an equilibrium will exist over mixed strategies. A mixed strategy for player (i, j) is a distribution over their set of pure strategies $S_{i,j}$, that is

$$\mathcal{S}_{i,j} = \left\{ \sigma_{i,j} : S_{i,j} \rightarrow [0, 1] : \sum_{\mathbf{s}_{i,j} \in S_{i,j}} \sigma_{i,j}(\mathbf{s}_{i,j}) = 1 \right\}.$$

The celebrated Nash theorem, recalled below for completeness, ensures that the game of interest has at least one equilibrium in mixed strategies.

Theorem 4.4.1. *Every finite game in strategic form, that has a finite number of players, has a Nash equilibrium in mixed strategies.*

Proof. See, for example, [Maschler et al. \(2020, Section 5\)](#). □

Conceptually speaking, the approach above endows Step 3 with a much broader range of possibilities on how to partition the sample space Ω . First, there are numerous ways in which the ‘incentives’ (utility functions) can be set, and in particular they can mimic the ones from Section 4.2. A refinement of Step 3 based on the principles outlined above is as follows: a) Compute a Nash equilibrium; b) Derive the cells of the conquerors resulting from such equilibrium. In terms of a) we note that computation of Nash equilibria is non-trivial in general, but it can be conducted using simplicial subdivision ([van der Laan et al., 1987](#)), a Newton method known as Govindan–Wilson algorithm ([Govindan & Wilson, 2003](#)), search methods ([Porter et al., 2008](#)), among other.

Keeping in mind the computational motivation of the section, in the numerical experiments to be reported below we focus on the computationally appealing approach from Section 4.3—that bypass the need for computing Nash equilibria in Step 3—but we aim to revisit the numerical performance of this game-theoretical variant of the proposed methods in future research.

4.5 Numerical Experiments on Artificial Data

4.5.1 Simulation Setup and One Shot Experiments

In this section we study the performance of the proposed methods via numerical experiments. An exhaustive Monte Carlo simulation study will be presented in Section 4.5.2. The Monte Carlo simulation will assess two data-driven approaches for setting the sieve size based on the conquering function as well as the strategy of setting a fixed low sieve size (e.g., $u = 0.1$). The data-driven approaches based on the conquering function will

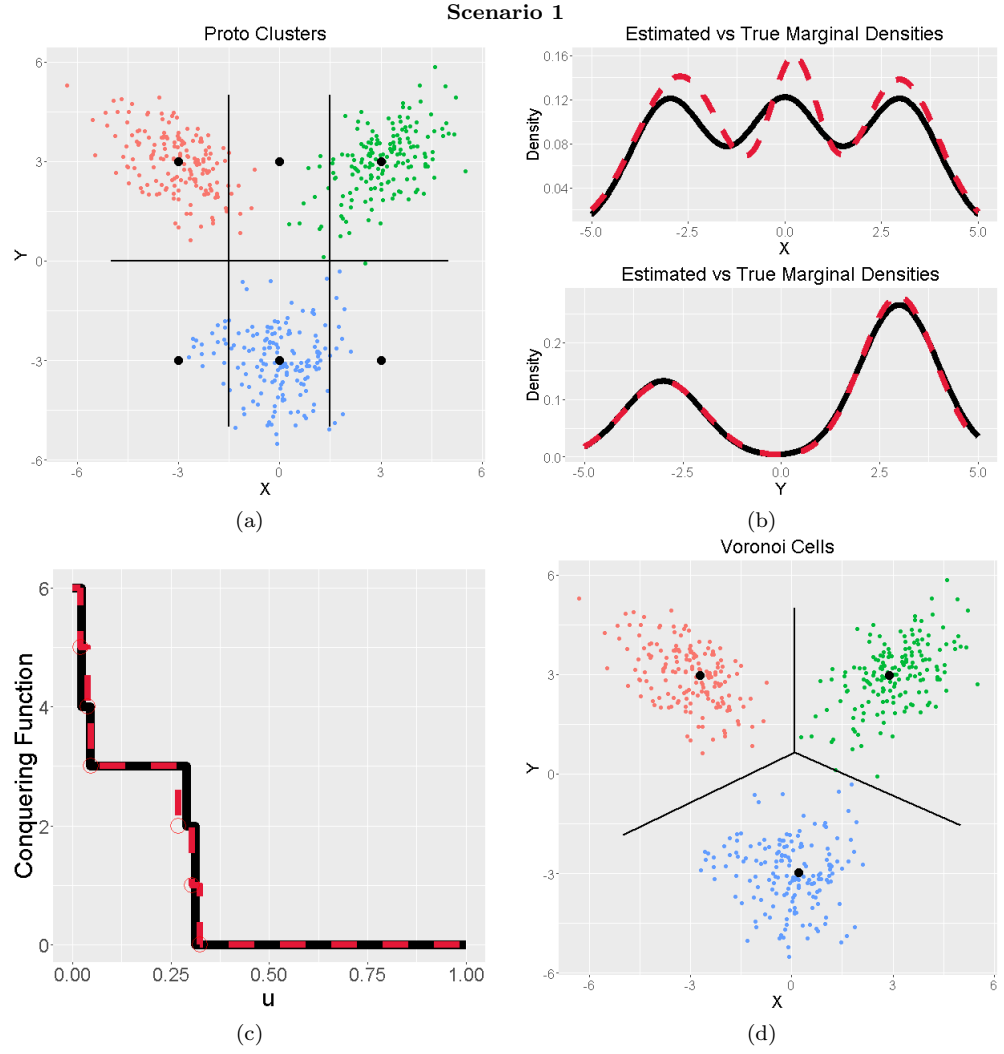


Figure 4.3: One shot experiments for Scenario 1. (a) Simulated data and protocusters (Step 1). (b) Estimated (dashed) vs. true (solid) marginal densities. (c) Estimated (dashed) vs true (solid) conquering functions. (d) Voronoi cells of the conquerors for $u = 0.1$ (Steps 2 and 3).

be called throughout as the *plateau* (u at which the longest plateau of $C(u)$ ends) and the *edge* (u at which the largest jump on $C(u)$ occurs); see Appendix B.2 for technical details.

Pitfalls 1–2 from Section 4.1 motivated us to design the following simulation scenarios:

- **Scenario 1:** Data are drawn from a mixture of $\mathcal{K} = 3$ bivariate Normal distributions with weights, mean vectors, and covariance matrices as in Example 4.2.1. To study the clustering performance as the sample size increases, we consider $n \in \{50, 100, 250, 500, 1000\}$. In Figure 4.3 we depict a one shot example of Reign-and-Conquer algorithm corresponding to a sample of size $n = 500$ and $u = 0.1$; as can be seen in Figure 4.3 (d), the proposed method suitably partitions the multivariate data.
- **Scenario 2:** Data are drawn from a mixture of $\mathcal{K} = 3$ Clayton copulas (Nelsen, 2007, Chapter 4.2) with Normal margins: $f_X(x) = \phi(x; -5, 4^2)/2 + \phi(x; 3, 4^2)/2$, and $f_Y(y) = \phi(x; -5, 1)/3 + \phi(x; 2.5, 1)/3 + \phi(x; 5, 1)/3$. Trivially, the joint distribution does not obey (4.1), and the number of clusters per margin is different ($K = 2$, $L = 3$). Here, we also consider sample sizes $n \in \{50, 100, 250, 500, 1000\}$, and in Figure 4.4 we depict the outcome of a one shot experiment with $n = 500$. As can be seen in Figure 4.4 (d), the proposed method suitably partitions the multivariate data.
- **Scenario 3:** Data are drawn from a mixture of d -variate Normal distributions in dimensions $d \in \{5, 10, 15, 20\}$ (moderate high dimensional data). In this scenario, the sample sizes and the number of clusters depend on d in the following way: $n_d \equiv n = \lfloor 10d^{3/2} \rfloor$ and $\mathcal{K}_d \equiv \mathcal{K} = \text{round}(\sqrt{d+1})$, where $\lfloor \cdot \rfloor$ and $\text{round}(\cdot)$ denotes the the floor and round functions respectively. The covariance matrices and mixing probabilities are specified as $\Sigma_k = \mathbf{I}_d$ and $\pi_k = 1/d$, for $k = 1, \dots, \mathcal{K}$, whereas the mean vectors $\mu_k = (\mu_1^{(k)}, \dots, \mu_d^{(k)})$ are sparsely defined: $\mu_i^{(k)} = 0$ for $i \neq k$ and $\mu_k^{(k)} = d/\sqrt{2}$. This scenario leads to several identical marginal distributions, and the mean vector components are constrained to be equidistant $\|\mu_i - \mu_j\|_2 = d$ for all $j \neq i$; therefore the separation between clusters increases linearly with the number of dimensions.

4.5.2 Monte Carlo Simulation Study

To assess the performance of the proposed clustering approach, for Scenarios 1–3 we redo the previous one shot analysis $M = 1000$ times so to estimate the following clustering agreement metrics: Rand and Adjusted Rand Index (RI and ARI respectively), Jaccard Index (JI), and Fowlkes–Mallows Index (FMI); see Pfitzner et al. (2009) and references

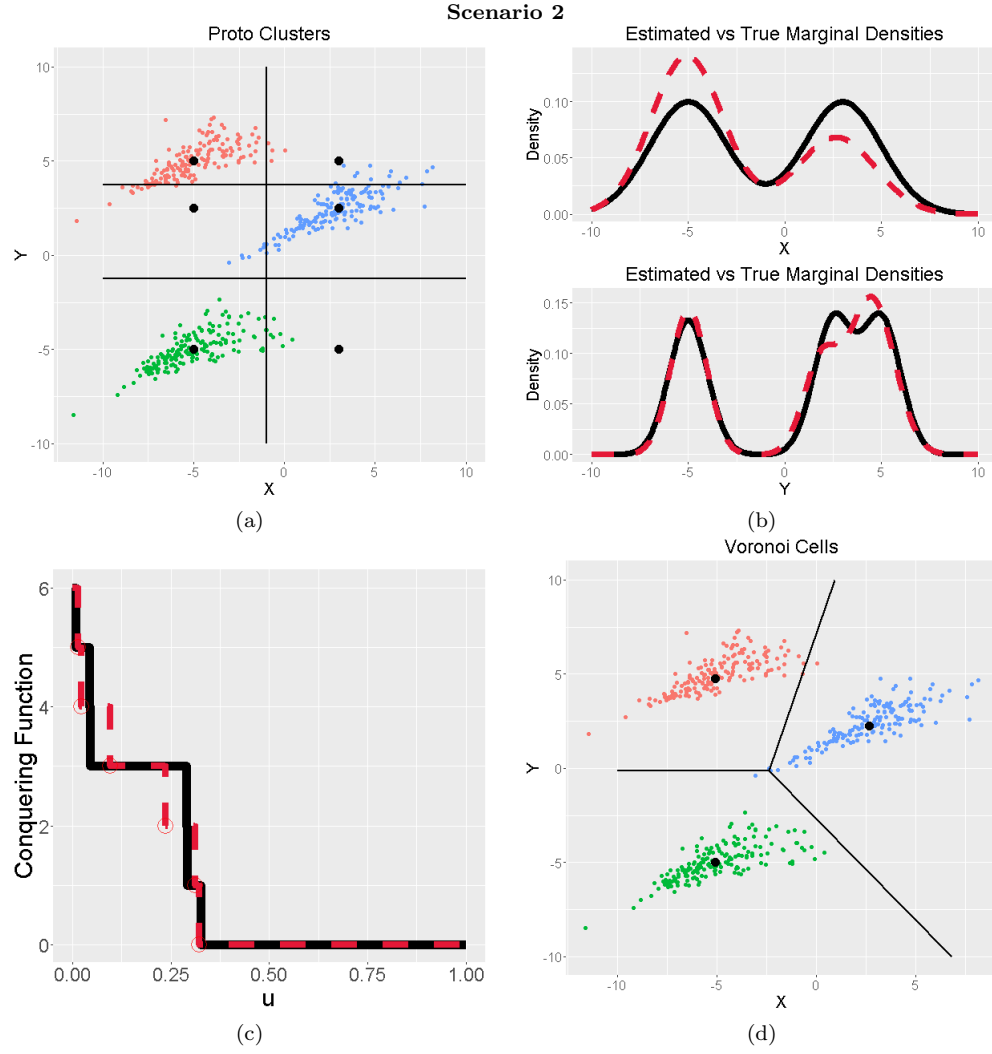


Figure 4.4: One shot experiments for Scenario 2. (a) Simulated data and protocluster (Step 1). (b) Estimated (dashed) vs true (solid) marginal densities. (c) Estimated (dashed) vs true (solid) conquering functions. (d) Voronoi cells of the conquerors for $u = 0.1$ (Steps 2 and 3).

therein. We also report the empirical distribution of the number of clusters detected by the proposed method, over different conquering strategies (i.e., fixed u , plateau, and edge), along with the same outputs for plain vanilla GMM (Gaussian Mixture Model)-based clustering. For GMM, model selection was conducted using the BIC, and the model was fitted using the `mclust` Fraley et al. (2012) package in R.

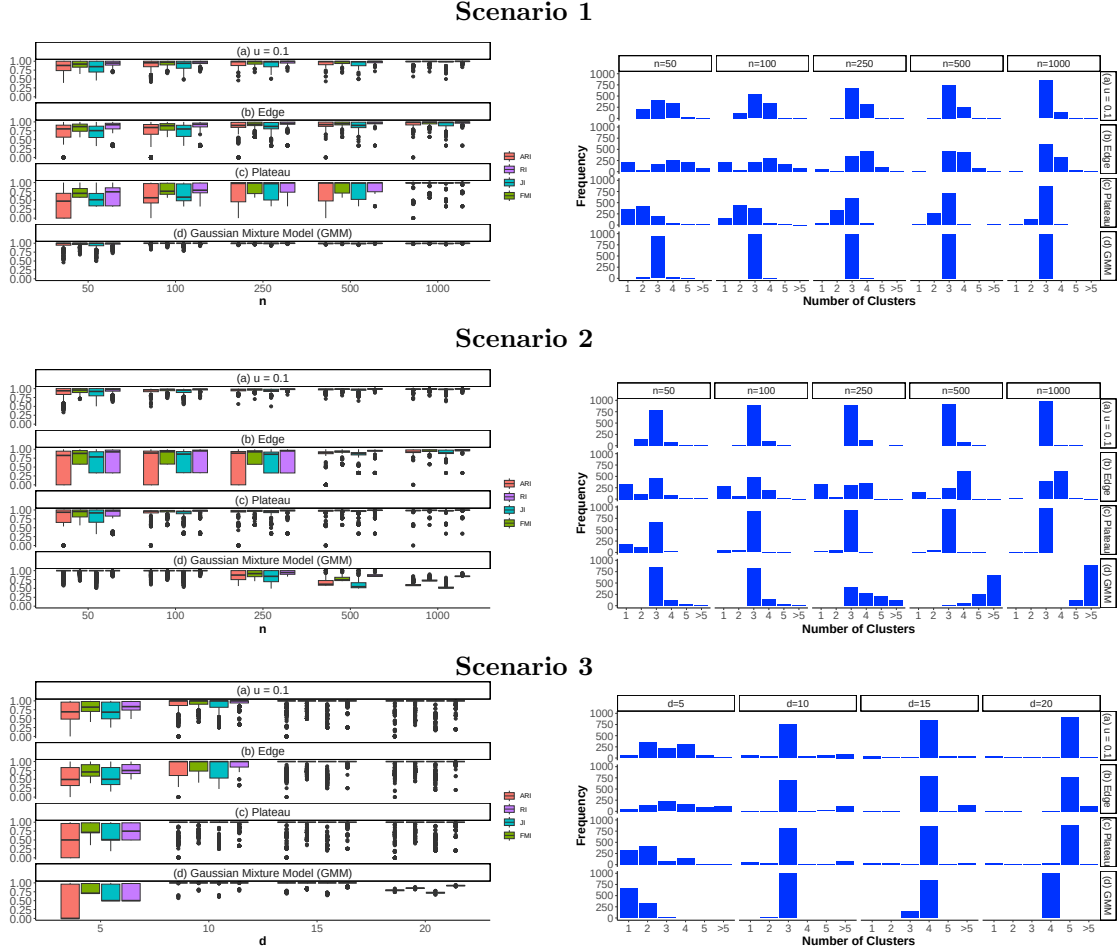


Figure 4.5: Monte Carlo simulation study: (Left) Performance metrics (ARI, RI, JI, FMI) (Right) Empirical distribution on the number of detected clusters.

In Figure 4.5 (top and middle), it can be seen that for Scenarios 1–2, as the sample size increases the performance metrics increase on average for all conquering strategies. Interestingly, for Scenario 1, the edge conquering strategy works better than the plateau for relatively small sample sizes. Conversely, in Scenario 2, the plateau yields on average better results than the edge for small sample sizes. In Scenarios 1–2, fixing a sieve size of $u = 0.1$, produces accurate clustering results on average even for small sample sizes. In addition, as the sample sizes increases the proposed method identifies most frequently the correct number of clusters $\mathcal{K} = 3$ for Scenarios 1–2—both when $u = 0.1$ as well as when u is set using the plateau.

In Figure 4.5 (bottom) we present the performance of the proposed method for Scenario 3. As can be seen in Figure 4.5 (bottom-left), as the dimension increases on average the proposed method presents better agreement metrics (recall that in Scenario 3 cluster separation grows linearly with data dimension). In Figure 4.5 (bottom-right), it can be seen that as the sample size increases, the proposed method most frequently captures the true number of clusters $\mathcal{K}_d \in \{2, 3, 4, 5\}$ for dimensions $d \in \{5, 10, 15, 20\}$ respectively.

Some final comments on the comparison of the Reign-and-Conquer clustering against GMM are in order. In Scenario 1 the data are from are simulated from a low-dimensional Gaussian mixture model, and hence perhaps not surprisingly GMM overperforms the proposed approach. Still, the performance of the proposed approach is still remarkable especially as we make no assumption on the joint. In addition, Reign-and-Conquer has a comparable, if not superior, performance than GMM over Scenarios 2–3.

4.6 Real Data Illustrations

Banknote Data

The first dataset to be analysed with the proposed methods contains $p = 6$ measurements made on 100 genuine and 100 counterfeit old-Swiss 1000-franc bank notes (Flury, 1988, pp. 5–8). The data consists of the following measurements (in millimetres): length of bill, width of left edge, width of right edge, bottom margin width, top margin width, and the width of diagonal. The data are available from the `mclust` R package (Fraley et al., 2012). In Figure 4.6 (a) we depict a biplot to represent the first two principal components of the data, along with the corresponding clustering yield by the proposed Reign-and-Conquer clustering. The sieve size was set using the Plateau conquering strategy that corresponds to the best average results on the simulations setting in Scenario 3. In Figure 4.6 (b) we depict the fitted conquering function, from where it can be seen that the plateau consists of $u = 0.36$. As can be noticed from the confusion matrix in Table 4.1, Reign-and-Conquer does an excellent job classifying counterfeit as well as genuine data. The GMM analysis combined with BIC suggests that there could be three clusters. The latter analysis is not least interesting from a forensic viewpoint, as it suggests that there might be two clusters of counterfeit banknotes. Finally, we note that the number of marginal clusters obtained using Reign-and-Conquer (i.e., $K_1, \dots, K_6$) ranges from 1–3 clusters.

Italian Wine Data

The second dataset on which the proposed approach will be illustrated contains the results of a chemical analysis of wines grown in the same region of Italy derived from three

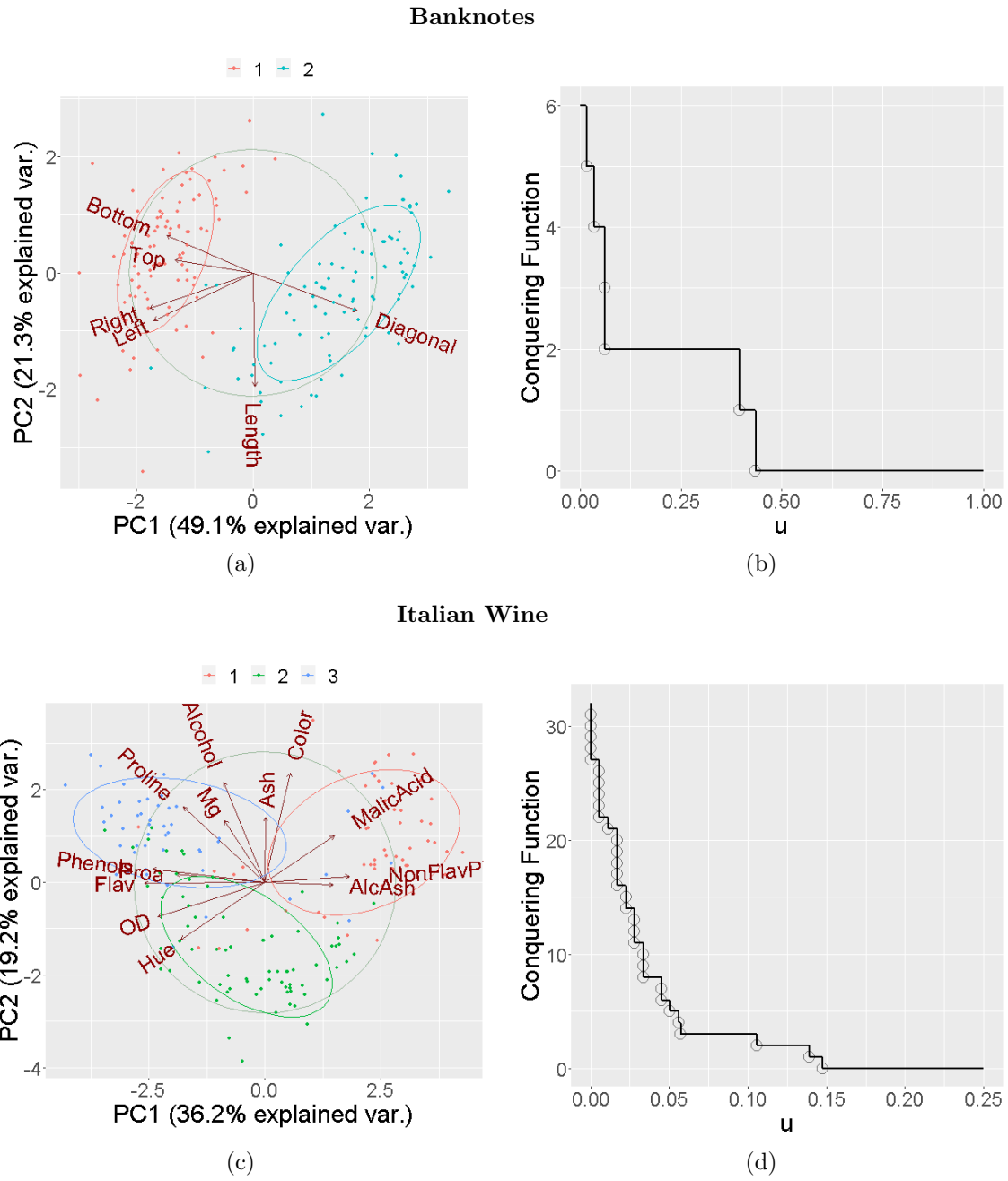


Figure 4.6: (a) Biplot banknote dataset, the color key corresponds to the cluster labels obtained for $u = 0.36$ (Plateau). (b) Conquering function estimate. (c) Biplot of wine dataset, the color key corresponds to the cluster labels obtained for $u = 0.105$ (Plateau). (d) Conquering function estimate (restricted to $[0, 0.25]$ for visualization purposes).

Method	Cluster	Counterfeit	Genuine	Method	Cluster	Barbera	Grignolino	Barolo
RC	1	99	0	RC	1	42	9	5
	2	1	100		2	1	56	16
GMM	1	16	2		3	5	7	37
	2	0	98	GMM	1	0	27	59
	3	84	0		2	48	44	0

Table 4.1: Confusion matrix for banknote (left) and wine (right) data. NOTE: RC = Reign-and-Conquer, GMM = Gaussian Mixture Model.

cultivars (Barbera, Grignolino, and Barolo); the data are available from [Vandeginste et al. \(1990\)](#). The chemical analysis includes the measurement of $d = 13$ continuous variables (such as Alcohol, Malic acid, Ash, Flavanoids, etc) on $n = 178$ instances. Similarly to the banknote data illustration, in Figure 4.6 (c) we depict a biplot to represent the first two principal components of the data, along with the corresponding clustering yield by the proposed method. The sieve size was set using the Plateau conquering strategy that corresponds to the best average results on the simulations setting in Scenario 3. In Figure 4.6 (d) we depict the fitted conquering function, the plateau consists of $u = 0.105$. As can be seen from the confusion matrix in Table 4.1, Reign-and-Conquer learns about the ‘right’ number of cultivars and the obtained clusters have a resemblance with the cultivars. The GMM analysis combined with BIC would offer another interesting outlook, suggesting that two of these clusters are so similar that they should perhaps be merged. Finally, we note that the number of marginal clusters (i.e., $K_1, \dots, K_{13}$) is 1 for 8 of the dimensions, and 2 for the remainder dimensions.

Application to Open Banking Data

In this section we illustrate the application of Reign-and-Conquer clustering to the OB-type data provided by MD. The data is a variant of the data used in Section 3.4. It contains 9,636 individuals for whom we have information about age, income, total expenses, asset contributions, and bank repayments in the year 2017. We used the total amounts over the year for each category to create a OB-type market segmentation. Keeping in mind the variables have different ranges we scaled the margins to perform the Voronoi tessellation using an Euclidean Distance. Table 4.2 depicts the number of marginal clusters detected by the Step 1 of the algorithm.

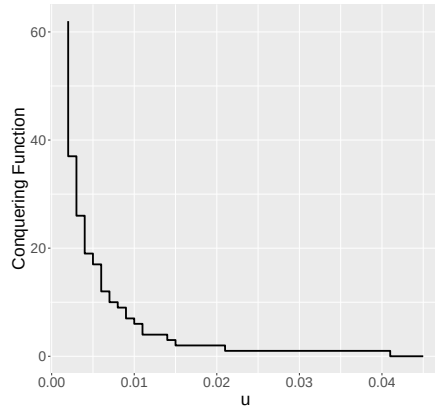
Variable	Age	Income	Expenses	Asset Contributions	Bank Repayments
Nr. of components	6	7	7	20	17

Table 4.2: Marginal modelling: Number of clusters on each margin.

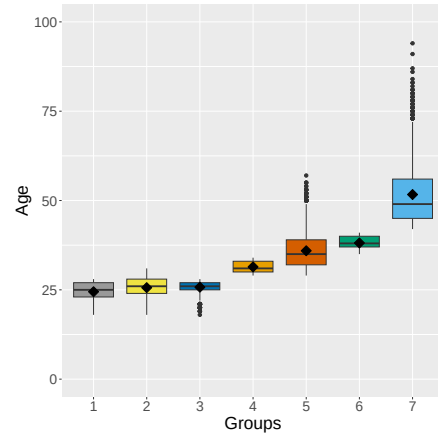
We observe that the number of components on each margin differs. Applying Step 2 of the algorithm results in $K_1 \times \dots \times K_5 = 99,960$ protoclusters. These combine the properties of groups on each respective margin. Therefore, the protoclusters form a convenient and easily interpretable grid for identifying detailed characteristics of individuals associated with a specific (non-empty) protocluster.

The conquer step provides the final customer segments. We chose a sieve size of $u = 0.01$, i.e. we discard protoclusters with less than 1% of all observations. Figure 4.7 (a) shows the corresponding conquering function $C(u)$. This step eliminates 99.953 protoclusters and results in 7 joint clusters, with the following number of individuals in each specified group, ordered as in Figure 4.7 from left to right: 473, 494, 2,071, 768, 436, 1,147 and 4,247. Figure 4.7 summarises the properties of each of these 7 groups. Since the first three groups have similar age distributions we focus on them to give an example of how to interpret the obtained results. In Figure 4.7 (c) we see that although the three groups are similar in age, the third group has a much higher annual income. Higher income likely leads to this group investing more in assets, as can be seen in Figure 4.7 (d), and to higher total annual expenditure, which can be seen in Figure 4.7 (f). The other two groups with similar ages are similar in their financial behaviour. However, there is a major difference in debt repayment. The first group has an average annual debt repayment of only £72, while the average annual debt repayment of the second group average yearly debt repayment is at £4,549 noticeably higher. We see that although the latter two groups are roughly the same, the proposed method is able to capture the differences that may be important in the development or provision of financial services.

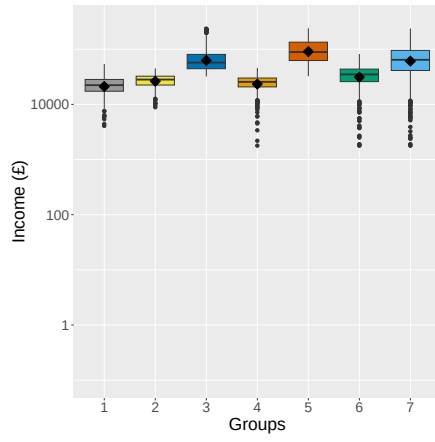
The example presented in this section shows that Reign-and-Conquer clustering provides an useful and interpretable clustering solution both at the protocluster level and at the joint. With the information provided by our method, institutions can precisely identify which market segment a specific customer belongs to. The protoclusters provide a detailed, grid-like analysis of the market, and the joint clusters presenting a complete picture of the market. This information can help in the development and provision of products and/or services for well-defined segments in the protocluster grid, as well as for the main market segments of the market given by the joint clusters.



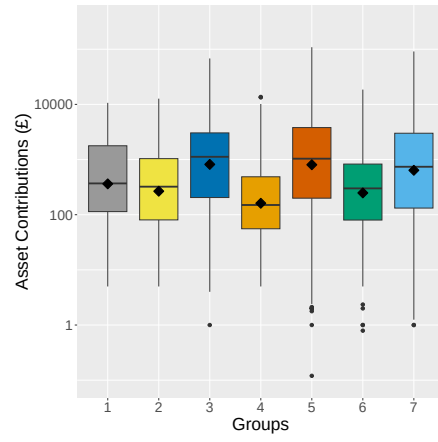
(a)



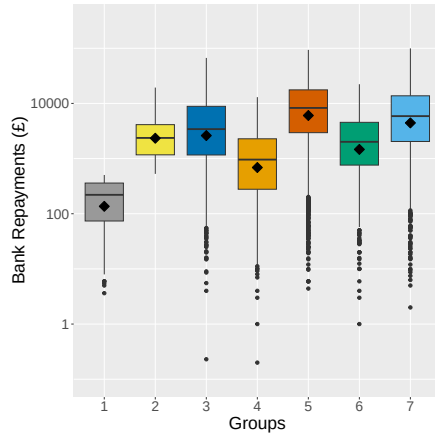
(b)



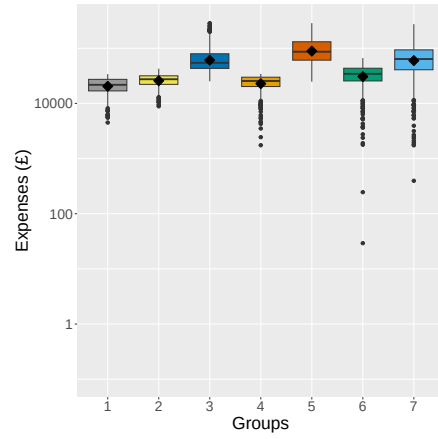
(c)



(d)



(e)



(f)

Figure 4.7: (a) Conquering function estimate (restricted to $[0, 0.045]$), (b) age distributions of identified clusters, (c) income distributions of identified clusters, (d) asset contribution distributions of identified clusters, (e) bank repayment distributions of identified clusters, (f) expense distributions of identified clusters. Subfigures (b), (c), (d), and (e) are represented on a log scale. In (a), (b), (c), (d), (e), and (f) each group is represented by a colour, the $\blacklozenge$ represents distribution means within each group.

4.7 Final Observations and Concluding Remarks

This section devises an unsupervised learning approach based on a marginal model-based specification followed-up by a strategy-game inspired algorithm that partitions the sample space. The approach was motivated from Pitfalls 1–2 from Section 4.1, and it can be used for clustering data—both in a multivariate manner as well as marginally. Pitfall 1—the single $\mathcal{K}$ problem—implies that fitting a plain vanilla multivariate Gaussian model might result in all margins and the joint having the same number of components, unless one is able to penalise for deviations from $\mu_{k,j} = \mu_{k',j}$ and $\sigma_{k,j} = \sigma_{k',j}$. Pitfall 2—the curse of dimensionality—is well known, and it implies that learning about a plain vanilla Gaussian mixture model implies learning about $O(d^2)$ parameters, when $d \rightarrow \infty$. Prompted by these concerns, the herein proposed clustering approach only specifies a model for the margins but leaves the joint unspecified, it has the advantage of being parallelisable, and bypasses the need to learn about $\mathcal{K}d(d+1)/2$ parameters used in the covariance matrices $\Sigma_1, \dots, \Sigma_{\mathcal{K}}$ required for a ‘full’ (joint) Gaussian model-based clustering approach. The conducted numerical experiments suggest that the proposed approach has a comparable performance, and even in some cases superior, than a plain vanilla Gaussian model-based approach.

The application to the OB-type data sample illustrates the uses and benefits of the here proposed method for creating a market segmentation that reflects the complex reality captured by OB-type data.

Alternative models for clustering OB-type data in an accurate and interpretable way, such as various copulas [Di Lascio et al. \(2011\)](#); [Nelsen \(2007\)](#) were considered. These can be used to reflect the complexity of OB-type data. However, they do not fit the goal of Chapter 4, aimed at addressing Pitfalls 1–2 from Section 4.1, while at the same time providing an interpretable unsupervised clustering method for OB-type data.

The obtained numerical evidence indicates a satisfactory performance of the proposed Reign-and-Conquer clustering method under a variety of situations, there is still room for improvement, open problems to be addressed as well as opportunities for future research. First, the geometry of the boundaries of the final kingdoms (i.e. the Voronoi Cells of the conquerors) could perhaps be bended so to better adapt to the structure of the data, to offer more flexibility to the resulting partitions, and ultimately to improve clustering results. Second, the game-theoretical variant from Section 4.4 opens a world of opportunities on ways to set the ‘incentives’ to conquer, via an utility function, to explored in a follow-up paper. In terms of the “single $\mathcal{K}$ problem”, an alternative to the path taken here would be to develop Bayesian regularization approaches that aim to penalise for deviations from $\mu_{k,j} = \mu_{k',j}$ and $\sigma_{k,j} = \sigma_{k',j}$. Computationally, the method is limited by the number of elements of the Cartesian product of component centers. These are necessary for Step 2

of the algorithm. High dimensional examples, such as the prostate cancer dataset ([Singh et al., 2002](#)) create a Cartesian product matrix, with approximately 2^{6033} elements. This is an computational limitation that can be handled by dimension reduction methods such as principal component analysis ([Hastie et al., 2009](#), Section 14.5). Recent advances regarding this topic can be found in [Johnson & Rossell \(2010\)](#). While useful for settings with a large number of variables dimension reduction methods make results harder to interpret. Finally, while here the focus has been on unsupervised learning, the potential of related strategy-game inspired approaches for supervised learning would seem natural.

Chapter 5

Closing Remarks

This section summarises, discusses and concludes the research presented in this thesis. It also addresses its limitations and points out possibilities for future research.

5.1 Summary

The research presented in this thesis developed new and relevant statistical learning methods tailored for a OB and the newly available OB-type data associated with it. Two main areas of methodological and applicational gaps were identified and addressed.

First, we highlight the lack and need for a statistical methodology to perform a subject-to-group comparison, i.e. the comparison of the characteristics of a specific reference subject to those of a group of similar subjects. We addressed this gap by proposing the time-varying comparative mean value, which pioneers statistical learning methods for performing subject-to-group comparison.

The second identified area we contribute to is the lack of suitable clustering solutions that reflect the detailed structure of OB-type data while being computationally feasible. We proposed the Reign-and-Conquer clustering algorithm, which, unlike joint mixture models, allows the number of clusters on each margin to differ. This better reflects the structure of OB-type data, as one can find different populations depending on the measures one focuses on. At the same time, modelling the margins separately mitigates the curse of dimensionality of joint mixture models, making the Reign-and-Conquer clustering method computationally attractive.

The presented statistical learning methodology contributes to the development of literature on statistical methods suitable for OB-type data analysis.

Contribution 1: Comparative Mean Value for Subject-to-Group Comparison and Truncated Covariate Regression

With the introduction of the time-varying comparative mean value we pioneered statistical methodology for subject-to-group comparison necessary for comparing a single reference subject with a group of similar subjects. Although we develop the concept for use in a OB context, the proposed framework is general and not limited to OB-type data applications. The time-varying comparative mean value can be used in any context where the aim is to perform a subject-to-group comparison. Examples of possible applications are specifying differences in patients with similar characteristics for a more personalised treatment in a medical setting, the analysis of differences between a specific individual and a group of similar individuals in various socio-economic settings, and the comparison of the characteristics of a company with those of a group of similar entities.

The wide range of possible applications is underlined by the fact that the time-varying comparative mean value is the first model for truncated covariate regression on an interval centred around a reference covariate value. This expands its potential use regression on relevant covariate values and ranges.

Contribution 2: Reign-and-Conquer Clustering as a Solution for the Single- $\mathcal{K}$ and Mitigation of the Curse of Dimensionality

This thesis introduces the Reign-and-Conquer algorithm, which is an unsupervised clustering method on the interface of model-based and similarity-based clustering approaches. Fitting joint mixture models likely result in all margins having the same number of component, that is the number of components on the joint distribution. We identify and describe this problem, which we refer to as the single- $\mathcal{K}$ problem. By specifying a model for each of the margins we allow the number of clusters on each margin to vary, and thus propose a solution to the single- $\mathcal{K}$ problem. Addressing the pitfall of mixture models is of great importance for creating meaningful market segmentation based on OB-type data, as it allows for different numbers of populations for each measure found in the data.

The separate modelling of the margins also makes the algorithm partially parallelisable. Unlike joint mixture models, the Reign-and-Conquer clustering algorithm does not suffer from the curse of dimensionality, which makes it computationally attractive.

By overcoming the single- $\mathcal{K}$ problem and the curse of dimensionality, the Reign-and-Conquer clustering algorithm contributes to the development of novel statistical learning solutions suitable for clustering OB-type data.

The last two steps of the Reign-and-Conquer algorithm, utilise similarity-based concepts to eliminate clusters with mass below a certain entry-level threshold u . This leads to a par-

tition of the sample space dominated by clusters with large mass. By modelling margins using mixture models and using similarity-based approaches to determine protoclusters and the joint clusters, Reign-and-Conquer clustering positions itself at the interface of model-based and similarity-based clustering methods.

The proposed Reign-and-Conquer clustering algorithm can be conceptualised in a game-theoretic context and thus also contributes to the literature on game-inspired clustering. Reign-and-Conquer clustering provides a rich framework on which to build and advance the fields of model-based clustering, similarity-based clustering, game theory-inspired clustering and customer segmentation using OB-type data.

5.2 Open Questions and Future Research

Contribution 1: Quantile Version of the Time-varying Comparative Mean Value and Truncated Covariate Quantile Regression

This thesis introduces the statistical learning method of the time-varying comparative mean value, which, as discussed, pioneers subject-to-group comparison and is the first model for performing truncated covariate regression. The conditional expected value of the time-varying comparative mean value can be replaced by a conditional quantile. This would introduce subject-to-group comparison based on quantiles to statistical methodology, as well as time-varying truncated covariate quantile regression.

Smoothing the time-varying comparative mean value provides a way to track its dynamics while at the same time obtaining a predictor. To this end, we take advantage of the Nadaraya-Watson kernel-weighted average smoother. Defining the smooth time-varying comparative mean value using higher degree local polynomial smoothers as well as other smoothing methods, such as splines and wavelets, could lead to performance improvements on noisy data, data with structural breaks or jumps and data concentrated at variable boundaries. Redefining the smooth time-varying comparative mean value using different smoothing methods and comparing their performance is an area for future research.

The time-varying comparative mean value is introduced in an univariate setting. Thanks to the general nature of the framework, extensions in multivariate settings are feasible and relatively straightforward. Such extensions would introduce the multivariate time-varying comparative mean value, the multivariate truncated covariate regression, the multivariate time-varying comparative quantile and the multivariate truncated quantile regression.

Contribution 2: Identification of Number of Joint Clusters via Conquering Function and Extensions of the Game-theory Framework of Reign-and-Conquer Clustering

The Reign-and-Conquer clustering method addresses the single- $\mathcal{K}$ problem and the curse of dimensionality of joint mixture models. Using a minimum entry-level threshold u , Step 3 of the algorithm sieves out low-density protoclusters to obtain final joint clusters. The conquering function shows the relationship between the minimum entry-level threshold and the final number of clusters on joint distribution. The proposed criteria for selecting the final number of clusters - edge and plateau - are initial approaches that highlight the largest change in the number of clusters given a marginal change of the entry-level threshold u and the longest ranges of u where the number of joint clusters does not change, respectively.

New criteria can be proposed and thus new ways of determining the number of joint clusters in the context of the Reign-and-Conquer clustering framework can be constructed. The development and study of these criteria is an area for future research. For example, an integration criterion can be constructed by considering the area under each step of the conquering function and choosing the number of joint clusters corresponding to the largest step area value. Such a criterion would lie midway between the edge and the plateau criteria, as it emphasises neither the greatest change in the number of clusters nor the ranges of longest stability, but rather finds a compromise between the two.

The similarity-based steps of the Reign-and-Conquer algorithm depend on the distance measures used in the Voronoi tessellation. All presented partitions obtained via Reign-and-Conquer clustering are determined using the Euclidean norm. However, one can also choose a different distance measure, which can lead to different geometries and thus to different clustering results. A simple example would be to replace the Euclidean distance with a Mahalanobis distance, which might be more suitable for data with large variance. The question of which distance measures are best suited for use in the Reign-and-Conquer clustering algorithm in different contexts still needs to be explored.

The conceptualisation of the Reign-and-Conquer algorithm within the framework of game theory opens up a large pool of possibilities for adapting the algorithm's outcome. This can be done, for example, by redefining the payoff function. Different payoff functions can lead to different clustering solutions that are suitable for different purposes.

One of the consequences of using model-based clustering for Step 1 of the Reign-and-Conquer clustering algorithm is that the method shares some of the beneficial properties of mixture models. In particular, it is possible to specify the uncertainty of the clustering solution. A simple way to do this is to take the MCMC output and perform the analysis of Steps 2 – 3 for each MCMC draw. In this way we can calculate the probability of

an observation belonging to a particular cluster. This aspect of the Reign-and-Conquer clustering algorithm has not been fully explored and is a possible area for future research.

Appendix A

Supplementary Information for Chapter 3

A.1 Properties and Proofs

A.1.1 Expected Residual Lifetime

Let $F_t(a) = P(Y_t \leq a)$ and $\bar{F}_t(a) = P(Y_t > a)$, and denote the right endpoint as $L_t = \sup\{x : F_t(x) < 1\}$. The expected residual lifetime is given by

$$m(a) = E(Y_t - a \mid Y_t > a) = \frac{\int_0^{L_t} \bar{F}_t(u) du}{\bar{F}_t(a)}.$$

Analogously

$$n(b) = E(Y_t \mid Y_t \leq a) = \frac{\int_0^a F_t(u) du}{F_t(a)} = \frac{\int_0^a u f_t(u) du}{F_t(a)}.$$

From the above we can derive the expression we are interested in. Denoting $b = \min\{y_{0,t} + \delta, L\}$ and $a = \max\{y_{0,t} - \delta, 0\}$ we can write:

$$r(a, b) = E(Y_t \mid a \leq Y_t < b) = \frac{\int_a^b F_t(u) du}{F_t(b) - F_t(a)} = \frac{\int_a^b u f_t(u) du}{F_t(b) - F_t(a)}.$$

Assuming $(y_{0,t} - \delta) \geq 0$ and $(y_{0,t} + \delta) \leq L$ and combining these results in this section we get

$$\begin{aligned} \mu_t(\delta) &= E(Z_t \mid Y_t \in W_{\delta,t}) = \alpha + \beta E(Y_t \mid Y_t \in W_{\delta,t}) \\ &= \alpha + \beta \left[\frac{\int_{y_{0,t}-\delta}^{y_{0,t}+\delta} F_t(u) du}{F_t(y_{0,t} + \delta) - F_t(y_{0,t} - \delta)} \right]. \end{aligned}$$

A.2 Parametric Examples of the Comparative Mean Value

Normal Distribution

Let $F(x) = \Phi(x)$ and $f(x) = \phi(x)$. Using the results from [Johnson et al. \(2005, p. 134\)](#), we can write the time-varying comparative mean value as

$$\mu_{\delta,t} = \alpha_t + \beta_t \left\{ \theta_t + \sigma \frac{\phi\left(\frac{x_{0,t} - \delta - \theta_t}{\sigma}\right) - \phi\left(\frac{x_{0,t} + \delta - \theta_t}{\sigma}\right)}{\Phi\left(\frac{x_{0,t} - \delta - \theta_t}{\sigma}\right) - \Phi\left(\frac{x_{0,t} + \delta - \theta_t}{\sigma}\right)} \right\}$$

for $\delta \neq 0$.

Uniform Distribution

Let $F(x) = (x - a)/(b - a)$ and $f(x) = 1/(b - a)$ be respectively the distribution function and density of a uniform (a, b) distribution. Then:

$$\begin{aligned} \mu_{\delta,t} &= \alpha_t + \beta_t \frac{1}{\frac{x_{0,t} + \delta - a}{b - a} - \frac{x_{0,t} - \delta - a}{b - a}} \int_{x_{0,t} - \delta}^{x_{0,t} + \delta} x \frac{1}{b - a} dx \\ &= \alpha_t + \beta_t \frac{1}{\frac{2\delta}{b - a}} \left[\frac{x^2}{2(b - a)} \right]_{x_{0,t} - \delta}^{x_{0,t} + \delta} \\ &= \alpha_t + \beta_t \frac{b - a}{2\delta} \left\{ \frac{(x_{0,t} + \delta)^2 - (x_{0,t} - \delta)^2}{2(b - a)} \right\} \\ &= \alpha_t + \beta_t \frac{(x_{0,t}^2 + 2x_{0,t}\delta + \delta^2) - (x_{0,t}^2 - 2x_{0,t}\delta + \delta^2)}{4\delta} \\ &= \alpha_t + \beta_t \frac{4x_{0,t}\delta}{4\delta} = \alpha_t + \beta_t x_{0,t} \end{aligned}$$

for $a < b$ and $\delta > 0$.

Exponential Distribution

Let $F(x) = 1 - \exp(-\lambda, x)$ and $f(x) = \lambda \exp(-\lambda, x)$ be respectively the distribution function and density of an exponential distribution. Then:

$$\begin{aligned}
\mu_{\delta,t} &= \alpha_t + \beta_t \left[\frac{1}{1 - \exp\{-\lambda(x_{0,t} + \delta)\} - [1 - \exp\{-\lambda(x_{0,t} - \delta)\}]} \right. \\
&\quad \left. \int_{x_{0,t}-\delta}^{x_{0,t}+\delta} x \lambda \exp(-\lambda x) dx \right] \\
&= \alpha_t + \beta_t \left\{ \frac{1}{-\exp(-\lambda x_{0,t}) \exp(-\lambda \delta) + \exp(-\lambda x_{0,t}) \exp(\lambda \delta)} \right. \\
&\quad \left. \left[-\frac{\exp(-\lambda x)(\lambda x + 1)}{\lambda} \right]_{x_{0,t}-\delta}^{x_{0,t}+\delta} \right\} \\
&= \alpha_t + \beta_t \left\{ \frac{\exp(-\lambda x_{0,t}) \exp(\lambda \delta)(\lambda x_{0,t} - \lambda \delta + 1)}{\lambda \exp(-\lambda x_{0,t})(\exp(\lambda \delta) - \exp(-\lambda \delta))} \right. \\
&\quad \left. - \frac{\exp(-\lambda x_{0,t}) \exp(-\lambda \delta)(\lambda x_{0,t} + \lambda \delta + 1)}{\lambda \exp(-\lambda x_{0,t})(\exp(\lambda \delta) - \exp(-\lambda \delta))} \right\} \\
&= \alpha_t + \beta_t \frac{\exp(\lambda \delta)(x_{0,t} - \delta + 1/\lambda) - \exp(-\lambda \delta)(x_{0,t} + \delta + 1/\lambda)}{\exp(\lambda \delta) - \exp(-\lambda \delta)} \\
&= \alpha_t + \beta_t \frac{\frac{\exp(2\lambda \delta)(x_{0,t} - \delta + 1/\lambda) - (x_{0,t} + \delta + 1/\lambda)}{\exp(\lambda \delta)}}{\frac{\exp(2\lambda \delta) - 1}{\exp(\lambda \delta)}} \\
&= \alpha_t + \beta_t \frac{\exp(2\lambda \delta)(x_{0,t} - \delta + 1/\lambda) - (x_{0,t} + \delta + 1/\lambda)}{\exp(2\lambda \delta) - 1}
\end{aligned}$$

for $\delta \neq 0$.

Power Distribution

Let $f(x) = \{c/(b-a)\}\{(x-a)/(b-a)\}^{(c-1)}$ be the density of a power distribution on $[a, b]$. The following formula was directly adapted from [Ruiz & Navarro \(1996, p. 570\)](#)

$$\begin{aligned}
\mu_{\delta,t} &= \alpha_t + \frac{\beta_t}{c+1} \left[\frac{\{c(x_{0,t} + \delta) + a\}(x_{0,t} + \delta - a)^{c_t}}{(x_{0,t} + \delta - a)^{c_t} - (x_{0,t} - \delta - a)^{c_t}} \right. \\
&\quad \left. - \frac{\{c(x_{0,t} - \delta) + a\}(x_{0,t} - \delta - a)^{c_t}}{(x_{0,t} + \delta - a)^{c_t} - (x_{0,t} - \delta - a)^{c_t}} \right]
\end{aligned}$$

for $a \leq x_{0,t} - \delta < x_{0,t} + \delta \leq b$ and $c > 0$.

Cauchy Distribution

Let $F(x) = (1/\pi) \arctan\{(x - \theta)/\sigma\} + 1/2$ and $f(x) = (1/\pi\sigma) \sigma^2 / \{(x - \theta)^2 + \sigma^2\}$ be respectively the distribution function and density of a Cauchy distribution.

$$\begin{aligned}
\mu_{\delta,t} &= \alpha_t + \beta_t \left[\frac{1}{\frac{1}{\pi} \arctan(\frac{x_{0,t}+\delta-\theta}{\sigma}) + \frac{1}{2} - \{\frac{1}{\pi} \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma}) + \frac{1}{2}\}} \right. \\
&\quad \left. \int_{x_{0,t}-\delta}^{x_{0,t}+\delta} x \frac{1}{\pi \sigma} \frac{\sigma^2}{(x-\theta)^2 + \sigma^2} dx \right] \\
&= \alpha_t + \beta_t \left\{ \frac{\pi}{\arctan(\frac{x_{0,t}+\delta-\theta}{\sigma}) - \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma})} \right. \\
&\quad \left. \left[\frac{\sigma \log\{\sigma^2 + (x-\theta)^2\} + 2\theta \arctan(\frac{x-\theta}{\sigma})}{2\pi} \right]_{x_{0,t}-\delta}^{x_{0,t}+\delta} \right\} \\
&= \alpha_t + \frac{\beta_t}{2} \left[\frac{\sigma [\log\{\sigma^2 + (x_{0,t} + \delta - \theta)^2\} - \log\{\sigma^2 + (x_{0,t} - \delta - \theta)^2\}]}{\arctan(\frac{x_{0,t}+\delta-\theta}{\sigma}) - \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma})} \right. \\
&\quad \left. + \frac{2\theta (\arctan \frac{x_{0,t}+\delta-\theta}{\sigma} - \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma}))}{\arctan(\frac{x_{0,t}+\delta-\theta}{\sigma}) - \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma})} \right] \\
&= \alpha_t + \beta_t \theta + \frac{\beta_t \sigma \log(\sigma^2 + (x_{0,t} + \delta - \theta)^2) - \log(\sigma^2 + (x_{0,t} - \delta - \theta)^2)}{2 \arctan(\frac{x_{0,t}+\delta-\theta}{\sigma}) - \arctan(\frac{x_{0,t}-\delta-\theta}{\sigma})}
\end{aligned}$$

for $\delta \neq 0$.

Pareto Distribution

Let

$$F(x) = \begin{cases} \frac{\theta \sigma^\theta}{x^{\theta+1}} & \text{if } x \geq \sigma, \\ 0 & \text{if } x < \sigma, \end{cases} \quad f(x) = \begin{cases} 1 - (\frac{\sigma}{x})^\theta & \text{if } x \geq \sigma, \\ 0, & \text{if } x < \sigma, \end{cases}$$

be respectively the distribution function and the density of a Pareto distribution.

Case 1: $\theta \neq 1$ and $x_{0,t} - \delta \geq \sigma$

$$\begin{aligned}
\delta_{\delta,t} &= \alpha_t + \beta_t \frac{1}{1 - (\frac{\sigma}{x_{0,t}+\delta})^\theta - \{1 - (\frac{\sigma}{x_{0,t}-\delta})^\theta\}} \int_{x_{0,t}-\delta}^{x_{0,t}+\delta} x \frac{\theta \sigma^\theta}{x^{\theta+1}} dx \\
&= \alpha_t + \beta_t \frac{1}{\frac{\sigma^\theta \{(x_{0,t}+\delta)^\theta - (x_{0,t}-\delta)^\theta\}}{(x_{0,t}+\delta)^\theta (x_{0,t}-\delta)^\theta}} \left[\frac{\theta \sigma^\theta x^{1-\theta}}{1-\theta} \right]_{x_{0,t}-\delta}^{x_{0,t}+\delta} \\
&= \alpha_t + \beta_t \frac{\theta}{1-\theta} \frac{(x_{0,t}+\delta)^\theta (x_{0,t}-\delta)^\theta}{(x_{0,t}+\delta)^\theta - (x_{0,t}-\delta)^\theta} \{ (x_{0,t}+\delta)^{1-\theta} - (x_{0,t}-\delta)^{1-\theta} \} \\
&= \alpha_t + \beta_t \frac{\theta}{1-\theta} \left\{ \frac{(x_{0,t}+\delta)^\theta (x_{0,t}-\delta)^\theta (x_{0,t}+\delta)^{1-\theta}}{(x_{0,t}+\delta)^\theta - (x_{0,t}-\delta)^\theta} \right. \\
&\quad \left. - \frac{(x_{0,t}+\delta)^\theta (x_{0,t}-\delta)^\theta (x_{0,t}-\delta)^{1-\theta}}{(x_{0,t}+\delta)^\theta - (x_{0,t}-\delta)^\theta} \right\} \\
&= \alpha_t + \beta_t \frac{\theta}{1-\theta} \frac{(x_{0,t}-\delta)^\theta (x_{0,t}+\delta) - (x_{0,t}+\delta)^\theta (x_{0,t}-\delta)}{(x_{0,t}+\delta)^\theta - (x_{0,t}-\delta)^\theta}
\end{aligned}$$

for $\delta \neq 0$.

Case 2: $\theta = 1$ and $x_{0,t} - \delta \geq \sigma$

$$\begin{aligned}
\delta_{\delta,t} &= \alpha_t + \beta_t \frac{1}{1 - \frac{\sigma}{x_{0,t} + \delta} - (1 - \frac{\sigma}{x_{0,t} - \delta})} \int_{x_{0,t} - \delta}^{x_{0,t} + \delta} x \frac{\sigma}{x^2} dx \\
&= \alpha_t + \beta_t \frac{1}{\frac{\sigma\{(x_{0,t} + \delta) - (x_{0,t} - \delta)\}}{(x_{0,t} + \delta)(x_{0,t} - \delta)}} \left[\sigma \log(x) \right]_{x_{0,t} - \delta}^{x_{0,t} + \delta} \\
&= \alpha_t + \beta_t \frac{(x_{0,t} + \delta)(x_{0,t} - \delta) \{\log(x_{0,t} + \delta) - \log(x_{0,t} - \delta)\}}{2\delta} \\
&= \alpha_t + \beta_t \frac{(x_{0,t}^2 - \delta^2) \log(\frac{x_{0,t} + \delta}{x_{0,t} - \delta})}{2\delta}
\end{aligned}$$

for $\delta \neq 0$ and $x_{0,t} \pm \delta \neq 0$.

Laplace Distribution

Let

$$F(x) = \begin{cases} \frac{1}{2} \exp(\frac{x - \theta_t}{\sigma_t}) & \text{if } x \leq \theta_t, \\ 1 - \frac{1}{2} \exp(-\frac{x - \theta_t}{\sigma_t}) & \text{if } x \geq \theta_t, \end{cases}, \quad f(x) = \frac{1}{2\sigma} \begin{cases} \exp(-\frac{\theta - x}{\sigma}) & \text{if } x < \theta, \\ \exp(-\frac{x - \theta}{\sigma}) & \text{if } x \geq \theta, \end{cases}$$

be respectively the distribution function and the density of a Laplace distribution.

Case 1: $x_{0,t} + \delta < \theta$

$$\begin{aligned}
\mu_{\delta,t} &= \alpha_t + \beta_t \frac{1}{\frac{1}{2} \exp(\frac{x_{0,t} + \delta - \theta}{\sigma}) - \frac{1}{2} \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})} \int_{x_{0,t} - \delta}^{x_{0,t} + \delta} x \frac{1}{2\sigma} \exp(-\frac{\theta - x}{\sigma}) dx \\
&= \alpha_t + \beta_t \frac{2}{\exp(\frac{x_{0,t} - \theta}{\sigma}) \{\exp(\frac{\delta}{\sigma}) - \exp(\frac{-\delta}{\sigma})\}} \left[\frac{(x - \sigma) \exp(\frac{x - \theta}{\sigma})}{2} \right]_{x_{0,t} - \delta}^{x_{0,t} + \delta} \\
&= \alpha_t + \beta_t \frac{(x_{0,t} + \delta - \sigma) \exp(\frac{x_{0,t} + \delta - \theta}{\sigma}) - (x_{0,t} - \delta - \sigma) \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})}{\exp(\frac{x_{0,t} - \theta}{\sigma}) \{\exp(\frac{\delta}{\sigma}) - \exp(\frac{-\delta}{\sigma})\}} \\
&= \alpha_t + \beta_t \frac{\exp(\frac{\delta}{\sigma})(x_{0,t} + \delta - \sigma) - \exp(\frac{-\delta}{\sigma})(x_{0,t} - \delta - \sigma)}{\exp(\frac{\delta}{\sigma}) - \exp(\frac{-\delta}{\sigma})} \\
&= \alpha_t + \beta_t \frac{\exp(\frac{2\delta}{\sigma})(x_{0,t} + \delta - \sigma) - x_{0,t} + \delta + \sigma}{\exp(\frac{2\delta}{\sigma}) - 1} \text{ for } \delta \neq 0.
\end{aligned}$$

Case 2: $x_{0,t} - \delta < \theta \leq x_{0,t} + \delta$

$$\begin{aligned}
\delta, t &= \alpha_t + \beta_t \frac{1}{1 - \frac{1}{2} \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \frac{1}{2} \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})} \\
&\quad \left\{ \int_{x_{0,t} - \delta}^{\theta} x \frac{1}{2\sigma} \exp(-\frac{\theta - x}{\sigma}) dx + \int_{\theta}^{x_{0,t} + \delta} x \frac{1}{2\sigma} \exp(-\frac{x - \theta}{\sigma}) dx \right\} \\
&= \alpha_t + \beta_t \frac{2}{2 - \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})} \\
&\quad \left\{ \left[\frac{(x - \sigma) \exp(\frac{x - \theta}{\sigma})}{2} \right]_{x_{0,t} - \delta}^{\theta} + \left[-\frac{(x + \sigma) \exp(\frac{\theta - x}{\sigma})}{2} \right]_{\theta}^{x_{0,t} + \delta} \right\} \\
&= \alpha_t + \beta_t \left\{ \frac{(\theta - \sigma) - (x_{0,t} - \delta - \sigma) \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})}{2 - \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})} \right. \\
&\quad \left. - \frac{(x_{0,t} + \delta + \sigma) \exp(\frac{\theta - x_{0,t} - \delta}{\sigma}) - (\theta + \sigma)}{2 - \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})} \right\} \\
&= \alpha_t + \beta_t \frac{2\theta - (x_{0,t} - \delta - \sigma) \exp(\frac{x_{0,t} - \delta - \theta}{\sigma}) - (x_{0,t} + \delta + \sigma) \exp(\frac{\theta - x_{0,t} - \delta}{\sigma})}{2 - \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \exp(\frac{x_{0,t} - \delta - \theta}{\sigma})}
\end{aligned}$$

for $\delta \neq 0$.

Case 3: $\theta \leq x_{0,t} - \delta$

$$\begin{aligned}
\delta, t &= \alpha_t + \beta_t \left\{ \frac{1}{1 - \frac{1}{2} \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \{1 - \frac{1}{2} \exp(-\frac{x_{0,t} - \delta - \theta}{\sigma})\}} \right. \\
&\quad \left. \int_{x_{0,t} - \delta}^{x_{0,t} + \delta} x \frac{1}{2\sigma} \exp(-\frac{x - \theta}{\sigma}) dx \right\} \\
&= \alpha_t + \beta_t \frac{2}{\exp(-\frac{x_{0,t} + \delta - \theta}{\sigma}) - \exp(-\frac{x_{0,t} + \delta - \theta}{\sigma})} \left[-\frac{(\sigma + x) \exp(\frac{x - \theta}{\sigma})}{2} \right]_{x_{0,t} - \delta}^{x_{0,t} + \delta} \\
&= \alpha_t + \beta_t \frac{(x_{0,t} + \delta - \sigma) \exp(\frac{\theta - x_{0,t} + \delta}{\sigma}) - (x_{0,t} + \delta + \sigma) \exp(\frac{\theta - x_{0,t} - \delta}{\sigma})}{\exp(\frac{\theta - x_{0,t} + \delta}{\sigma}) - \exp(\frac{\theta - x_{0,t} - \delta}{\sigma})} \\
&= \alpha_t + \beta_t \frac{(x_{0,t} + \delta - \sigma) \exp(\frac{\delta}{\sigma}) - (x_{0,t} + \delta + \sigma) \exp(-\frac{\delta}{\sigma})}{\exp(\frac{\delta}{\sigma}) - \exp(-\frac{\delta}{\sigma})} \\
&= \alpha_t + \beta_t \frac{\exp(\frac{2\delta}{\sigma})(\sigma + x_{0,t} - \delta) - \sigma - x_{0,t} - \delta}{\exp(\frac{2\delta}{\sigma}) - 1}
\end{aligned}$$

for $\delta \neq 0$.

A.3 Further Monte Carlo Evidence

In this section we assess the finite sample performance of the time-varying comparative sample mean and the smooth time-varying comparative sample mean in two scenarios where the response is subject to a jump or discontinuity. The data generating processes

are described in Table A.3.1. The simulated data are essentially modifications of the data used in scenarios from Section 3.3.1. This section introduces a jump at time point $t = 70$. All fits presented in this section follow the same of attack as in Section 3.3 (e.g. choice of bandwidth, number of Monte Carlo runs). As it can be seen from Figs A.1–A.2, despite the presence of the jump, the proposed approach still fits the true targets quite closely. Despite the good performance of the proposed methods under such discontinuity, as mentioned in the section, if jumps are to be expected then versions of the proposed time-varying comparative value that resort to models for curves with jumps (e.g. Gijbels et al., 2004; Kang, 2020) or to wavelets (Abramovich et al., 2000)—rather than local polynomial methods—may be more natural.

Table A.3.1: Data generating processes used over the simulation study with jumps; we use the shape-scale parametrisation of the Gamma distribution.

Data	Data generating process	
	$t \leq 70$	$t > 70$
Jump scenario A		
Covariate ($X_{i,t}$)	$\text{Gamma}(\exp(t/100), \exp(t/100))$	$\text{Gamma}(\exp(t/100), \exp(t/100))$
Response ($Y_{i,t}$)	$\text{Gamma}(\{\sin(t/10) + t/10\}^2/t, 2)$	$\text{Gamma}(\{\sin(t/10) + t/10\}^2 + t/t, 2)$
Jump scenario B		
Covariate ($X_{i,t}$)	$N(\sin(t/10) + 1 , 5)$	$N(\sin(t/10) + 1 , 5)$
Response ($Y_{i,t}$)	$5 + 2 X_{i,t}$	$12 + 2 X_{i,t}$

A.4 Non-statistical Comparison: Expense to Income Ratio

In this section we present the expense to income ratios for MD users for whom the time-varying comparative mean expenses were discussed in Section 3.4. The calculated ratios can be found in Figure A.3. While financial ratios are commonly used by academics, regulatory institutions and financial institutions (e.g. see Bae et al., 2005; Greninger et al., 1996; Melzer, 2017; OECD, 2020)—for a review see Harness et al. (2008)—financial ratios were constructed for other purposes than for subject-to-group comparison and are not based on any statistical methodology.

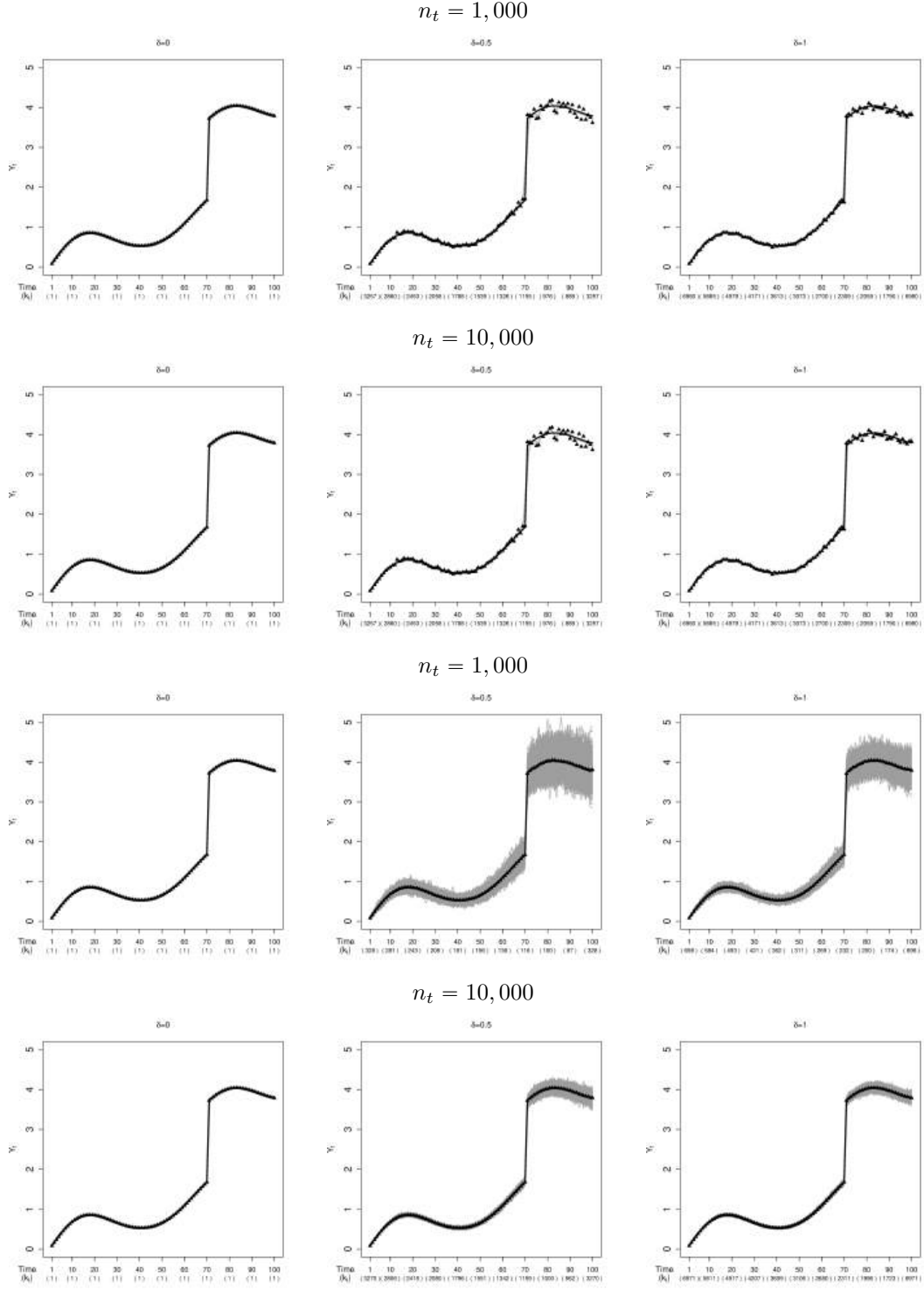


Figure A.1: Jump Scenario A: Top two rows show the single run experiment, and the bottom two rows the Monte Carlo simulation. The time-varying comparative sample mean 3.7 is represented by the triangles ($\blacktriangle$). The grey curves represent the customer smooth time-varying comparative sample means 3.9, and the black curve is the true comparative mean value 3.6. The x -axis denotes time, and the number of subjects in the window of comparison, k_t .

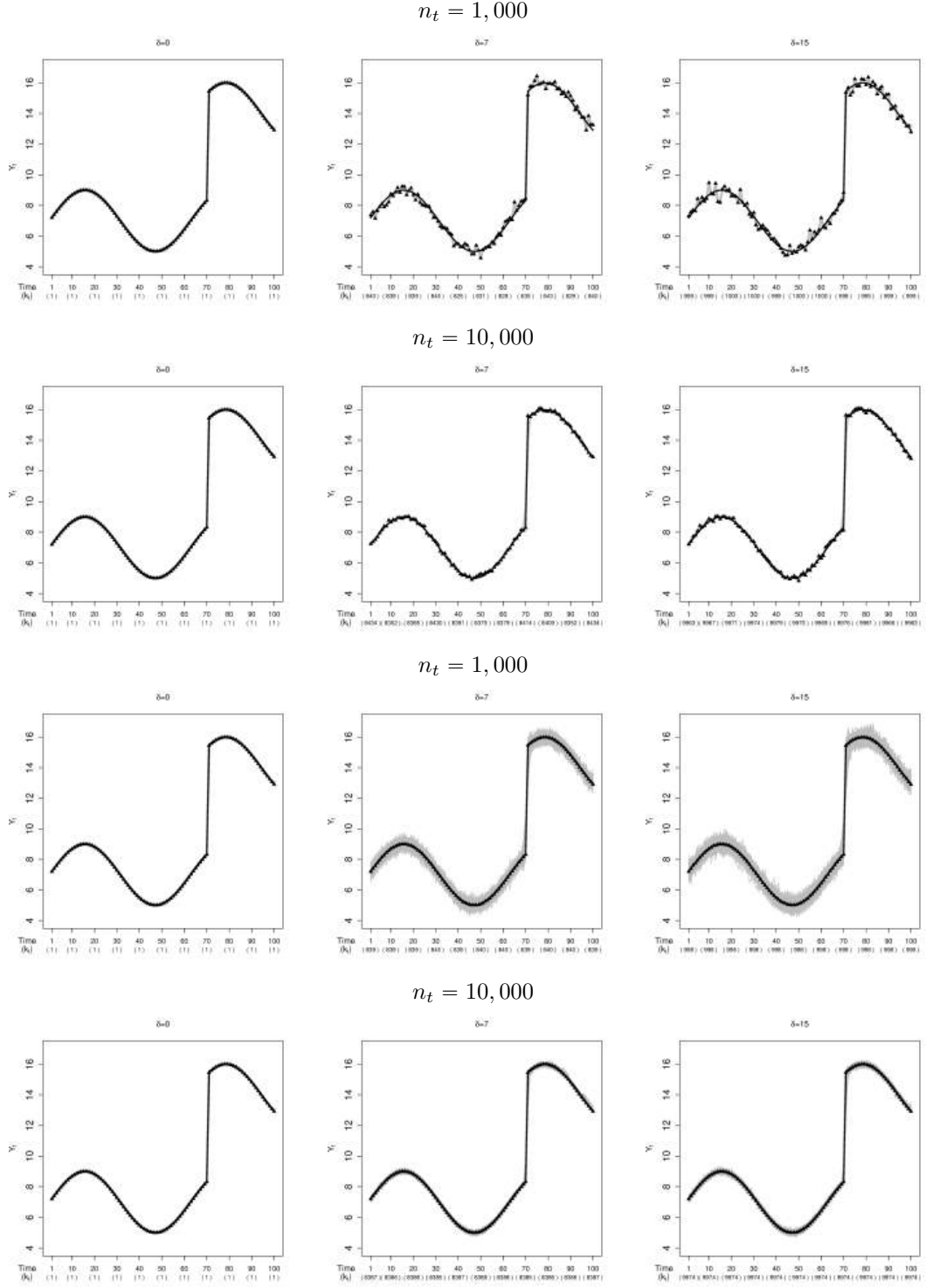


Figure A.2: Jump Scenario B: Top two rows show the single run experiment, and the bottom two rows the Monte Carlo simulation. The time-varying comparative sample mean 3.7 is represented by the triangles ($\blacktriangle$). The grey curves represent the customer smooth time-varying comparative sample means 3.9, and the black curve is the true comparative mean value 3.6. The x -axis denotes time, and the number of subjects in the window of comparison, k_t .

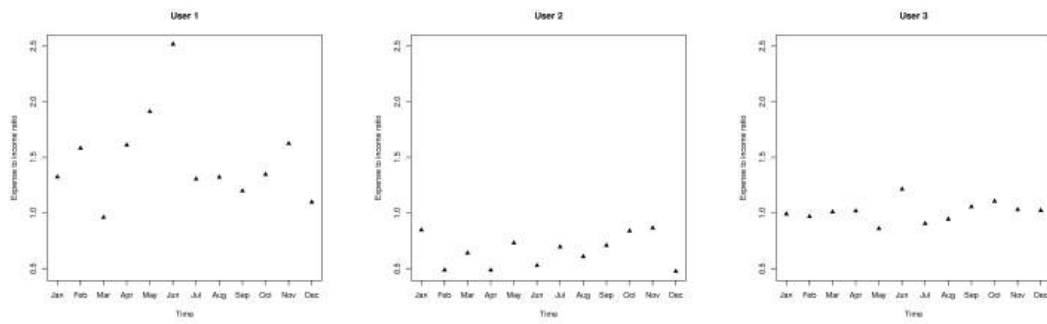


Figure A.3: Expense to income ratio: selected MD users for 2017.

Appendix B

Supplementary Information for Chapter 4

B.1 Preparation

Before getting started with the proofs we lay the groundwork. The proof of Theorem 4.2.2 uses the following representation of the conquering function

$$C(u) = \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{D_u}(\mathbf{i}), \quad (\text{B.1})$$

which follows directly from (4.10). Here, 1_A is the indicator of set A and in the proof we will make use of some of its well-known properties (e.g., Rasmussen & Williams, 2019, Chapter 1), such as

$$\limsup_{n \rightarrow \infty} 1_{A_n} = 1_{\limsup_{n \rightarrow \infty} A_n}, \quad \liminf_{n \rightarrow \infty} 1_{A_n} = 1_{\liminf_{n \rightarrow \infty} A_n}. \quad (\text{B.2})$$

Since (B.2) holds for both $\limsup$ and $\liminf$ it follows that $\lim_{n \rightarrow \infty} 1_{A_n} = 1_{\lim_{n \rightarrow \infty} A_n}$. Recall in addition that if $\{A_n\}$ is an nondecreasing sequence of sets, then its limit is the infinite union, that is

$$A_n \subseteq A_{n+1} \implies \lim_{n \rightarrow \infty} P(A_n) = \bigcup_{n=1}^{\infty} A_n. \quad (\text{B.3})$$

See, for instance, Rasmussen & Williams (2019, Proposition 1.4.1). Finally, the proof of Claim d) in Theorem 4.2.2 will make use of the Lebesgue measure over the unit interval, $\lambda([a, b]) = b - a$, for $[a, b] \subseteq [0, 1]$.

Proof of Theorem 4.2.2.

- a) Consider $(u, v) \in [0, 1]^2$ such that $u \leq v$. Then, whenever $P(A_{\mathbf{i}}) \leq u$ it follows that $P(A_{\mathbf{i}}) \leq v$; or in other words $D_u \subseteq D_v$, which in turn implies that $1_{D_u}(\mathbf{i}) \leq 1_{D_v}(\mathbf{i})$. Hence,

$$\begin{aligned} -\sum_{\mathbf{i} \in I} 1_{D_u}(\mathbf{i}) &\geq -\sum_{\mathbf{i} \in I} 1_{D_v}(\mathbf{i}) \implies \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{D_u}(\mathbf{i}) \geq \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{D_v}(\mathbf{i}) \\ &\implies C(u) \geq C(v), \end{aligned}$$

from where the final result follows. $\square$

- b) First note that the proof of Claim a) implies that D_u is a nondecreasing, in the sense $D_u \subseteq D_v$, for any $u \leq v$ with $(u, v) \in [0, 1]^2$. Next, consider an arbitrary $u \in [0, 1]$ and a sequence u_n such that $u_n \rightarrow u$, with $u_n \leq u$ for every $n \in \mathbb{N}$. Then, for a sufficiently large n it holds that $u_n \leq u_{n+1}$ which in turn implies that $D_{u_n} \subseteq D_{u_{n+1}}$. This, along with (B.3) and the fact that D_u is nondecreasing, implies that $\lim_{n \rightarrow \infty} D_{u_n} = \bigcup_{n=1}^{\infty} D_{u_n} = D_u$. Finally, (B.1) and (B.2) then yield that

$$\begin{aligned} \lim_{n \rightarrow \infty} C(u_n) &= \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} \lim_{n \rightarrow \infty} 1_{D_n}(\mathbf{i}) \\ &= \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{\lim_{n \rightarrow \infty} D_n}(\mathbf{i}) \\ &= \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{D_u}(\mathbf{i}) = C(u), \end{aligned}$$

which concludes the proof. $\square$

- c) Trivially, since Claim a) shows that $C(u)$ is nonincreasing it follows that for every $u \in [0, 1]$,

$$C(u) \geq C(1) = \prod_{j=1}^d K_j - |D_1| = 0,$$

where the final equality is a consequence of the fact that

$|D_1| = |\{\mathbf{i} : P(A_{\mathbf{i}}) \leq 1\}| = |\{A_{\mathbf{i}} : \mathbf{i} \in I\}| = \prod_{j=1}^d K_j$. The final result then follows from (B.1) and (B.2) by noting that for every $u \in [0, 1]$,

$$C(u) \leq C(0) = \prod_{j=1}^d K_j - |D_0| = \prod_{j=1}^d K_j,$$

since $|D_0| = |\{\mathbf{i} : P(A_{i,j}) \leq 0\}| = |\emptyset| = 0$. $\square$

d) First note that,

$$1_{D_u^c}(\mathbf{i}) = \begin{cases} 1, & \mathbf{i} \in D_u^c, \\ 0, & \mathbf{i} \in D_u, \end{cases} = \begin{cases} 1, & 0 \leq u < P(A_{\mathbf{i}}), \\ 0, & \text{otherwise.} \end{cases} \quad (\text{B.4})$$

Next, observe that $|D_u^c| = \sum_{\mathbf{i} \in I} 1_{D_u^c}(\mathbf{i})$ which along with (B.4) yields

$$\begin{aligned} \int_0^1 C(u) \, du &= \int_0^1 |D_u^c| \, du = \sum_{\mathbf{i} \in I} \int_0^1 1_{D_u^c}(\mathbf{i}) \, du = \sum_{\mathbf{i} \in I} \int_{[0, P(A_{\mathbf{i}})]} du \\ &= \sum_{\mathbf{i} \in I} \lambda([0, P(A_{\mathbf{i}})]) = \sum_{\mathbf{i} \in I} P(A_{\mathbf{i}}) = 1, \end{aligned}$$

which concludes the proof. $\square$

B.2 Step Function Representation, Plateau, and Edge

This part of the appendix shows formally that the conquering function is a step function with a finite number of steps (provided that $K_1, \dots, K_d$ are finite), and it uses that representation so to formally define the plateau and the edge. As a consequence of (B.1) and of (B.4) in Appendix B.1 it holds that

$$\begin{aligned} C(u) &= \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{D_u}(\mathbf{i}) = \prod_{j=1}^d K_j - \sum_{\mathbf{i} \in I} 1_{[0, P(A_{\mathbf{i}}))}(u) \\ &= \sum_{\mathbf{i} \in I} \{1 - 1_{[0, P(A_{\mathbf{i}}))}(u)\} = \sum_{\mathbf{i} \in I} 1_{[P(A_{\mathbf{i}}), 1]}(u), \end{aligned} \quad (\text{B.5})$$

where the final equality follows from the well-known property of the indicator, $1 - 1_B = 1_{B^c}$. Hence, Equation (B.5) shows that $C(u)$ is a step function with a maximum of $|I| = \prod_{j=1}^d K_j$ steps. Given this representation, it follows that

$$\text{plateau} = \sup \left\{ u : C(u) = \max_{\mathbf{i} \in I} [P(A_{\mathbf{i}}^c)] \right\}, \quad \text{edge} = \arg \max_u C(u) - C(u^+).$$

In words, the plateau is the value at which the longest plateau of $C(u)$ ends, and the edge is the value at which the largest jump on $C(u)$ occurs.

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